Raytracer

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- file cpu_renderer.h

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- file gpu_realtime.h
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- file gpu_renderer.h
- file gpu_types.cuh

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- file hybrid_renderer.cu
- file hybrid_renderer.h

5.4 src/math Directory Reference

Files

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- file interval.cuh
- file random.cuh
- file ray.cuh
- file vec3.cuh

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- file camera.cpp
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- directory hybrid
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- directory scene
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Files

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Files

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- file image.cpp
- file image.h
- file timer.h

Chapter 6

Class Documentation

6.1 AABB Class Reference

Public Member Functions

- RT_HOSTDEV AABB ()
- RT_HOSTDEV AABB (const Interval &x, const Interval &y, const Interval &z)
- RT_HOSTDEV AABB (const Point3 &a, const Point3 &b)
- RT_HOSTDEV const Interval & axis_interval (int n) const
- RT_HOSTDEV Point3 min_point () const
- RT_HOSTDEV Point3 max_point () const
- RT_HOSTDEV bool hit (const Ray &r, double t_min, double t_max) const
- RT_HOSTDEV bool hit (const Ray &r, Interval ray_t) const
- RT_HOSTDEV double surface_area () const
- RT_HOSTDEV int longest_axis () const

Public Attributes

- Interval x
- Interval y
- Interval z

Static Public Attributes

- static const AABB empty = AABB(Interval::empty, Interval::empty, Interval::empty)
- static const AABB universe = AABB(Interval::universe, Interval::universe, Interval::universe)

6.1.1 Detailed Description

Definition at line 16 of file aabb.cuh.

6.1.2 Constructor & Destructor Documentation

6.1.2.1 AABB() [1/3]

```
RT_HOSTDEV AABB::AABB () [inline]
```

Definition at line 20 of file aabb.cuh.

6.1.2.2 AABB() [2/3]

```
RT_HOSTDEV AABB::AABB (
            const Interval & x,
            const Interval & y,
            const Interval & z) [inline]
```

Definition at line 22 of file aabb.cuh.

6.1.2.3 AABB() [3/3]

```
RT_HOSTDEV AABB::AABB (
            const Point3 & a,
            const Point3 & b) [inline]
```

Definition at line 26 of file aabb.cuh.

6.1.3 Member Function Documentation

6.1.3.1 axis_interval()

```
RT_HOSTDEV const Interval & AABB::axis_interval (
            int n) const [inline]
```

Definition at line 34 of file aabb.cuh.

6.1.3.2 hit() [1/2]

```
RT_HOSTDEV bool AABB::hit (
            const Ray & r,
            double t_min,
            double t_max) const [inline]
```

Definition at line 47 of file aabb.cuh.

6.1.3.3 hit() [2/2]

```
RT_HOSTDEV bool AABB::hit (
            const Ray & r,
            Interval ray_t) const [inline]
```

Definition at line 79 of file aabb.cuh.

6.1.3.4 longest_axis()

```
RT_HOSTDEV int AABB::longest_axis () const  [inline]
```

Definition at line 92 of file aabb.cuh.

6.1.3.5 max_point()

```
RT_HOSTDEV Point3 AABB::max_point () const  [inline]
```

Definition at line 41 of file aabb.cuh.

6.1.3.6 min_point()

```
RT_HOSTDEV Point3 AABB::min_point () const  [inline]
```

Definition at line 40 of file aabb.cuh.

6.1.3.7 surface_area()

```
RT_HOSTDEV double AABB::surface_area () const  [inline]
```

Definition at line 86 of file aabb.cuh.

6.1.4 Member Data Documentation

6.1.4.1 empty

```
const AABB AABB::empty = AABB(Interval::empty, Interval::empty, Interval::empty)  [inline],
[static]
```

Definition at line 98 of file aabb.cuh.

6.1.4.2 universe

```
const AABB AABB::universe = AABB(Interval::universe, Interval::universe, Interval::universe)
[inline], [static]
```

Definition at line 99 of file aabb.cuh.

6.1.4.3 x

```
Interval AABB::x
```

Definition at line 18 of file aabb.cuh.

6.1.4.4 y

```
Interval AABB::y
```

Definition at line 18 of file aabb.cuh.

6.1.4.5 z

```
Interval AABB::z
```

Definition at line 18 of file aabb.cuh.

The documentation for this class was generated from the following file:

- src/math/aabb.cuh

6.2 BVHAccel::BuildNode Struct Reference

Public Attributes

- AABB bbox
- BuildNode ∗ children [2] = {nullptr, nullptr}
- int split_axis = 0
- int first_prim_offset = 0
- int num_primitives = 0

6.2.1 Detailed Description

Definition at line 28 of file bvh_builder.h.

6.2.2 Member Data Documentation

6.2.2.1 bbox

```
AABB BVHAccel::BuildNode::bbox
```

Definition at line 29 of file bvh_builder.h.

6.2.2.2 children

```
BuildNode* BVHAccel::BuildNode::children[2] = {nullptr, nullptr}
```

Definition at line 30 of file bvh_builder.h.

6.2.2.3 first_prim_offset

```
int BVHAccel::BuildNode::first_prim_offset = 0
```

Definition at line 32 of file bvh_builder.h.

6.2.2.4 num_primitives

```
int BVHAccel::BuildNode::num_primitives = 0
```

Definition at line 33 of file bvh_builder.h.

6.2.2.5 split_axis

```
int BVHAccel::BuildNode::split_axis = 0
```

Definition at line 31 of file bvh_builder.h.

The documentation for this struct was generated from the following file:

- src/scene/bvh_builder.h

6.3 BVHAccel::BuildPrimInfo Struct Reference

Public Attributes

- size_t prim_index
- AABB bbox
- Point3 centroid

6.3.1 Detailed Description

Definition at line 22 of file bvh_builder.h.

6.3.2 Member Data Documentation

6.3.2.1 bbox

```
AABB BVHAccel::BuildPrimInfo::bbox
```

Definition at line 24 of file bvh_builder.h.

6.3.2.2 centroid

```
Point3 BVHAccel::BuildPrimInfo::centroid
```

Definition at line 25 of file bvh_builder.h.

6.3.2.3 prim_index

```
size_t BVHAccel::BuildPrimInfo::prim_index
```

Definition at line 23 of file bvh_builder.h.

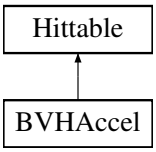
The documentation for this struct was generated from the following file:

- src/scene/bvh_builder.h

6.4 BVHAccel Class Reference

```
#include <bvh_builder.h>
```

Inheritance diagram for BVHAccel:



Classes

- struct BuildPrimInfo
- struct BuildNode

Public Member Functions

- BVHAccel (const HittableList &list)
- bool hit (const Ray &r, Interval ray_t, HitRecord &rec) const override
- AABB bounding_box () const override

Public Member Functions inherited from Hittable

- virtual ~Hittable ()=default

Private Member Functions

- BuildNode * recursive_build (std::vector< BuildPrimInfo > &prim_info, int start, int end, std::vector< std::↩ shared_ptr< Hittable > > &src_prims, int &total_nodes)
- int flatten_tree (BuildNode *node, int &offset)
- void delete_tree (BuildNode *node)

Private Attributes

- std::vector< FlatBVHNode > nodes_
- std::vector< std::shared_ptr< Hittable > > ordered_prims_

6.4.1 Detailed Description

Definition at line 11 of file bvh_builder.h.

6.4.2 Constructor & Destructor Documentation

6.4.2.1 BVHAccel()

```
BVHAccel::BVHAccel (
            const HittableList & list)
```

Definition at line 11 of file bvh_builder.cpp.

6.4.3 Member Function Documentation

6.4.3.1 bounding_box()

```
AABB BVHAccel::bounding_box () const  [override], [virtual]
```

Implements Hittable.

Definition at line 250 of file bvh_builder.cpp.

6.4.3.2 delete_tree()

```
void BVHAccel::delete_tree (
            BuildNode * node)  [private]
```

Definition at line 196 of file bvh_builder.cpp.

6.4.3.3 flatten_tree()

```
int BVHAccel::flatten_tree (
            BuildNode * node,
            int & offset)  [private]
```

Definition at line 174 of file bvh_builder.cpp.

6.4.3.4 hit()

```
bool BVHAccel::hit (
            const Ray & r,
            Interval ray_t,
            HitRecord & rec) const  [override], [virtual]
```

Implements Hittable.

Definition at line 203 of file bvh_builder.cpp.

6.4.3.5 recursive_build()

```
BVHAccel::BuildNode * BVHAccel::recursive_build (
            std::vector< BuildPrimInfo > & prim_info,
            int start,
            int end,
            std::vector< std::shared_ptr< Hittable > > & src_prims,
            int & total_nodes)  [private]
```

Definition at line 37 of file bvh_builder.cpp.

6.4.4 Member Data Documentation

6.4.4.1 nodes_

```
std::vector<FlatBVHNode> BVHAccel::nodes_  [private]
```

Definition at line 19 of file bvh_builder.h.

6.4.4.2 ordered_prims_

```
std::vector<std::shared_ptr<Hittable> > BVHAccel::ordered_prims_  [private]
```

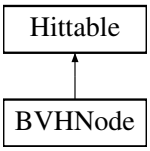
Definition at line 20 of file bvh_builder.h.

The documentation for this class was generated from the following files:

- src/scene/bvh_builder.h
- src/scene/bvh_builder.cpp

6.5 BVHNode Class Reference

Inheritance diagram for BVHNode:



Public Member Functions

- BVHNode (std::vector< std::shared_ptr< Hittable > > &objects, size_t start, size_t end)
- BVHNode (const class HittableList &list)
- bool hit (const Ray &r, Interval ray_t, HitRecord &rec) const override
- AABB bounding_box () const override

Public Member Functions inherited from Hittable

- virtual ~Hittable ()=default

Private Attributes

- std::shared_ptr< Hittable > left_
- std::shared_ptr< Hittable > right_
- AABB bbox_
- int split_axis_ = 0

6.5.1 Detailed Description

Definition at line 23 of file bvh_node.cuh.

6.5.2 Constructor & Destructor Documentation

6.5.2.1 BVHNode() [1/2]

```
BVHNode::BVHNode (
            std::vector< std::shared_ptr< Hittable > > & objects,
            size_t start,
            size_t end)
```

Definition at line 264 of file bvh_builder.cpp.

6.5.2.2 BVHNode() [2/2]

```
BVHNode::BVHNode (
            const class HittableList & list)
```

6.5.3 Member Function Documentation

6.5.3.1 bounding_box()

```
AABB BVHNode::bounding_box () const  [override], [virtual]
```

Implements Hittable.

Definition at line 308 of file bvh_builder.cpp.

6.5.3.2 hit()

```
bool BVHNode::hit (
            const Ray & r,
            Interval ray_t,
            HitRecord & rec) const  [override], [virtual]
```

Implements Hittable.

Definition at line 293 of file bvh_builder.cpp.

6.5.4 Member Data Documentation

6.5.4.1 bbox_

```
AABB BVHNode::bbox_  [private]
```

Definition at line 34 of file bvh_node.cuh.

6.5.4.2 left_

```
std::shared_ptr<Hittable> BVHNode::left_  [private]
```

Definition at line 32 of file bvh_node.cuh.

6.5.4.3 right_

```
std::shared_ptr<Hittable> BVHNode::right_  [private]
```

Definition at line 33 of file bvh_node.cuh.

6.5.4.4 split_axis_

```
int BVHNode::split_axis_ = 0  [private]
```

Definition at line 35 of file bvh_node.cuh.

The documentation for this class was generated from the following files:

- src/scene/bvh_node.cuh
- src/scene/bvh_builder.cpp

6.6 Camera Class Reference

Public Member Functions

- void initialize ()
- int get_image_width () const
- int get_image_height () const
- Ray get_ray (int i, int j) const

Public Attributes

- double aspect_ratio = 16.0 / 9.0
- int image_width = 400
- int samples_per_pixel = 100
- int max_depth = 50
- double vfov = 90.0
- Point3 lookfrom = Point3(0, 0, 0)
- Point3 lookat = Point3(0, 0, -1)
- Vec3 vup = Vec3(0, 1, 0)
- double defocus_angle = 0.0
- double focus_dist = 10.0

Private Member Functions

- Vec3 sample_square () const
- Point3 defocus_disk_sample () const

Private Attributes

- int image_height_
- Point3 center_
- Point3 pixel00_loc_
- Vec3 pixel_delta_u_
- Vec3 pixel_delta_v_
- Vec3 u_
- Vec3 v_
- Vec3 w_
- Vec3 defocus_disk_u_
- Vec3 defocus_disk_v_

6.6.1 Detailed Description

Definition at line 7 of file camera.cuh.

6.6.2 Member Function Documentation

6.6.2.1 defocus_disk_sample()

```
Point3 Camera::defocus_disk_sample () const  [private]
```

Definition at line 58 of file camera.cpp.

6.6.2.2 get_image_height()

```
int Camera::get_image_height () const  [inline]
```

Definition at line 28 of file camera.cuh.

6.6.2.3 get_image_width()

```
int Camera::get_image_width () const  [inline]
```

Definition at line 27 of file camera.cuh.

6.6.2.4 get_ray()

```
Ray Camera::get_ray (
            int i,
            int j) const
```

Definition at line 42 of file camera.cpp.

6.6.2.5 initialize()

```
void Camera::initialize ()
```

Definition at line 6 of file camera.cpp.

6.6.2.6 sample_square()

```
Vec3 Camera::sample_square () const  [private]
```

Definition at line 54 of file camera.cpp.

6.6.3 Member Data Documentation

6.6.3.1 aspect_ratio

```
double Camera::aspect_ratio = 16.0 / 9.0
```

Definition at line 10 of file camera.cuh.

6.6.3.2 center_

```
Point3 Camera::center_  [private]
```

Definition at line 35 of file camera.cuh.

6.6.3.3 defocus_angle

```
double Camera::defocus_angle = 0.0
```

Definition at line 20 of file camera.cuh.

6.6.3.4 defocus_disk_u_

```
Vec3 Camera::defocus_disk_u_ [private]
```

Definition at line 40 of file camera.cuh.

6.6.3.5 defocus_disk_v_

```
Vec3 Camera::defocus_disk_v_ [private]
```

Definition at line 41 of file camera.cuh.

6.6.3.6 focus_dist

```
double Camera::focus_dist = 10.0
```

Definition at line 21 of file camera.cuh.

6.6.3.7 image_height_

```
int Camera::image_height_ [private]
```

Definition at line 34 of file camera.cuh.

6.6.3.8 image_width

```
int Camera::image_width = 400
```

Definition at line 11 of file camera.cuh.

6.6.3.9 lookat

```
Point3 Camera::lookat = Point3(0, 0, -1)
```

Definition at line 17 of file camera.cuh.

6.6.3.10 lookfrom

```
Point3 Camera::lookfrom = Point3(0, 0, 0)
```

Definition at line 16 of file camera.cuh.

6.6.3.11 max_depth

```
int Camera::max_depth = 50
```

Definition at line 13 of file camera.cuh.

6.6.3.12 pixel00_loc_

```
Point3 Camera::pixel00_loc_ [private]
```

Definition at line 36 of file camera.cuh.

6.6.3.13 pixel_delta_u_

```
Vec3 Camera::pixel_delta_u_ [private]
```

Definition at line 37 of file camera.cuh.

6.6.3.14 pixel_delta_v_

```
Vec3 Camera::pixel_delta_v_ [private]
```

Definition at line 38 of file camera.cuh.

6.6.3.15 samples_per_pixel

```
int Camera::samples_per_pixel = 100
```

Definition at line 12 of file camera.cuh.

6.6.3.16 u_

```
Vec3 Camera::u_ [private]
```

Definition at line 39 of file camera.cuh.

6.6.3.17 v_

```
Vec3 Camera::v_ [private]
```

Definition at line 39 of file camera.cuh.

6.6.3.18 vfov

```
double Camera::vfov = 90.0
```

Definition at line 15 of file camera.cuh.

6.6.3.19 vup

```
Vec3 Camera::vup = Vec3(0, 1, 0)
```

Definition at line 18 of file camera.cuh.

6.6.3.20 w_

```
Vec3 Camera::w_ [private]
```

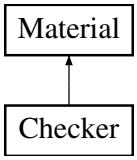
Definition at line 39 of file camera.cuh.

The documentation for this class was generated from the following files:

- src/scene/camera.cuh
- src/scene/camera.cpp

6.7 Checker Class Reference

Inheritance diagram for Checker:



Public Member Functions

- Checker (const Color &color1, const Color &color2, double scale=1.0)
- bool scatter (const Ray &r_in, const HitRecord &rec, Color &attenuation, Ray &scattered) const override
- const Color & color1 () const
- const Color & color2 () const
- double scale () const

Public Member Functions inherited from Material

- virtual ~Material ()=default
- Type type () const

Private Attributes

- Color color1_
- Color color2_
- double scale_

Additional Inherited Members

Public Types inherited from Material

- enum class Type : uint8_t {
  Lambertian , Metal , Dielectric , Checker ,
  Emissive , Unknown }

Protected Attributes inherited from Material

- Type type_ = Type::Unknown

6.7.1 Detailed Description

Definition at line 80 of file material.cuh.

6.7.2 Constructor & Destructor Documentation

6.7.2.1 Checker()

```
Checker::Checker (
            const Color & color1,
            const Color & color2,
            double scale = 1.0)
```

Definition at line 89 of file material.cpp.

6.7.3 Member Function Documentation

6.7.3.1 color1()

```
const Color & Checker::color1 () const  [inline]
```

Definition at line 87 of file material.cuh.

6.7.3.2 color2()

```
const Color & Checker::color2 () const  [inline]
```

Definition at line 88 of file material.cuh.

6.7.3.3 scale()

```
double Checker::scale () const  [inline]
```

Definition at line 89 of file material.cuh.

6.7.3.4 scatter()

```
bool Checker::scatter (
            const Ray & r_in,
            const HitRecord & rec,
            Color & attenuation,
            Ray & scattered) const  [override], [virtual]
```

Implements Material.

Definition at line 94 of file material.cpp.

6.7.4 Member Data Documentation

6.7.4.1 color1_

```
Color Checker::color1_ [private]
```

Definition at line 92 of file material.cuh.

6.7.4.2 color2_

```
Color Checker::color2_ [private]
```

Definition at line 92 of file material.cuh.

6.7.4.3 scale_

```
double Checker::scale_ [private]
```

Definition at line 93 of file material.cuh.

The documentation for this class was generated from the following files:

- src/scene/material.cuh
- src/scene/material.cpp

6.8 CpuRenderer Class Reference

```
#include <cpu_renderer.h>
```

Public Member Functions

- CpuRenderer (const Scene &scene, ImageBuffer &image)
- void render ()
- void render_threaded (int num_threads=0)
- void enable_bvh ()
- void render_tile (int x0, int y0, int x1, int y1) const

Private Member Functions

- const Hittable & get_world () const
- Color ray_color (const Ray &r, const Hittable &world, int depth) const

Private Attributes

- const Scene & scene_
- ImageBuffer & image_
- std::shared_ptr< Hittable > bvh_root_
- bool use_bvh_ = false

6.8.1 Detailed Description

Definition at line 10 of file cpu_renderer.h.

6.8.2 Constructor & Destructor Documentation

6.8.2.1 CpuRenderer()

```
CpuRenderer::CpuRenderer (
            const Scene & scene,
            ImageBuffer & image)
```

Definition at line 14 of file cpu_renderer.cpp.

6.8.3 Member Function Documentation

6.8.3.1 enable_bvh()

```
void CpuRenderer::enable_bvh ()
```

Definition at line 17 of file cpu_renderer.cpp.

6.8.3.2 get_world()

```
const Hittable & CpuRenderer::get_world () const  [private]
```

Definition at line 22 of file cpu_renderer.cpp.

6.8.3.3 ray_color()

```
Color CpuRenderer::ray_color (
            const Ray & r,
            const Hittable & world,
            int depth) const  [private]
```

Definition at line 188 of file cpu_renderer.cpp.

6.8.3.4 render()

```
void CpuRenderer::render ()
```

Definition at line 75 of file cpu_renderer.cpp.

6.8.3.5 render_threaded()

```
void CpuRenderer::render_threaded (
            int num_threads = 0)
```

Definition at line 130 of file cpu_renderer.cpp.

6.8.3.6 render_tile()

```
void CpuRenderer::render_tile (
            int x0,
            int y0,
            int x1,
            int y1) const
```

Definition at line 98 of file cpu_renderer.cpp.

6.8.4 Member Data Documentation

6.8.4.1 bvh_root_

```
std::shared_ptr<Hittable> CpuRenderer::bvh_root_ [private]
```

Definition at line 29 of file cpu_renderer.h.

6.8.4.2 image_

```
ImageBuffer& CpuRenderer::image_ [private]
```

Definition at line 28 of file cpu_renderer.h.

6.8.4.3 scene_

```
const Scene& CpuRenderer::scene_ [private]
```

Definition at line 27 of file cpu_renderer.h.

6.8.4.4 use_bvh_

```
bool CpuRenderer::use_bvh_ = false [private]
```

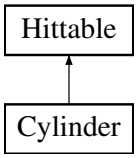
Definition at line 30 of file cpu_renderer.h.

The documentation for this class was generated from the following files:

- src/cpu/cpu_renderer.h
- src/cpu/cpu_renderer.cpp

6.9 Cylinder Class Reference

Inheritance diagram for Cylinder:



Public Member Functions

- Cylinder (const Point3 &base, double radius, double height, const Material ∗material)
- bool hit (const Ray &r, Interval ray_t, HitRecord &rec) const override
- AABB bounding_box () const override

Public Member Functions inherited from Hittable

- virtual ∼Hittable ()=default

Private Attributes

- Point3 base_
- double radius_
- double height_
- const Material ∗ material_

6.9.1 Detailed Description

Definition at line 74 of file primitives.cuh.

6.9.2 Constructor & Destructor Documentation

6.9.2.1 Cylinder()

```
Cylinder::Cylinder (
            const Point3 & base,
            double radius,
            double height,
            const Material * material)
```

Definition at line 95 of file primitives.cpp.

6.9.3 Member Function Documentation

6.9.3.1 bounding_box()

```
AABB Cylinder::bounding_box () const  [override], [virtual]
```

Implements Hittable.

Definition at line 134 of file primitives.cpp.

6.9.3.2 hit()

```
bool Cylinder::hit (
            const Ray & r,
            Interval ray_t,
            HitRecord & rec) const  [override], [virtual]
```

Implements Hittable.

Definition at line 98 of file primitives.cpp.

6.9.4 Member Data Documentation

6.9.4.1 base_

```
Point3 Cylinder::base_  [private]
```

Definition at line 82 of file primitives.cuh.

6.9.4.2 height_

```
double Cylinder::height_  [private]
```

Definition at line 84 of file primitives.cuh.

6.9.4.3 material_

```
const Material* Cylinder::material_  [private]
```

Definition at line 85 of file primitives.cuh.

6.9.4.4 radius_

```
double Cylinder::radius_  [private]
```

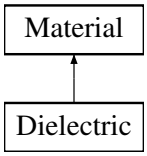
Definition at line 83 of file primitives.cuh.

The documentation for this class was generated from the following files:

- src/scene/primitives.cuh
- src/scene/primitives.cpp

6.10 Dielectric Class Reference

Inheritance diagram for Dielectric:



Public Member Functions

- Dielectric (double refraction_index)
- bool scatter (const Ray &r_in, const HitRecord &rec, Color &attenuation, Ray &scattered) const override
- double refraction_index () const

Public Member Functions inherited from Material

- virtual ~Material ()=default
- Type type () const

Static Public Member Functions

- static double reflectance (double cosine, double refraction_index)

Private Attributes

- double refraction_index_

Additional Inherited Members

Public Types inherited from Material

- enum class Type : uint8_t {
  Lambertian , Metal , Dielectric , Checker ,
  Emissive , Unknown }

Protected Attributes inherited from Material

- Type type_ = Type::Unknown

6.10.1 Detailed Description

Definition at line 62 of file material.cuh.

6.10.2 Constructor & Destructor Documentation

6.10.2.1 Dielectric()

```
Dielectric::Dielectric (
            double refraction_index)  [explicit]
```

Definition at line 50 of file material.cpp.

6.10.3 Member Function Documentation

6.10.3.1 reflectance()

```
double Dielectric::reflectance (
            double cosine,
            double refraction_index)  [static]
```

Definition at line 76 of file material.cpp.

6.10.3.2 refraction_index()

```
double Dielectric::refraction_index () const  [inline]
```

Definition at line 69 of file material.cuh.

6.10.3.3 scatter()

```
bool Dielectric::scatter (
            const Ray & r_in,
            const HitRecord & rec,
            Color & attenuation,
            Ray & scattered) const  [override], [virtual]
```

Implements Material.

Definition at line 55 of file material.cpp.

6.10.4 Member Data Documentation

6.10.4.1 refraction_index_

```
double Dielectric::refraction_index_  [private]
```

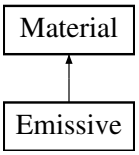
Definition at line 75 of file material.cuh.

The documentation for this class was generated from the following files:

- src/scene/material.cuh
- src/scene/material.cpp

6.11 Emissive Class Reference

Inheritance diagram for Emissive:



Public Member Functions

- Emissive (const Color &emit_color, double intensity=1.0)
- bool scatter (const Ray &r_in, const HitRecord &rec, Color &attenuation, Ray &scattered) const override
- Color emitted () const
- const Color & emit_color () const
- double intensity () const

Public Member Functions inherited from Material

- virtual ~Material ()=default
- Type type () const

Private Attributes

- Color emit_
- double intensity_

Additional Inherited Members

Public Types inherited from Material

- enum class Type : uint8_t {
  Lambertian , Metal , Dielectric , Checker ,
  Emissive , Unknown }

Protected Attributes inherited from Material

- Type type_ = Type::Unknown

6.11.1 Detailed Description

Definition at line 98 of file material.cuh.

6.11.2 Constructor & Destructor Documentation

6.11.2.1 Emissive()

```
Emissive::Emissive (
            const Color & emit_color,
            double intensity = 1.0)
```

Definition at line 115 of file material.cpp.

6.11.3 Member Function Documentation

6.11.3.1 emit_color()

```
const Color & Emissive::emit_color () const  [inline]
```

Definition at line 106 of file material.cuh.

6.11.3.2 emitted()

```
Color Emissive::emitted () const  [inline]
```

Definition at line 105 of file material.cuh.

6.11.3.3 intensity()

```
double Emissive::intensity () const  [inline]
```

Definition at line 107 of file material.cuh.

6.11.3.4 scatter()

```
bool Emissive::scatter (
            const Ray & r_in,
            const HitRecord & rec,
            Color & attenuation,
            Ray & scattered) const  [override], [virtual]
```

Implements Material.

Definition at line 120 of file material.cpp.

6.11.4 Member Data Documentation

6.11.4.1 emit_

```
Color Emissive::emit_  [private]
```

Definition at line 110 of file material.cuh.

6.11.4.2 intensity_

```
double Emissive::intensity_ [private]
```

Definition at line 111 of file material.cuh.

The documentation for this class was generated from the following files:

- src/scene/material.cuh
- src/scene/material.cpp

6.12 FlatBVHNode Struct Reference

Public Attributes

- AABB bbox
- union {
  uint32_t primitives_offset
  uint32_t second_child_offset
  };
- uint16_t num_primitives
- uint8_t split_axis
- uint8_t pad

6.12.1 Detailed Description

Definition at line 11 of file bvh_node.cuh.

6.12.2 Member Data Documentation

6.12.2.1 [union]

```
union { ... } FlatBVHNode
```

6.12.2.2 bbox

```
AABB FlatBVHNode::bbox
```

Definition at line 12 of file bvh_node.cuh.

6.12.2.3 num_primitives

```
uint16_t FlatBVHNode::num_primitives
```

Definition at line 17 of file bvh_node.cuh.

6.12.2.4 pad

```
uint8_t FlatBVHNode::pad
```

Definition at line 19 of file bvh_node.cuh.

6.12.2.5 primitives_offset

```
uint32_t FlatBVHNode::primitives_offset
```

Definition at line 14 of file bvh_node.cuh.

6.12.2.6 second_child_offset

```
uint32_t FlatBVHNode::second_child_offset
```

Definition at line 15 of file bvh_node.cuh.

6.12.2.7 split_axis

```
uint8_t FlatBVHNode::split_axis
```

Definition at line 18 of file bvh_node.cuh.

The documentation for this struct was generated from the following file:

- src/scene/bvh_node.cuh

6.13 GAABB Struct Reference

Public Member Functions

- __host__ __device__ GAABB ()
- __host__ __device__ GAABB (const GVec3 &a, const GVec3 &b)
- __host__ __device__ bool hit (const GRay &r, float t_min, float t_max) const

Public Attributes

- GVec3 mn
- GVec3 mx

6.13.1 Detailed Description

Definition at line 77 of file gpu_types.cuh.

6.13.2 Constructor & Destructor Documentation

6.13.2.1 GAABB() [1/2]

```
__host__ __device__ GAABB::GAABB () [inline]
```

Definition at line 80 of file gpu_types.cuh.

6.13.2.2 GAABB() [2/2]

```
__host__ __device__ GAABB::GAABB (
            const GVec3 & a,
            const GVec3 & b) [inline]
```

Definition at line 81 of file gpu_types.cuh.

6.13.3 Member Function Documentation

6.13.3.1 hit()

```
__host__ __device__ bool GAABB::hit (
            const GRay & r,
            float t_min,
            float t_max) const [inline]
```

Definition at line 83 of file gpu_types.cuh.

6.13.4 Member Data Documentation

6.13.4.1 mn

```
GVec3 GAABB::mn
```

Definition at line 78 of file gpu_types.cuh.

6.13.4.2 mx

```
GVec3 GAABB::mx
```

Definition at line 78 of file gpu_types.cuh.

The documentation for this struct was generated from the following file:

- src/gpu/gpu_types.cuh

6.14 GBuildInfo Struct Reference

Public Attributes

- int sphere_idx
- GAABB bbox
- GVec3 centroid

6.14.1 Detailed Description

Definition at line 723 of file gpu_renderer.cu.

6.14.2 Member Data Documentation

6.14.2.1 bbox

```
GAABB GBuildInfo::bbox
```

Definition at line 723 of file gpu_renderer.cu.

6.14.2.2 centroid

```
GVec3 GBuildInfo::centroid
```

Definition at line 723 of file gpu_renderer.cu.

6.14.2.3 sphere_idx

```
int GBuildInfo::sphere_idx
```

Definition at line 723 of file gpu_renderer.cu.

The documentation for this struct was generated from the following file:

- src/gpu/gpu_renderer.cu

6.15 GBVHNode Struct Reference

Public Attributes

- GAABB bbox
- uint32_t data
- uint16_t num_prims
- uint8_t split_axis
- uint8_t prim_type

6.15.1 Detailed Description

Definition at line 160 of file gpu_types.cuh.

6.15.2 Member Data Documentation

6.15.2.1 bbox

```
GAABB GBVHNode::bbox
```

Definition at line 161 of file gpu_types.cuh.

6.15.2.2 data

```
uint32_t GBVHNode::data
```

Definition at line 162 of file gpu_types.cuh.

6.15.2.3 num_prims

```
uint16_t GBVHNode::num_prims
```

Definition at line 163 of file gpu_types.cuh.

6.15.2.4 prim_type

```
uint8_t GBVHNode::prim_type
```

Definition at line 165 of file gpu_types.cuh.

6.15.2.5 split_axis

```
uint8_t GBVHNode::split_axis
```

Definition at line 164 of file gpu_types.cuh.

The documentation for this struct was generated from the following file:

- src/gpu/gpu_types.cuh

6.16 GCamera Struct Reference

Public Attributes

- GVec3 origin
- GVec3 pixel00_loc
- GVec3 pixel_delta_u
- GVec3 pixel_delta_v
- GVec3 defocus_disk_u
- GVec3 defocus_disk_v
- float defocus_angle
- int image_width
- int image_height
- int samples_per_pixel
- int max_depth

6.16.1 Detailed Description

Definition at line 171 of file gpu_types.cuh.

6.16.2 Member Data Documentation

6.16.2.1 defocus_angle

```
float GCamera::defocus_angle
```

Definition at line 178 of file gpu_types.cuh.

6.16.2.2 defocus_disk_u

```
GVec3 GCamera::defocus_disk_u
```

Definition at line 176 of file gpu_types.cuh.

6.16.2.3 defocus_disk_v

```
GVec3 GCamera::defocus_disk_v
```

Definition at line 177 of file gpu_types.cuh.

6.16.2.4 image_height

```
int GCamera::image_height
```

Definition at line 180 of file gpu_types.cuh.

6.16.2.5 image_width

```
int GCamera::image_width
```

Definition at line 179 of file gpu_types.cuh.

6.16.2.6 max_depth

```
int GCamera::max_depth
```

Definition at line 182 of file gpu_types.cuh.

6.16.2.7 origin

```
GVec3 GCamera::origin
```

Definition at line 172 of file gpu_types.cuh.

6.16.2.8 pixel00_loc

```
GVec3 GCamera::pixel00_loc
```

Definition at line 173 of file gpu_types.cuh.

6.16.2.9 pixel_delta_u

```
GVec3 GCamera::pixel_delta_u
```

Definition at line 174 of file gpu_types.cuh.

6.16.2.10 pixel_delta_v

```
GVec3 GCamera::pixel_delta_v
```

Definition at line 175 of file gpu_types.cuh.

6.16.2.11 samples_per_pixel

```
int GCamera::samples_per_pixel
```

Definition at line 181 of file gpu_types.cuh.

The documentation for this struct was generated from the following file:

- src/gpu/gpu_types.cuh

6.17 GHitRecord Struct Reference

Public Member Functions

- __host__ __device__ void set_face_normal (const GRay &r, const GVec3 &outward_normal)

Public Attributes

- GVec3 p
- GVec3 normal
- float t
- int material_id
- bool front_face

6.17.1 Detailed Description

Definition at line 144 of file gpu_types.cuh.

6.17.2 Member Function Documentation

6.17.2.1 set_face_normal()

```
__host__ __device__ void GHitRecord::set_face_normal (
            const GRay & r,
            const GVec3 & outward_normal)  [inline]
```

Definition at line 151 of file gpu_types.cuh.

6.17.3 Member Data Documentation

6.17.3.1 front_face

```
bool GHitRecord::front_face
```

Definition at line 149 of file gpu_types.cuh.

6.17.3.2 material_id

```
int GHitRecord::material_id
```

Definition at line 148 of file gpu_types.cuh.

6.17.3.3 normal

```
GVec3 GHitRecord::normal
```

Definition at line 146 of file gpu_types.cuh.

6.17.3.4 p

```
GVec3 GHitRecord::p
```

Definition at line 145 of file gpu_types.cuh.

6.17.3.5 t

```
float GHitRecord::t
```

Definition at line 147 of file gpu_types.cuh.

The documentation for this struct was generated from the following file:

- src/gpu/gpu_types.cuh

6.18 GMaterial Struct Reference

Public Member Functions

- __host__ __device__ GMaterial ()

Public Attributes

- GVec3 albedo
- float param
- GMaterialType type

6.18.1 Detailed Description

Definition at line 104 of file gpu_types.cuh.

6.18.2 Constructor & Destructor Documentation

6.18.2.1 GMaterial()

```
__host__ __device__ GMaterial::GMaterial () [inline]
```

Definition at line 109 of file gpu_types.cuh.

6.18.3 Member Data Documentation

6.18.3.1 albedo

```
GVec3 GMaterial::albedo
```

Definition at line 105 of file gpu_types.cuh.

6.18.3.2 param

```
float GMaterial::param
```

Definition at line 106 of file gpu_types.cuh.

6.18.3.3 type

```
GMaterialType GMaterial::type
```

Definition at line 107 of file gpu_types.cuh.

The documentation for this struct was generated from the following file:

- src/gpu/gpu_types.cuh

6.19 GPlane Struct Reference

Public Attributes

- GVec3 point
- GVec3 normal
- int material_id

6.19.1 Detailed Description

Definition at line 135 of file gpu_types.cuh.

6.19.2 Member Data Documentation

6.19.2.1 material_id

```
int GPlane::material_id
```

Definition at line 138 of file gpu_types.cuh.

6.19.2.2 normal

```
GVec3 GPlane::normal
```

Definition at line 137 of file gpu_types.cuh.

6.19.2.3 point

```
GVec3 GPlane::point
```

Definition at line 136 of file gpu_types.cuh.

The documentation for this struct was generated from the following file:

- src/gpu/gpu_types.cuh

6.20 GpuRenderer Class Reference

```
#include <gpu_renderer.h>
```

Public Member Functions

- GpuRenderer ()
- ~GpuRenderer ()
- void upload_scene (const GSphere *spheres, int num_spheres, const GPlane *planes, int num_planes, const GMaterial *materials, int num_materials, const GBVHNode *bvh_nodes, int num_bvh_nodes)
- void render (const GCamera &camera, float *h_output, int width, int height)
- float last_render_ms () const
- void ensure_framebuffer (int width, int height)

Private Member Functions

- void free_scene ()
- void free_framebuffer ()

Private Attributes

- GSphere * d_spheres_ = nullptr
- float * d_soa_cx_ = nullptr
- float * d_soa_cy_ = nullptr
- float * d_soa_cz_ = nullptr
- float * d_soa_r_ = nullptr
- int * d_soa_mat_ = nullptr
- GPlane * d_planes_ = nullptr
- GMaterial * d_materials_ = nullptr
- GBVHNode * d_bvh_nodes_ = nullptr
- float * d_framebuffer_ = nullptr
- int num_spheres_ = 0
- int num_planes_ = 0
- int num_materials_ = 0
- int num_bvh_nodes_ = 0
- int fb_width_ = 0
- int fb_height_ = 0
- float last_ms_ = 0.0f

Friends

- float gpu_realtime_render_frame (const Scene &scene)
- void gpu_render_scene_denoised (const Scene &scene, ImageBuffer &image, float &kernel_ms, float sigma_color=0.1f)

6.20.1 Detailed Description

Definition at line 18 of file gpu_renderer.h.

6.20.2 Constructor & Destructor Documentation

6.20.2.1 GpuRenderer()

```
GpuRenderer::GpuRenderer ()
```

Definition at line 443 of file gpu_renderer.cu.

6.20.2.2 ~GpuRenderer()

```
GpuRenderer::~GpuRenderer ()
```

Definition at line 444 of file gpu_renderer.cu.

6.20.3 Member Function Documentation

6.20.3.1 ensure_framebuffer()

```
void GpuRenderer::ensure_framebuffer (
            int width,
            int height)
```

Definition at line 463 of file gpu_renderer.cu.

6.20.3.2 free_framebuffer()

```
void GpuRenderer::free_framebuffer () [private]
```

Definition at line 458 of file gpu_renderer.cu.

6.20.3.3 free_scene()

```
void GpuRenderer::free_scene () [private]
```

Definition at line 446 of file gpu_renderer.cu.

6.20.3.4 last_render_ms()

```
float GpuRenderer::last_render_ms () const  [inline]
```

Definition at line 33 of file gpu_renderer.h.

6.20.3.5 render()

```
void GpuRenderer::render (
            const GCamera & camera,
            float * h_output,
            int width,
            int height)
```

Definition at line 521 of file gpu_renderer.cu.

6.20.3.6 upload_scene()

```
void GpuRenderer::upload_scene (
            const GSphere * spheres,
            int num_spheres,
            const GPlane * planes,
            int num_planes,
            const GMaterial * materials,
            int num_materials,
            const GBVHNode * bvh_nodes,
            int num_bvh_nodes)
```

Definition at line 470 of file gpu_renderer.cu.

6.20.4 Friends And Related Symbol Documentation

6.20.4.1 gpu_realtime_render_frame

```
float gpu_realtime_render_frame (
            const Scene & scene)  [friend]
```

Definition at line 948 of file gpu_renderer.cu.

6.20.4.2 gpu_render_scene_denoised

```
void gpu_render_scene_denoised (
            const Scene & scene,
            ImageBuffer & image,
            float & kernel_ms,
            float sigma_color = 0.1f)  [friend]
```

Definition at line 861 of file gpu_renderer.cu.

6.20.5 Member Data Documentation

6.20.5.1 d_bvh_nodes_

```
GBVHNode* GpuRenderer::d_bvh_nodes_ = nullptr  [private]
```

Definition at line 49 of file gpu_renderer.h.

6.20.5.2 d_framebuffer_

```
float* GpuRenderer::d_framebuffer_ = nullptr  [private]
```

Definition at line 50 of file gpu_renderer.h.

6.20.5.3 d_materials_

```
GMaterial* GpuRenderer::d_materials_ = nullptr  [private]
```

Definition at line 48 of file gpu_renderer.h.

6.20.5.4 d_planes_

```
GPlane* GpuRenderer::d_planes_ = nullptr  [private]
```

Definition at line 47 of file gpu_renderer.h.

6.20.5.5 d_soa_cx_

```
float* GpuRenderer::d_soa_cx_ = nullptr  [private]
```

Definition at line 42 of file gpu_renderer.h.

6.20.5.6 d_soa_cy_

```
float* GpuRenderer::d_soa_cy_ = nullptr  [private]
```

Definition at line 43 of file gpu_renderer.h.

6.20.5.7 d_soa_cz_

```
float* GpuRenderer::d_soa_cz_ = nullptr  [private]
```

Definition at line 44 of file gpu_renderer.h.

6.20.5.8 d_soa_mat_

```
int* GpuRenderer::d_soa_mat_ = nullptr  [private]
```

Definition at line 46 of file gpu_renderer.h.

6.20.5.9 d_soa_r_

```
float* GpuRenderer::d_soa_r_ = nullptr  [private]
```

Definition at line 45 of file gpu_renderer.h.

6.20.5.10 d_spheres_

```
GSphere* GpuRenderer::d_spheres_ = nullptr  [private]
```

Definition at line 40 of file gpu_renderer.h.

6.20.5.11 fb_height_

```
int GpuRenderer::fb_height_ = 0  [private]
```

Definition at line 57 of file gpu_renderer.h.

6.20.5.12 fb_width_

```
int GpuRenderer::fb_width_ = 0  [private]
```

Definition at line 56 of file gpu_renderer.h.

6.20.5.13 last_ms_

```
float GpuRenderer::last_ms_ = 0.0f  [private]
```

Definition at line 59 of file gpu_renderer.h.

6.20.5.14 num_bvh_nodes_

```
int GpuRenderer::num_bvh_nodes_ = 0  [private]
```

Definition at line 55 of file gpu_renderer.h.

6.20.5.15 num_materials_

```
int GpuRenderer::num_materials_ = 0  [private]
```

Definition at line 54 of file gpu_renderer.h.

6.20.5.16 num_planes_

```
int GpuRenderer::num_planes_ = 0  [private]
```

Definition at line 53 of file gpu_renderer.h.

6.20.5.17 num_spheres_

```
int GpuRenderer::num_spheres_ = 0  [private]
```

Definition at line 52 of file gpu_renderer.h.

The documentation for this class was generated from the following files:

- src/gpu/gpu_renderer.h
- src/gpu/gpu_renderer.cu

6.21 GRay Struct Reference

Public Member Functions

- __host__ __device__ GRay ()
- __host__ __device__ GRay (const GVec3 &o, const GVec3 &d)
- __host__ __device__ GVec3 at (float t) const

Public Attributes

- GVec3 orig
- GVec3 dir
- GVec3 inv_dir
- GVec3 orig_mul_inv

6.21.1 Detailed Description

Definition at line 59 of file gpu_types.cuh.

6.21.2 Constructor & Destructor Documentation

6.21.2.1 GRay() [1/2]

```
__host__ __device__ GRay::GRay ()  [inline]
```

Definition at line 65 of file gpu_types.cuh.

6.21.2.2 GRay() [2/2]

```
__host__ __device__ GRay::GRay (
            const GVec3 & o,
            const GVec3 & d)  [inline]
```

Definition at line 66 of file gpu_types.cuh.

6.21.3 Member Function Documentation

6.21.3.1 at()

```
__host__ __device__ GVec3 GRay::at (
            float t) const  [inline]
```

Definition at line 71 of file gpu_types.cuh.

6.21.4 Member Data Documentation

6.21.4.1 dir

```
GVec3 GRay::dir
```

Definition at line 61 of file gpu_types.cuh.

6.21.4.2 inv_dir

```
GVec3 GRay::inv_dir
```

Definition at line 62 of file gpu_types.cuh.

6.21.4.3 orig

```
GVec3 GRay::orig
```

Definition at line 60 of file gpu_types.cuh.

6.21.4.4 orig_mul_inv

```
GVec3 GRay::orig_mul_inv
```

Definition at line 63 of file gpu_types.cuh.

The documentation for this struct was generated from the following file:

- src/gpu/gpu_types.cuh

6.22 GSphere Struct Reference

Public Attributes

- GVec3 center
- float radius
- int material_id

6.22.1 Detailed Description

Definition at line 116 of file gpu_types.cuh.

6.22.2 Member Data Documentation

6.22.2.1 center

```
GVec3 GSphere::center
```

Definition at line 117 of file gpu_types.cuh.

6.22.2.2 material_id

```
int GSphere::material_id
```

Definition at line 119 of file gpu_types.cuh.

6.22.2.3 radius

```
float GSphere::radius
```

Definition at line 118 of file gpu_types.cuh.

The documentation for this struct was generated from the following file:

- src/gpu/gpu_types.cuh

6.23 GSphereSoA Struct Reference

Public Attributes

- float * cx
- float * cy
- float * cz
- float * r
- int * mat
- int count

6.23.1 Detailed Description

Definition at line 123 of file gpu_types.cuh.

6.23.2 Member Data Documentation

6.23.2.1 count

```
int GSphereSoA::count
```

Definition at line 129 of file gpu_types.cuh.

6.23.2.2 cx

```
float* GSphereSoA::cx
```

Definition at line 124 of file gpu_types.cuh.

6.23.2.3 cy

```
float* GSphereSoA::cy
```

Definition at line 125 of file gpu_types.cuh.

6.23.2.4 cz

```
float* GSphereSoA::cz
```

Definition at line 126 of file gpu_types.cuh.

6.23.2.5 mat

```
int* GSphereSoA::mat
```

Definition at line 128 of file gpu_types.cuh.

6.23.2.6 r

```
float* GSphereSoA::r
```

Definition at line 127 of file gpu_types.cuh.

The documentation for this struct was generated from the following file:

- src/gpu/gpu_types.cuh

6.24 HitRecord Struct Reference

Public Member Functions

- void set_face_normal (const Ray &r, const Vec3 &outward_normal)

Public Attributes

- Point3 p
- Vec3 normal
- const Material ∗ material = nullptr
- double t = 0.0
- bool front_face = true

6.24.1 Detailed Description

Definition at line 15 of file primitives.cuh.

6.24.2 Member Function Documentation

6.24.2.1 set_face_normal()

```
void HitRecord::set_face_normal (
            const Ray & r,
            const Vec3 & outward_normal)  [inline]
```

Definition at line 23 of file primitives.cuh.

6.24.3 Member Data Documentation

6.24.3.1 front_face

```
bool HitRecord::front_face = true
```

Definition at line 20 of file primitives.cuh.

6.24.3.2 material

```
const Material* HitRecord::material = nullptr
```

Definition at line 18 of file primitives.cuh.

6.24.3.3 normal

```
Vec3 HitRecord::normal
```

Definition at line 17 of file primitives.cuh.

6.24.3.4 p

```
Point3 HitRecord::p
```

Definition at line 16 of file primitives.cuh.

6.24.3.5 t

```
double HitRecord::t = 0.0
```

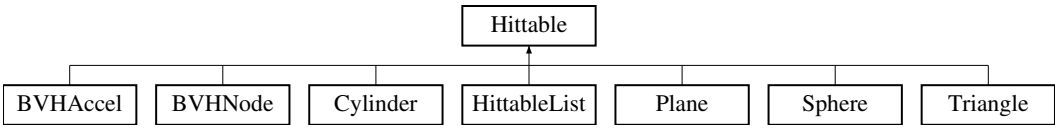
Definition at line 19 of file primitives.cuh.

The documentation for this struct was generated from the following file:

- src/scene/primitives.cuh

6.25 Hittable Class Reference

Inheritance diagram for Hittable:



Public Member Functions

- virtual ~Hittable ()=default
- virtual bool hit (const Ray &r, Interval ray_t, HitRecord &rec) const =0
- virtual AABB bounding_box () const =0

6.25.1 Detailed Description

Definition at line 31 of file primitives.cuh.

6.25.2 Constructor & Destructor Documentation

6.25.2.1 ~Hittable()

```
virtual Hittable::~Hittable () [virtual], [default]
```

6.25.3 Member Function Documentation

6.25.3.1 bounding_box()

```
virtual AABB Hittable::bounding_box () const  [pure virtual]
```

Implemented in BVHAccel, BVHNode, Cylinder, HittableList, Plane, Sphere, and Triangle.

6.25.3.2 hit()

```
virtual bool Hittable::hit (
            const Ray & r,
            Interval ray_t,
            HitRecord & rec) const  [pure virtual]
```

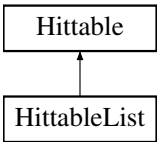
Implemented in BVHAccel, BVHNode, Cylinder, HittableList, Plane, Sphere, and Triangle.

The documentation for this class was generated from the following file:

- src/scene/primitives.cuh

6.26 HittableList Class Reference

Inheritance diagram for HittableList:



Public Member Functions

- HittableList ()=default
- void add (std::shared_ptr< Hittable > object)
- void clear ()
- bool hit (const Ray &r, Interval ray_t, HitRecord &rec) const override
- AABB bounding_box () const override
- const std::vector< std::shared_ptr< Hittable > > & objects () const

Public Member Functions inherited from Hittable

- virtual ~Hittable ()=default

Private Attributes

- std::vector< std::shared_ptr< Hittable > > objects_
- AABB bbox_

6.26.1 Detailed Description

Definition at line 105 of file primitives.cuh.

6.26.2 Constructor & Destructor Documentation

6.26.2.1 HittableList()

```
HittableList::HittableList () [default]
```

6.26.3 Member Function Documentation

6.26.3.1 add()

```
void HittableList::add (
            std::shared_ptr< Hittable > object)
```

Definition at line 199 of file primitives.cpp.

6.26.3.2 bounding_box()

```
AABB HittableList::bounding_box () const [override], [virtual]
```

Implements Hittable.

Definition at line 224 of file primitives.cpp.

6.26.3.3 clear()

```
void HittableList::clear ()
```

Definition at line 204 of file primitives.cpp.

6.26.3.4 hit()

```
bool HittableList::hit (
            const Ray & r,
            Interval ray_t,
            HitRecord & rec) const [override], [virtual]
```

Implements Hittable.

Definition at line 208 of file primitives.cpp.

6.26.3.5 objects()

```
const std::vector< std::shared_ptr< Hittable > > & HittableList::objects () const [inline]
```

Definition at line 115 of file primitives.cuh.

6.26.4 Member Data Documentation

6.26.4.1 bbox_

```
AABB HittableList::bbox_ [private]
```

Definition at line 119 of file primitives.cuh.

6.26.4.2 objects_

```
std::vector<std::shared_ptr<Hittable> > HittableList::objects_ [private]
```

Definition at line 118 of file primitives.cuh.

The documentation for this class was generated from the following files:

- src/scene/primitives.cuh
- src/scene/primitives.cpp

6.27 ImageBuffer Class Reference

```
#include <image.h>
```

Public Member Functions

- ImageBuffer (int width, int height)
- int width () const
- int height () const
- void set_pixel (int x, int y, const Color &color)
- Color get_pixel (int x, int y) const
- void write_ppm (std::ostream &out) const
- void write_ppm (const std::string &filename) const
- void write_ppm_binary (const std::string &filename) const

Private Member Functions

- int index (int x, int y) const

Static Private Member Functions

- static int to_byte (double linear)

Private Attributes

- int width_
- int height_
- std::vector< Color > pixels_

6.27.1 Detailed Description

Definition at line 9 of file image.h.

6.27.2 Constructor & Destructor Documentation

6.27.2.1 ImageBuffer()

```
ImageBuffer::ImageBuffer (
            int width,
            int height)
```

Definition at line 7 of file image.cpp.

6.27.3 Member Function Documentation

6.27.3.1 get_pixel()

```
Color ImageBuffer::get_pixel (
            int x,
            int y) const
```

Definition at line 14 of file image.cpp.

6.27.3.2 height()

```
int ImageBuffer::height () const  [inline]
```

Definition at line 14 of file image.h.

6.27.3.3 index()

```
int ImageBuffer::index (
            int x,
            int y) const  [inline], [private]
```

Definition at line 31 of file image.h.

6.27.3.4 set_pixel()

```
void ImageBuffer::set_pixel (
            int x,
            int y,
            const Color & color)
```

Definition at line 10 of file image.cpp.

6.27.3.5 to_byte()

```
int ImageBuffer::to_byte (
            double linear)  [static], [private]
```

Definition at line 26 of file image.cpp.

6.27.3.6 width()

```
int ImageBuffer::width () const  [inline]
```

Definition at line 13 of file image.h.

6.27.3.7 write_ppm() [1/2]

```
void ImageBuffer::write_ppm (
            const std::string & filename) const
```

Definition at line 45 of file image.cpp.

6.27.3.8 write_ppm() [2/2]

```
void ImageBuffer::write_ppm (
            std::ostream & out) const
```

Definition at line 33 of file image.cpp.

6.27.3.9 write_ppm_binary()

```
void ImageBuffer::write_ppm_binary (
            const std::string & filename) const
```

Definition at line 52 of file image.cpp.

6.27.4 Member Data Documentation

6.27.4.1 height_

```
int ImageBuffer::height_  [private]
```

Definition at line 28 of file image.h.

6.27.4.2 pixels_

```
std::vector<Color> ImageBuffer::pixels_ [private]
```

Definition at line 29 of file image.h.

6.27.4.3 width_

```
int ImageBuffer::width_ [private]
```

Definition at line 27 of file image.h.

The documentation for this class was generated from the following files:

- src/utils/image.h
- src/utils/image.cpp

6.28 Interval Class Reference

Public Member Functions

- RT_HOSTDEV Interval ()
- RT_HOSTDEV Interval (double min, double max)
- RT_HOSTDEV double size () const
- RT_HOSTDEV bool contains (double x) const
- RT_HOSTDEV bool surrounds (double x) const
- RT_HOSTDEV double clamp (double x) const

Public Attributes

- double min
- double max

Static Public Attributes

- static const Interval empty
- static const Interval universe

6.28.1 Detailed Description

Definition at line 13 of file interval.cuh.

6.28.2 Constructor & Destructor Documentation

6.28.2.1 Interval() [1/2]

```
RT_HOSTDEV Interval::Interval () [inline]
```

Definition at line 18 of file interval.cuh.

6.28.2.2 Interval() [2/2]

```
RT_HOSTDEV Interval::Interval (
            double min,
            double max)  [inline]
```

Definition at line 22 of file interval.cuh.

6.28.3 Member Function Documentation

6.28.3.1 clamp()

```
RT_HOSTDEV double Interval::clamp (
            double x) const  [inline]
```

Definition at line 33 of file interval.cuh.

6.28.3.2 contains()

```
RT_HOSTDEV bool Interval::contains (
            double x) const  [inline]
```

Definition at line 27 of file interval.cuh.

6.28.3.3 size()

```
RT_HOSTDEV double Interval::size () const  [inline]
```

Definition at line 24 of file interval.cuh.

6.28.3.4 surrounds()

```
RT_HOSTDEV bool Interval::surrounds (
            double x) const  [inline]
```

Definition at line 31 of file interval.cuh.

6.28.4 Member Data Documentation

6.28.4.1 empty

```
const Interval Interval::empty  [inline], [static]
```

Initial value:

```
= Interval(+std::numeric_limits<double>::infinity(),
           -std::numeric_limits<double>::infinity())
```

Definition at line 39 of file interval.cuh.

6.28.4.2 max

```
double Interval::max
```

Definition at line 15 of file interval.cuh.

6.28.4.3 min

```
double Interval::min
```

Definition at line 15 of file interval.cuh.

6.28.4.4 universe

```
const Interval Interval::universe  [inline], [static]
```

Initial value:

```
= Interval(-std::numeric_limits<double>::infinity(),
           +std::numeric_limits<double>::infinity())
```

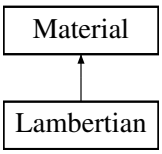
Definition at line 40 of file interval.cuh.

The documentation for this class was generated from the following file:

- src/math/interval.cuh

6.29 Lambertian Class Reference

Inheritance diagram for Lambertian:



Public Member Functions

- Lambertian (const Color &albedo)
- bool scatter (const Ray &r_in, const HitRecord &rec, Color &attenuation, Ray &scattered) const override
- const Color & albedo () const

Public Member Functions inherited from Material

- virtual ~Material ()=default
- Type type () const

Private Attributes

- Color albedo_

Additional Inherited Members

Public Types inherited from Material

- enum class Type : uint8_t {
  Lambertian , Metal , Dielectric , Checker ,
  Emissive , Unknown }

Protected Attributes inherited from Material

- Type type_ = Type::Unknown

6.29.1 Detailed Description

Definition at line 30 of file material.cuh.

6.29.2 Constructor & Destructor Documentation

6.29.2.1 Lambertian()

```
Lambertian::Lambertian (
            const Color & albedo)  [explicit]
```

Definition at line 11 of file material.cpp.

6.29.3 Member Function Documentation

6.29.3.1 albedo()

```
const Color & Lambertian::albedo () const  [inline]
```

Definition at line 37 of file material.cuh.

6.29.3.2 scatter()

```
bool Lambertian::scatter (
            const Ray & r_in,
            const HitRecord & rec,
            Color & attenuation,
            Ray & scattered) const  [override], [virtual]
```

Implements Material.

Definition at line 15 of file material.cpp.

6.29.4 Member Data Documentation

6.29.4.1 albedo_

```
Color Lambertian::albedo_ [private]
```

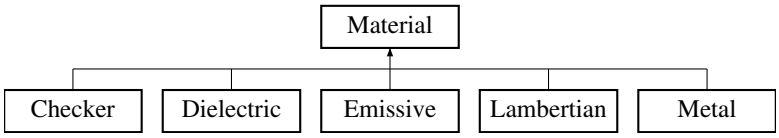
Definition at line 40 of file material.cuh.

The documentation for this class was generated from the following files:

- src/scene/material.cuh
- src/scene/material.cpp

6.30 Material Class Reference

Inheritance diagram for Material:



Public Types

- enum class Type : uint8_t {
  Lambertian , Metal , Dielectric , Checker ,
  Emissive , Unknown }

Public Member Functions

- virtual ~Material ()=default
- virtual bool scatter (const Ray &r_in, const HitRecord &rec, Color &attenuation, Ray &scattered) const =0
- Type type () const

Protected Attributes

- Type type_ = Type::Unknown

6.30.1 Detailed Description

Definition at line 12 of file material.cuh.

6.30.2 Member Enumeration Documentation

6.30.2.1 Type

```
enum class Material::Type : uint8_t [strong]
```

Enumerator

Lambertian	
Metal	
Dielectric	
Checker	
Emissive	
Unknown	

Definition at line 15 of file material.cuh.

6.30.3 Constructor & Destructor Documentation

6.30.3.1 ~Material()

```
virtual Material::~Material () [virtual], [default]
```

6.30.4 Member Function Documentation

6.30.4.1 scatter()

```
virtual bool Material::scatter (
            const Ray & r_in,
            const HitRecord & rec,
            Color & attenuation,
            Ray & scattered) const [pure virtual]
```

Implemented in Checker, Dielectric, Emissive, Lambertian, and Metal.

6.30.4.2 type()

```
Type Material::type () const [inline]
```

Definition at line 22 of file material.cuh.

6.30.5 Member Data Documentation

6.30.5.1 type_

```
Type Material::type_ = Type::Unknown [protected]
```

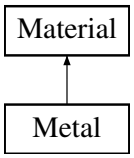
Definition at line 25 of file material.cuh.

The documentation for this class was generated from the following file:

- src/scene/material.cuh

6.31 Metal Class Reference

Inheritance diagram for Metal:



Public Member Functions

- Metal (const Color &albedo, double fuzz)
- bool scatter (const Ray &r_in, const HitRecord &rec, Color &attenuation, Ray &scattered) const override
- const Color & albedo () const
- double fuzz () const

Public Member Functions inherited from Material

- virtual ~Material ()=default
- Type type () const

Private Attributes

- Color albedo_
- double fuzz_

Additional Inherited Members

Public Types inherited from Material

- enum class Type : uint8_t {
  Lambertian , Metal , Dielectric , Checker ,
  Emissive , Unknown }

Protected Attributes inherited from Material

- Type type_ = Type::Unknown

6.31.1 Detailed Description

Definition at line 45 of file material.cuh.

6.31.2 Constructor & Destructor Documentation

6.31.2.1 Metal()

```
Metal::Metal (
            const Color & albedo,
            double fuzz)
```

Definition at line 32 of file material.cpp.

6.31.3 Member Function Documentation

6.31.3.1 albedo()

```
const Color & Metal::albedo () const  [inline]
```

Definition at line 52 of file material.cuh.

6.31.3.2 fuzz()

```
double Metal::fuzz () const  [inline]
```

Definition at line 53 of file material.cuh.

6.31.3.3 scatter()

```
bool Metal::scatter (
            const Ray & r_in,
            const HitRecord & rec,
            Color & attenuation,
            Ray & scattered) const  [override], [virtual]
```

Implements Material.

Definition at line 37 of file material.cpp.

6.31.4 Member Data Documentation

6.31.4.1 albedo_

```
Color Metal::albedo_  [private]
```

Definition at line 56 of file material.cuh.

6.31.4.2 fuzz_

```
double Metal::fuzz_  [private]
```

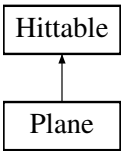
Definition at line 57 of file material.cuh.

The documentation for this class was generated from the following files:

- src/scene/material.cuh
- src/scene/material.cpp

6.32 Plane Class Reference

Inheritance diagram for Plane:



Public Member Functions

- Plane (const Point3 &point, const Vec3 &normal, const Material ∗material)
- bool hit (const Ray &r, Interval ray_t, HitRecord &rec) const override
- AABB bounding_box () const override

Public Member Functions inherited from Hittable

- virtual ∼Hittable ()=default

Private Attributes

- Point3 point_
- Vec3 normal_
- const Material ∗ material_

6.32.1 Detailed Description

Definition at line 59 of file primitives.cuh.

6.32.2 Constructor & Destructor Documentation

6.32.2.1 Plane()

```
Plane::Plane (
            const Point3 & point,
            const Vec3 & normal,
            const Material * material)
```

Definition at line 50 of file primitives.cpp.

6.32.3 Member Function Documentation

6.32.3.1 bounding_box()

```
AABB Plane::bounding_box () const  [override], [virtual]
```

Implements Hittable.

Definition at line 73 of file primitives.cpp.

6.32.3.2 hit()

```
bool Plane::hit (
            const Ray & r,
            Interval ray_t,
            HitRecord & rec) const  [override], [virtual]
```

Implements Hittable.

Definition at line 53 of file primitives.cpp.

6.32.4 Member Data Documentation

6.32.4.1 material_

```
const Material* Plane::material_  [private]
```

Definition at line 69 of file primitives.cuh.

6.32.4.2 normal_

```
Vec3 Plane::normal_  [private]
```

Definition at line 68 of file primitives.cuh.

6.32.4.3 point_

```
Point3 Plane::point_  [private]
```

Definition at line 67 of file primitives.cuh.

The documentation for this class was generated from the following files:

- src/scene/primitives.cuh
- src/scene/primitives.cpp

6.33 Ray Class Reference

Public Member Functions

- RT_HOSTDEV Ray ()
- RT_HOSTDEV Ray (const Point3 &origin, const Vec3 &direction)
- RT_HOSTDEV const Point3 & origin () const
- RT_HOSTDEV const Vec3 & direction () const
- RT_HOSTDEV Point3 at (double t) const

Public Attributes

- Point3 orig
- Vec3 dir
- Vec3 inv_dir
- int dir_sign [3]

6.33.1 Detailed Description

Definition at line 11 of file ray.cuh.

6.33.2 Constructor & Destructor Documentation

6.33.2.1 Ray() [1/2]

```
RT_HOSTDEV Ray::Ray ()  [inline]
```

Definition at line 18 of file ray.cuh.

6.33.2.2 Ray() [2/2]

```
RT_HOSTDEV Ray::Ray (
            const Point3 & origin,
            const Vec3 & direction)  [inline]
```

Definition at line 19 of file ray.cuh.

6.33.3 Member Function Documentation

6.33.3.1 at()

```
RT_HOSTDEV Point3 Ray::at (
            double t) const  [inline]
```

Definition at line 33 of file ray.cuh.

6.33.3.2 direction()

```
RT_HOSTDEV const Vec3 & Ray::direction () const  [inline]
```

Definition at line 30 of file ray.cuh.

6.33.3.3 origin()

```
RT_HOSTDEV const Point3 & Ray::origin () const  [inline]
```

Definition at line 29 of file ray.cuh.

6.33.4 Member Data Documentation

6.33.4.1 dir

```
Vec3 Ray::dir
```

Definition at line 14 of file ray.cuh.

6.33.4.2 dir_sign

```
int Ray::dir_sign[3]
```

Definition at line 16 of file ray.cuh.

6.33.4.3 inv_dir

```
Vec3 Ray::inv_dir
```

Definition at line 15 of file ray.cuh.

6.33.4.4 orig

```
Point3 Ray::orig
```

Definition at line 13 of file ray.cuh.

The documentation for this class was generated from the following file:

- src/math/ray.cuh

6.34 Scene Class Reference

```
#include <scene.h>
```

Public Member Functions

- template<typename T, typename... Args>
  T * add_material (Args &&... args)

Static Public Member Functions

- static Scene build_three_spheres ()
- static Scene build_final_scene ()
- static Scene build_triangle_scene ()
- static Scene build_cornell_box ()
- static Scene build_showcase ()
- static Scene build_hall ()
- static Scene build_galaxy ()
- static Scene build_minimal ()

Public Attributes

- Camera camera
- HittableList world

Private Attributes

- std::vector< std::unique_ptr< Material > > materials_

6.34.1 Detailed Description

Definition at line 10 of file scene.h.

6.34.2 Member Function Documentation

6.34.2.1 add_material()

```
template<typename T, typename... Args>
T * Scene::add_material (
            Args &&... args) [inline]
```

Definition at line 17 of file scene.h.

6.34.2.2 build_cornell_box()

```
Scene Scene::build_cornell_box () [static]
```

Definition at line 153 of file scene.cpp.

6.34.2.3 build_final_scene()

```
Scene Scene::build_final_scene () [static]
```

Definition at line 45 of file scene.cpp.

6.34.2.4 build_galaxy()

```
Scene Scene::build_galaxy () [static]
```

Definition at line 298 of file scene.cpp.

6.34.2.5 build_hall()

```
Scene Scene::build_hall () [static]
```

Definition at line 259 of file scene.cpp.

6.34.2.6 build_minimal()

```
Scene Scene::build_minimal () [static]
```

Definition at line 335 of file scene.cpp.

6.34.2.7 build_showcase()

```
Scene Scene::build_showcase () [static]
```

Definition at line 208 of file scene.cpp.

6.34.2.8 build_three_spheres()

```
Scene Scene::build_three_spheres () [static]
```

Definition at line 9 of file scene.cpp.

6.34.2.9 build_triangle_scene()

```
Scene Scene::build_triangle_scene () [static]
```

Definition at line 108 of file scene.cpp.

6.34.3 Member Data Documentation

6.34.3.1 camera

```
Camera Scene::camera
```

Definition at line 12 of file scene.h.

6.34.3.2 materials_

```
std::vector<std::unique_ptr<Material> > Scene::materials_ [private]
```

Definition at line 49 of file scene.h.

6.34.3.3 world

```
HittableList Scene::world
```

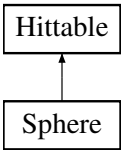
Definition at line 13 of file scene.h.

The documentation for this class was generated from the following files:

- src/scene/scene.h
- src/scene/scene.cpp

6.35 Sphere Class Reference

Inheritance diagram for Sphere:



Public Member Functions

- Sphere (const Point3 &center, double radius, const Material *material)
- bool hit (const Ray &r, Interval ray_t, HitRecord &rec) const override
- AABB bounding_box () const override
- const Point3 & center () const
- double radius () const

Public Member Functions inherited from Hittable

- virtual ~Hittable ()=default

Private Attributes

- Point3 center_
- double radius_
- const Material * material_

6.35.1 Detailed Description

Definition at line 41 of file primitives.cuh.

6.35.2 Constructor & Destructor Documentation

6.35.2.1 Sphere()

```
Sphere::Sphere (
            const Point3 & center,
            double radius,
            const Material * material)
```

Definition at line 8 of file primitives.cpp.

6.35.3 Member Function Documentation

6.35.3.1 bounding_box()

```
AABB Sphere::bounding_box () const  [override], [virtual]
```

Implements Hittable.

Definition at line 41 of file primitives.cpp.

6.35.3.2 center()

```
const Point3 & Sphere::center () const  [inline]
```

Definition at line 48 of file primitives.cuh.

6.35.3.3 hit()

```
bool Sphere::hit (
            const Ray & r,
            Interval ray_t,
            HitRecord & rec) const  [override], [virtual]
```

Implements Hittable.

Definition at line 11 of file primitives.cpp.

6.35.3.4 radius()

```
double Sphere::radius () const  [inline]
```

Definition at line 49 of file primitives.cuh.

6.35.4 Member Data Documentation

6.35.4.1 center_

```
Point3 Sphere::center_  [private]
```

Definition at line 52 of file primitives.cuh.

6.35.4.2 material_

```
const Material* Sphere::material_  [private]
```

Definition at line 54 of file primitives.cuh.

6.35.4.3 radius_

```
double Sphere::radius_  [private]
```

Definition at line 53 of file primitives.cuh.

The documentation for this class was generated from the following files:

- src/scene/primitives.cuh
- src/scene/primitives.cpp

6.36 Timer Class Reference

```
#include <timer.h>
```

Public Member Functions

- Timer ()
- void reset ()
- double elapsed () const
- double elapsed_ms () const

Private Attributes

- std::chrono::high_resolution_clock::time_point start_

6.36.1 Detailed Description

Definition at line 6 of file timer.h.

6.36.2 Constructor & Destructor Documentation

6.36.2.1 Timer()

```
Timer::Timer () [inline]
```

Definition at line 8 of file timer.h.

6.36.3 Member Function Documentation

6.36.3.1 elapsed()

```
double Timer::elapsed () const [inline]
```

Definition at line 15 of file timer.h.

6.36.3.2 elapsed_ms()

```
double Timer::elapsed_ms () const [inline]
```

Definition at line 21 of file timer.h.

6.36.3.3 reset()

```
void Timer::reset () [inline]
```

Definition at line 10 of file timer.h.

6.36.4 Member Data Documentation

6.36.4.1 start_

```
std::chrono::high_resolution_clock::time_point Timer::start_ [private]
```

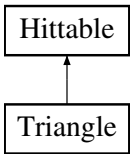
Definition at line 26 of file timer.h.

The documentation for this class was generated from the following file:

- src/utils/timer.h

6.37 Triangle Class Reference

Inheritance diagram for Triangle:



Public Member Functions

- Triangle (const Point3 &v0, const Point3 &v1, const Point3 &v2, const Material *material)
- bool hit (const Ray &r, Interval ray_t, HitRecord &rec) const override
- AABB bounding_box () const override

Public Member Functions inherited from Hittable

- virtual ~Hittable ()=default

Private Attributes

- Point3 v0_
- Point3 v1_
- Point3 v2_
- Vec3 normal_
- const Material * material_

6.37.1 Detailed Description

Definition at line 90 of file primitives.cuh.

6.37.2 Constructor & Destructor Documentation

6.37.2.1 Triangle()

```
Triangle::Triangle (
            const Point3 & v0,
            const Point3 & v1,
            const Point3 & v2,
            const Material * material)
```

Definition at line 145 of file primitives.cpp.

6.37.3 Member Function Documentation

6.37.3.1 bounding_box()

```
AABB Triangle::bounding_box () const  [override], [virtual]
```

Implements Hittable.

Definition at line 181 of file primitives.cpp.

6.37.3.2 hit()

```
bool Triangle::hit (
            const Ray & r,
            Interval ray_t,
            HitRecord & rec) const  [override], [virtual]
```

Implements Hittable.

Definition at line 150 of file primitives.cpp.

6.37.4 Member Data Documentation

6.37.4.1 material_

```
const Material* Triangle::material_  [private]
```

Definition at line 100 of file primitives.cuh.

6.37.4.2 normal_

```
Vec3 Triangle::normal_  [private]
```

Definition at line 99 of file primitives.cuh.

6.37.4.3 v0_

```
Point3 Triangle::v0_ [private]
```

Definition at line 98 of file primitives.cuh.

6.37.4.4 v1_

```
Point3 Triangle::v1_ [private]
```

Definition at line 98 of file primitives.cuh.

6.37.4.5 v2_

```
Point3 Triangle::v2_ [private]
```

Definition at line 98 of file primitives.cuh.

The documentation for this class was generated from the following files:

- src/scene/primitives.cuh
- src/scene/primitives.cpp

6.38 Vec3 Class Reference

Public Member Functions

- RT_HOSTDEV Vec3 ()
- RT_HOSTDEV Vec3 (double x, double y, double z)
- RT_HOSTDEV double x () const
- RT_HOSTDEV double y () const
- RT_HOSTDEV double z () const
- RT_HOSTDEV Vec3 operator- () const
- RT_HOSTDEV double operator[ ] (int i) const
- RT_HOSTDEV double & operator[ ] (int i)
- RT_HOSTDEV Vec3 & operator+= (const Vec3 &v)
- RT_HOSTDEV Vec3 & operator-= (const Vec3 &v)
- RT_HOSTDEV Vec3 & operator*= (double t)
- RT_HOSTDEV Vec3 & operator/= (double t)
- RT_HOSTDEV double length_squared () const
- RT_HOSTDEV double length () const
- RT_HOSTDEV bool near_zero () const

Public Attributes

- double e [3]

6.38.1 Detailed Description

Definition at line 19 of file vec3.cuh.

6.38.2 Constructor & Destructor Documentation

6.38.2.1 Vec3() [1/2]

```
RT_HOSTDEV Vec3::Vec3 ()  [inline]
```

Definition at line 23 of file vec3.cuh.

6.38.2.2 Vec3() [2/2]

```
RT_HOSTDEV Vec3::Vec3 (
            double x,
            double y,
            double z)  [inline]
```

Definition at line 24 of file vec3.cuh.

6.38.3 Member Function Documentation

6.38.3.1 length()

```
RT_HOSTDEV double Vec3::length () const  [inline]
```

Definition at line 58 of file vec3.cuh.

6.38.3.2 length_squared()

```
RT_HOSTDEV double Vec3::length_squared () const  [inline]
```

Definition at line 54 of file vec3.cuh.

6.38.3.3 near_zero()

```
RT_HOSTDEV bool Vec3::near_zero () const  [inline]
```

Definition at line 64 of file vec3.cuh.

6.38.3.4 operator*=()

```
RT_HOSTDEV Vec3 & Vec3::operator*= (
            double t)  [inline]
```

Definition at line 44 of file vec3.cuh.

6.38.3.5 operator+=()

```
RT_HOSTDEV Vec3 & Vec3::operator+= (
            const Vec3 & v)  [inline]
```

Definition at line 34 of file vec3.cuh.

6.38.3.6 operator-()

```
RT_HOSTDEV Vec3 Vec3::operator- () const  [inline]
```

Definition at line 30 of file vec3.cuh.

6.38.3.7 operator-=()

```
RT_HOSTDEV Vec3 & Vec3::operator-= (
            const Vec3 & v)  [inline]
```

Definition at line 39 of file vec3.cuh.

6.38.3.8 operator/=()

```
RT_HOSTDEV Vec3 & Vec3::operator/= (
            double t)  [inline]
```

Definition at line 50 of file vec3.cuh.

6.38.3.9 operator[]() [1/2]

```
RT_HOSTDEV double & Vec3::operator[] (
            int i)  [inline]
```

Definition at line 32 of file vec3.cuh.

6.38.3.10 operator[]() [2/2]

```
RT_HOSTDEV double Vec3::operator[] (
            int i) const  [inline]
```

Definition at line 31 of file vec3.cuh.

6.38.3.11 x()

```
RT_HOSTDEV double Vec3::x () const  [inline]
```

Definition at line 26 of file vec3.cuh.

6.38.3.12 y()

```
RT_HOSTDEV double Vec3::y () const  [inline]
```

Definition at line 27 of file vec3.cuh.

6.38.3.13 z()

```
RT_HOSTDEV double Vec3::z () const  [inline]
```

Definition at line 28 of file vec3.cuh.

6.38.4 Member Data Documentation

6.38.4.1 e

```
double Vec3::e[3]
```

Definition at line 21 of file vec3.cuh.

The documentation for this class was generated from the following file:

- src/math/vec3.cuh

Chapter 7

File Documentation

7.1 src/cpu/cpu_renderer.cpp File Reference

```
#include "cpu/cpu_renderer.h"
#include "math/random.cuh"
#include "math/interval.cuh"
#include "scene/material.cuh"
#include "scene/primitives.cuh"
#include "scene/bvh_builder.h"
#include <iostream>
#include <limits>
#include <thread>
#include <atomic>
#include <vector>
#include <cmath>
```

Functions

- static bool scatter_inline (const Material *mat, const Ray &r_in, const HitRecord &rec, Color &attenuation, Ray &scattered)

7.1.1 Function Documentation

7.1.1.1 scatter_inline()

```
bool scatter_inline (
            const Material * mat,
            const Ray & r_in,
            const HitRecord & rec,
            Color & attenuation,
            Ray & scattered)  [inline], [static]
```

Definition at line 29 of file cpu_renderer.cpp.

7.2 cpu_renderer.cpp

Go to the documentation of this file.

```
#include "cpu/cpu_renderer.h"
#include "math/random.cuh"
#include "math/interval.cuh"
#include "scene/material.cuh"
#include "scene/primitives.cuh"
#include "scene/bvh_builder.h"
#include <iostream>
#include <limits>
#include <thread>
#include <atomic>
#include <vector>
#include <cmath>

CpuRenderer::CpuRenderer(const Scene& scene, ImageBuffer& image)
    : scene_(scene), image_(image) {}

void CpuRenderer::enable_bvh() {
    bvh_root_ = std::make_shared<BVHAccel>(scene_.world);
    use_bvh_ = true;
}

const Hittable& CpuRenderer::get_world() const {
    if (use_bvh_ && bvh_root_)
        return *bvh_root_;
    return scene_.world;
}

// Inline scatter to avoid virtual dispatch in the hot path
static inline bool scatter_inline(const Material* mat, const Ray& r_in,
                                  const HitRecord& rec,
                                  Color& attenuation, Ray& scattered) {
    switch (mat->type()) {
    case Material::Type::Lambertian: {
        const auto* lam = static_cast<const Lambertian*>(mat);
        Vec3 scatter_dir = rec.normal + random_unit_vector();
        if (scatter_dir.near_zero()) scatter_dir = rec.normal;
        scattered = Ray(rec.p, scatter_dir);
        attenuation = lam->albedo();
        return true;
    }
    case Material::Type::Metal: {
        const auto* met = static_cast<const Metal*>(mat);
        Vec3 reflected = reflect(unit_vector(r_in.direction()), rec.normal);
        reflected = reflected + met->fuzz() * random_in_unit_sphere();
        scattered = Ray(rec.p, reflected);
        attenuation = met->albedo();
        return (dot(scattered.direction(), rec.normal) > 0.0);
    }
    case Material::Type::Dielectric: {
        const auto* die = static_cast<const Dielectric*>(mat);
        attenuation = Color(1.0, 1.0, 1.0);
        double ri_val = die->refraction_index();
        double ri = rec.front_face ? (1.0 / ri_val) : ri_val;
        Vec3 unit_dir = unit_vector(r_in.direction());
        double cos_theta = std::fmin(dot(-unit_dir, rec.normal), 1.0);
        double sin_theta = std::sqrt(1.0 - cos_theta * cos_theta);
        bool cannot_refract = ri * sin_theta > 1.0;
        Vec3 direction;
        if (cannot_refract || Dielectric::reflectance(cos_theta, ri) > random_double())
            direction = reflect(unit_dir, rec.normal);
        else
            direction = refract(unit_dir, rec.normal, ri);
        scattered = Ray(rec.p, direction);
        return true;
    }
    case Material::Type::Checker: {
        // Devirtualise comme Lambertian mais avec couleur alternee
        return mat->scatter(r_in, rec, attenuation, scattered);
    }
    default:
        return mat->scatter(r_in, rec, attenuation, scattered);
    }
}

void CpuRenderer::render() {
    const int width = image_.width();
    const int height = image_.height();
    const int spp = scene_.camera.samples_per_pixel;
    const int max_depth = scene_.camera.max_depth;
    const double inv_spp = 1.0 / spp;
    const Hittable& world = get_world();
    const Camera& cam = scene_.camera;
```

```cpp

    for (int j = 0; j < height; ++j) {
        std::cerr << "\rScanlines remaining: " << (height - j) << ' ' << std::flush;
        for (int i = 0; i < width; ++i) {
            Color pixel_color(0, 0, 0);
            for (int s = 0; s < spp; ++s) {
                Ray r = cam.get_ray(i, j);
                pixel_color += ray_color(r, world, max_depth);
            }
            image_.set_pixel(i, j, pixel_color * inv_spp);
        }
    }
    std::cerr << "\rDone.                          \n";
}

void CpuRenderer::render_tile(int x0, int y0, int x1, int y1) const {
    const int spp = scene_.camera.samples_per_pixel;
    const int max_depth = scene_.camera.max_depth;
    const double inv_spp = 1.0 / spp;
    const Hittable& world = get_world();
    const Camera& cam = scene_.camera;

    // Process samples in pairs for better ILP
    const int spp_pairs = spp / 2;
    const int spp_remainder = spp % 2;

    for (int j = y0; j < y1; ++j) {
        for (int i = x0; i < x1; ++i) {
            Color acc0(0, 0, 0);
            Color acc1(0, 0, 0);

            for (int s = 0; s < spp_pairs; ++s) {
                Ray r0 = cam.get_ray(i, j);
                acc0 += ray_color(r0, world, max_depth);
                Ray r1 = cam.get_ray(i, j);
                acc1 += ray_color(r1, world, max_depth);
            }
            if (spp_remainder) {
                Ray r = cam.get_ray(i, j);
                acc0 += ray_color(r, world, max_depth);
            }

            image_.set_pixel(i, j, (acc0 + acc1) * inv_spp);
        }
    }
}

void CpuRenderer::render_threaded(int num_threads) {
    if (num_threads <= 0)
        num_threads = static_cast<int>(std::thread::hardware_concurrency());
    if (num_threads <= 0)
        num_threads = 4;

    const int width = image_.width();
    const int height = image_.height();
    constexpr int TILE_SIZE = 16;

    const int tiles_x = (width + TILE_SIZE - 1) / TILE_SIZE;
    const int tiles_y = (height + TILE_SIZE - 1) / TILE_SIZE;
    const int total_tiles = tiles_x * tiles_y;

    std::atomic<int> next_tile(0);
    std::atomic<int> completed_tiles(0);

    auto worker = [&]() {
        while (true) {
            int tile_idx = next_tile.fetch_add(1);
            if (tile_idx >= total_tiles)
                break;

            int tx = tile_idx % tiles_x;
            int ty = tile_idx / tiles_x;
            int x0 = tx * TILE_SIZE;
            int y0 = ty * TILE_SIZE;
            int x1 = std::min(x0 + TILE_SIZE, width);
            int y1 = std::min(y0 + TILE_SIZE, height);

            render_tile(x0, y0, x1, y1);

            int done = completed_tiles.fetch_add(1) + 1;
            if (done % 10 == 0 || done == total_tiles) {
                std::cerr << "\rTiles: " << done << "/" << total_tiles
                          << " (" << (100 * done / total_tiles) << "%)" << std::flush;
            }
        }
    };

```

```
    std::cerr « "Rendering with " « num_threads « " threads, "
              « total_tiles « " tiles (" « TILE_SIZE « "x" « TILE_SIZE « ")...\n";

    std::vector<std::thread> threads;
    threads.reserve(num_threads);
    for (int i = 0; i < num_threads; ++i)
        threads.emplace_back(worker);

    for (auto& t : threads)
        t.join();

    std::cerr « "\rDone.                                        \n";
}

// CS:APP 5.5 + 5.6: Iterative ray_color with devirtualized scatter
// - No recursion (eliminates stack frame per bounce)
// - Inline scatter avoids vtable lookup per bounce
// - Local variables eliminate repeated memory dereferences
Color CpuRenderer::ray_color(const Ray& r, const Hittable& world, int depth) const {
    Color throughput(1, 1, 1);
    Ray current_ray = r;
    constexpr double T_MIN = 0.001;
    constexpr double T_MAX = 1e30; // CS:APP 5.6: avoid std::numeric_limits call

    for (int d = 0; d < depth; ++d) {
        HitRecord rec;
        if (!world.hit(current_ray, Interval(T_MIN, T_MAX), rec)) {
            // Sky: CS:APP 5.6 - compute with local values
            const Vec3& dir = current_ray.direction();
            // Inline unit_vector Y component to avoid full normalization
            // We only need the Y component for the sky gradient
            double len_sq = dir.length_squared();
            double inv_len = 1.0 / std::sqrt(len_sq);
            double unit_y = dir.y() * inv_len;
            double a = 0.5 * (unit_y + 1.0);
            // Sky color: lerp white->blue
            Color sky(1.0 - 0.5 * a, 1.0 - 0.3 * a, 1.0);
            return throughput * sky;
        }

        // CS:APP 5.5: Devirtualized scatter - switch on type tag, no vtable
        const Material* mat = rec.material;
        if (!mat) break;

        // Area light : si emissif, retourner la lumiere emise
        if (mat->type() == Material::Type::Emissive) {
            const auto* em = static_cast<const Emissive*>(mat);
            return throughput * em->emitted();
        }

        Ray scattered;
        Color attenuation;
        if (!scatter_inline(mat, current_ray, rec, attenuation, scattered))
            break; // absorbed

        // CS:APP 5.6: Multiply throughput components directly
        throughput = throughput * attenuation;

        // Early termination + Russian Roulette (RTIOW CUDA article Opt#4+5)
        double max_comp = std::fmax(throughput.x(), std::fmax(throughput.y(), throughput.z()));
        if (max_comp < 0.001)
            break;
        if (d > 3) {
            if (random_double() > max_comp)
                break;
            throughput = throughput / max_comp;
        }

        current_ray = scattered;
    }

    return Color(0, 0, 0);
}
```

7.3 src/cpu/cpu_renderer.h File Reference

```
#include "scene/scene.h"
#include "utils/image.h"
#include <memory>
```

Classes

- class CpuRenderer

7.4 cpu_renderer.h

Go to the documentation of this file.

```
#ifndef RT_CPU_RENDERER_H
#define RT_CPU_RENDERER_H

#include "scene/scene.h"
#include "utils/image.h"
#include <memory>

class Hittable;

class CpuRenderer {
public:
    CpuRenderer(const Scene& scene, ImageBuffer& image);

    // Single-threaded render
    void render();

    // Multi-threaded render with tile decomposition
    void render_threaded(int num_threads = 0);  // 0 = auto-detect

    // Enable BVH acceleration (builds BVH from scene)
    void enable_bvh();

    // Render a single tile (used by hybrid renderer)
    void render_tile(int x0, int y0, int x1, int y1) const;

private:
    const Scene& scene_;
    ImageBuffer& image_;
    std::shared_ptr<Hittable> bvh_root_;
    bool use_bvh_ = false;

    const Hittable& get_world() const;
    Color ray_color(const Ray& r, const Hittable& world, int depth) const;
};

#endif // RT_CPU_RENDERER_H
```

7.5 src/gpu/gpu_realtime.cpp File Reference

```
#include "gpu/gpu_realtime.h"
#include <cstdio>
```

Functions

- void gpu_realtime_render (const Scene &, int, int)

7.5.1 Function Documentation

7.5.1.1 gpu_realtime_render()

```
void gpu_realtime_render (
            const Scene & scene,
            int width,
            int height)
```

Definition at line 173 of file gpu_realtime.cpp.

7.6 gpu_realtime.cpp

Go to the documentation of this file.

```
// Real-time GPU raytracer with SDL3
// Pipeline GPU-side : render -> accumulate -> tonemap -> ARGB, tout sur GPU
// Seulement le buffer ARGB 8-bit est transfere au host (4 bytes/px vs 12 bytes/px)

#ifdef RT_REALTIME_ENABLED

#include "gpu/gpu_realtime.h"
#include "gpu/gpu_renderer.h"
#include <SDL3/SDL.h>
#include <cstdio>
#include <cmath>
#include <algorithm>
#include <chrono>
#include <cstring>

void gpu_realtime_render(const Scene& scene, int width, int height) {
    if (!SDL_Init(SDL_INIT_VIDEO)) {
        fprintf(stderr, "SDL_Init error: %s\n", SDL_GetError());
        return;
    }

    Scene& rt_scene = const_cast<Scene&>(scene);
    int orig_w = rt_scene.camera.image_width;
    int orig_spp = rt_scene.camera.samples_per_pixel;
    int orig_depth = rt_scene.camera.max_depth;
    Point3 orig_from = rt_scene.camera.lookfrom;
    Point3 orig_at = rt_scene.camera.lookat;

    rt_scene.camera.image_width = width;
    // Le spp demande par l'utilisateur est respecte
    // Pas de changement dynamique : si --spp 500, c'est 500
    rt_scene.camera.max_depth = 50;
    rt_scene.camera.initialize();
    int h = rt_scene.camera.get_image_height();

    SDL_Window* window = SDL_CreateWindow("Raytracer [GPU Real-time]", width, h, 0);
    if (!window) { fprintf(stderr, "SDL error: %s\n", SDL_GetError()); SDL_Quit(); return; }

    SDL_Renderer* sdl_renderer = SDL_CreateRenderer(window, NULL);
    SDL_Texture* texture = SDL_CreateTexture(sdl_renderer,
        SDL_PIXELFORMAT_ARGB8888, SDL_TEXTUREACCESS_STREAMING, width, h);

    // Init GPU-side pipeline
    gpu_realtime_init(rt_scene, width, h);

    // Host ARGB buffer (pinned, alloue par gpu_realtime_init)
    unsigned char* display = new unsigned char[width * h * 4];

    Vec3 cam_pos = rt_scene.camera.lookfrom;
    Vec3 cam_dir = unit_vector(rt_scene.camera.lookat - cam_pos);
    double cam_yaw = std::atan2(cam_dir.x(), -cam_dir.z());
    double cam_pitch = std::asin(std::clamp(cam_dir.y(), -0.99, 0.99));
    double move_speed = 2.0, mouse_sens = 0.003;
    bool mouse_captured = false, camera_moved = false;

    fprintf(stderr, "Real-time GPU-side: %dx%d | WASD + Souris | S=screenshot | ESC=quit\n", width, h);

    bool running = true;
    int frame = 0;
    int frames_since_move = 0;
    auto t_last_fps = std::chrono::high_resolution_clock::now();
    int frames_since_fps = 0;
    float current_fps = 0, last_km = 0;

    while (running) {
        camera_moved = false;

        SDL_Event event;
        while (SDL_PollEvent(&event)) {
            switch (event.type) {
            case SDL_EVENT_QUIT: running = false; break;
            case SDL_EVENT_KEY_DOWN:
                if (event.key.key == SDLK_ESCAPE) running = false;
                if (event.key.key == SDLK_S) {
                    // Screenshot : sauve l'image actuelle en PPM
                    char fname[128];
                    snprintf(fname, sizeof(fname), "output/screenshot_%d.ppm", frame);
                    ImageBuffer screenshot(width, h);
                    // Copier depuis le display ARGB vers ImageBuffer
                    for (int y = 0; y < h; y++)
                        for (int x = 0; x < width; x++) {
```

```
                            int idx = (y * width + x) * 4;
                            double b = display[idx] / 255.0;
                            double g = display[idx+1] / 255.0;
                            double r = display[idx+2] / 255.0;
                            // Inverse gamma+ACES pour stocker lineaire dans ImageBuffer
                            screenshot.set_pixel(x, y, Color(r*r, g*g, b*b));
                        }
                    screenshot.write_ppm(fname);
                    fprintf(stderr, "Screenshot saved: %s\n", fname);
                }
                break;
            case SDL_EVENT_MOUSE_BUTTON_DOWN:
                if (!mouse_captured) { SDL_SetWindowRelativeMouseMode(window, true); mouse_captured = true; }
                break;
            case SDL_EVENT_MOUSE_MOTION:
                if (mouse_captured && (event.motion.xrel != 0 || event.motion.yrel != 0)) {
                    cam_yaw += event.motion.xrel * mouse_sens;
                    cam_pitch -= event.motion.yrel * mouse_sens;
                    cam_pitch = std::clamp(cam_pitch, -1.5, 1.5);
                    camera_moved = true;
                }
                break;
            }
        }

        const bool* keys = SDL_GetKeyboardState(NULL);
        Vec3 forward(std::sin(cam_yaw)*std::cos(cam_pitch), std::sin(cam_pitch), -std::cos(cam_yaw)*std::cos(cam_pitch));
        Vec3 right_vec = unit_vector(cross(forward, Vec3(0,1,0)));
        double speed = move_speed * (keys[SDL_SCANCODE_LSHIFT] ? 5.0 : 1.0);
        double dt = (current_fps > 0) ? 1.0 / current_fps : 0.016;

        Vec3 old_pos = cam_pos;
        if (keys[SDL_SCANCODE_W]) cam_pos = cam_pos + forward * (speed * dt);
        if (keys[SDL_SCANCODE_S] && !keys[SDL_SCANCODE_LCTRL]) cam_pos = cam_pos - forward * (speed * dt);
        if (keys[SDL_SCANCODE_A]) cam_pos = cam_pos - right_vec * (speed * dt);
        if (keys[SDL_SCANCODE_D]) cam_pos = cam_pos + right_vec * (speed * dt);
        if (keys[SDL_SCANCODE_SPACE]) cam_pos = cam_pos + Vec3(0,1,0) * (speed * dt);
        if (keys[SDL_SCANCODE_LCTRL]) cam_pos = cam_pos - Vec3(0,1,0) * (speed * dt);
        if ((cam_pos - old_pos).length_squared() > 1e-10) camera_moved = true;

        if (camera_moved)
            gpu_realtime_reset_accum();

        rt_scene.camera.lookfrom = cam_pos;
        rt_scene.camera.lookat = cam_pos + forward;
        rt_scene.camera.initialize();

        last_km = gpu_realtime_render_frame(rt_scene);
        frame++;
        frames_since_fps++;

        // Tonemap GPU -> ARGB host (seulement 4 bytes/px transferes)
        int total_spp = gpu_realtime_tonemap(display, width, h);

        SDL_UpdateTexture(texture, NULL, display, width * 4);
        SDL_RenderTexture(sdl_renderer, texture, NULL, NULL);
        SDL_RenderPresent(sdl_renderer);

        auto t_now = std::chrono::high_resolution_clock::now();
        double elapsed = std::chrono::duration<double>(t_now - t_last_fps).count();
        if (elapsed >= 0.5) {
            current_fps = (float)(frames_since_fps / elapsed);
            frames_since_fps = 0;
            t_last_fps = t_now;
            char title[256];
            snprintf(title, sizeof(title),
                "Raytracer [GPU] %dx%d | %.1f FPS | %.1fms | %d spp | pos(%.1f,%.1f,%.1f) | S=screenshot",
                width, h, current_fps, last_km,
                total_spp * rt_scene.camera.samples_per_pixel,
                cam_pos.x(), cam_pos.y(), cam_pos.z());
            SDL_SetWindowTitle(window, title);
        }
    }

    rt_scene.camera.image_width = orig_w;
    rt_scene.camera.samples_per_pixel = orig_spp;
    rt_scene.camera.max_depth = orig_depth;
    rt_scene.camera.lookfrom = orig_from;
    rt_scene.camera.lookat = orig_at;
    rt_scene.camera.initialize();

    delete[] display;
    SDL_DestroyTexture(texture);
```

```
    SDL_DestroyRenderer(sdl_renderer);
    SDL_DestroyWindow(window);
    SDL_Quit();
}

#else
#include "gpu/gpu_realtime.h"
#include <cstdio>
void gpu_realtime_render(const Scene&, int, int) {
    fprintf(stderr, "Real-time requires SDL3\n");
}
#endif
```

7.7 src/gpu/gpu_realtime.h File Reference

```
#include "scene/scene.h"
#include "utils/image.h"
```

Functions

- void gpu_realtime_render (const Scene &scene, int width, int height)

7.7.1 Function Documentation

7.7.1.1 gpu_realtime_render()

```
void gpu_realtime_render (
            const Scene & scene,
            int width,
            int height)
```

Definition at line 173 of file gpu_realtime.cpp.

7.8 gpu_realtime.h

Go to the documentation of this file.

```
#ifndef RT_GPU_REALTIME_H
#define RT_GPU_REALTIME_H

#include "scene/scene.h"
#include "utils/image.h"

// Lance le rendu GPU temps reel dans une fenetre Win32
// Boucle infinie : 1 spp par frame, affiche FPS
// Appuyer sur ESC pour quitter
void gpu_realtime_render(const Scene& scene, int width, int height);

#endif
```

7.9 src/gpu/gpu_renderer.cu File Reference

```
#include "gpu/gpu_types.cuh"
#include "gpu/gpu_renderer.h"
#include "scene/primitives.cuh"
#include "scene/material.cuh"
#include <cstdio>
#include <cmath>
#include <vector>
#include <curand_kernel.h>
```

Classes

- struct GBuildInfo

Macros

- #define BVH_SHARED_NODES 7
- #define d_camera (∗(const GCamera∗)d_camera_bytes)
- #define CUDA_CHECK(call)

Typedefs

- typedef curandState GRng

Functions

- __device__ void grng_init (GRng &state, uint32_t tid, uint32_t seed)
- __device__ float grng_float (GRng &state)
- __device__ GVec3 random_in_unit_sphere (GRng &rng)
- __device__ GVec3 random_unit_vector (GRng &rng)
- __device__ GVec3 random_in_unit_disk (GRng &rng)
- __device__ bool intersect_sphere (const GSphere &s, const GRay &r, float t_min, float t_max, GHitRecord &rec)
- __device__ bool intersect_sphere_soa (const GSphereSoA &soa, int idx, const GRay &r, float t_min, float t_max, GHitRecord &rec)
- __device__ bool intersect_plane (const GPlane &pl, const GRay &r, float t_min, float t_max, GHitRecord &rec)
- __device__ bool trace_bvh (const GBVHNode ∗__restrict__ nodes, int num_nodes, const GSphereSoA &soa, const GPlane ∗__restrict__ planes, const GRay &r, float t_min, float t_max, GHitRecord &rec, const GBVHNode ∗__restrict__ sh_bvh, int sh_count)
- __device__ bool trace_brute (const GSphere ∗spheres, int num_spheres, const GPlane ∗planes, int num_↩ planes, const GRay &r, float t_min, float t_max, GHitRecord &rec)
- __device__ bool scatter_mat (const GMaterial &mat, const GRay &r_in, const GHitRecord &rec, GVec3 &attenuation, GRay &scattered, GRng &rng)
- __device__ GVec3 ray_color (GRay r, const GBVHNode ∗bvh, int num_bvh, const GSphereSoA &soa, const GSphere ∗spheres, int num_spheres, const GPlane ∗planes, int num_planes, const GMaterial ∗materials, int max_depth, GRng &rng, const GBVHNode ∗sh_bvh, int sh_count)

- __global__ void render_kernel (float *__restrict__ framebuffer, int width, int height, const GBVHNode *__restrict__ bvh, int num_bvh, const GSphere *__restrict__ spheres, int num_spheres, const GPlane *__restrict__ planes, int num_planes, const GMaterial *__restrict__ materials, const float *__restrict__ soa_cx, const float *__restrict__ soa_cy, const float *__restrict__ soa_cz, const float *__restrict__ soa_r, const int *__restrict__ soa_mat)
- __device__ float aces_tonemap_gpu (float x)
- __global__ void tonemap_kernel (const float *__restrict__ linear_fb, unsigned char *__restrict__ argb_out, int width, int height, int total_samples)
- __global__ void accumulate_kernel (const float *__restrict__ frame_fb, float *__restrict__ accum_fb, int total_pixels)
- __global__ void denoise_bilateral_kernel (const float *__restrict__ input, float *__restrict__ output, int width, int height, float sigma_color)
- static void convert_scene (const Scene &scene, std::vector< GSphere > &out_spheres, std::vector< GPlane > &out_planes, std::vector< GMaterial > &out_materials, GCamera &out_camera)
- static void convert_camera_only (const Camera &cam, GCamera &out)
- static void build_gpu_bvh (std::vector< GBuildInfo > &info, int start, int end, std::vector< GBVHNode > &nodes, int num_planes)
- void gpu_render_scene (const Scene &scene, ImageBuffer &image, float &kernel_ms)
- void gpu_render_scene_denoised (const Scene &scene, ImageBuffer &image, float &kernel_ms, float sigma_color)
- void gpu_realtime_init (const Scene &scene, int width, int height)
- float gpu_realtime_render_frame (const Scene &scene)
- int gpu_realtime_tonemap (unsigned char *argb_host, int width, int height)
- void gpu_realtime_reset_accum ()

Variables

- __constant__ char d_camera_bytes [sizeof(GCamera)]
- GpuRenderer * s_renderer = nullptr
- static float * s_rt_frame_fb = nullptr
- static float * s_rt_accum_fb = nullptr
- static unsigned char * s_rt_argb = nullptr
- static unsigned char * s_rt_argb_host = nullptr
- static int s_rt_w = 0
- static int s_rt_h = 0
- static int s_rt_accum_count = 0
- static cudaStream_t s_rt_stream = nullptr
- static dim3 s_rt_block
- static dim3 s_rt_grid
- static int s_rt_acc_grid = 0
- static cudaEvent_t s_rt_start = nullptr
- static cudaEvent_t s_rt_stop = nullptr
- static bool s_rt_config_done = false

7.9.1 Macro Definition Documentation

7.9.1.1 BVH_SHARED_NODES

```
#define BVH_SHARED_NODES 7
```

Definition at line 110 of file gpu_renderer.cu.

7.9.1.2 CUDA_CHECK

```
#define CUDA_CHECK(
            call)
```

Value:

```
    do { \
    cudaError_t err = (call); \
    if (err != cudaSuccess) { \
        fprintf(stderr, "CUDA error at %s:%d: %s\n", __FILE__, __LINE__, cudaGetErrorString(err)); \
        return; \
    } \
} while(0)
```

Definition at line 435 of file gpu_renderer.cu.

7.9.1.3 d_camera

```
#define d_camera (*(const GCamera*)d_camera_bytes)
```

Definition at line 286 of file gpu_renderer.cu.

7.9.2 Typedef Documentation

7.9.2.1 GRng

```
typedef curandState GRng
```

Definition at line 16 of file gpu_renderer.cu.

7.9.3 Function Documentation

7.9.3.1 accumulate_kernel()

```
__global__ void accumulate_kernel (
            const float *__restrict__ frame_fb,
            float *__restrict__ accum_fb,
            int total_pixels)
```

Definition at line 383 of file gpu_renderer.cu.

7.9.3.2 aces_tonemap_gpu()

```
__device__ float aces_tonemap_gpu (
            float x)
```

Definition at line 352 of file gpu_renderer.cu.

7.9.3.3 build_gpu_bvh()

```
void build_gpu_bvh (
            std::vector< GBuildInfo > & info,
            int start,
            int end,
            std::vector< GBVHNode > & nodes,
            int num_planes)  [static]
```

Definition at line 725 of file gpu_renderer.cu.

7.9.3.4 convert_camera_only()

```
void convert_camera_only (
            const Camera & cam,
            GCamera & out)  [static]
```

Definition at line 695 of file gpu_renderer.cu.

7.9.3.5 convert_scene()

```
void convert_scene (
            const Scene & scene,
            std::vector< GSphere > & out_spheres,
            std::vector< GPlane > & out_planes,
            std::vector< GMaterial > & out_materials,
            GCamera & out_camera)  [static]
```

Definition at line 593 of file gpu_renderer.cu.

7.9.3.6 denoise_bilateral_kernel()

```
__global__ void denoise_bilateral_kernel (
            const float *__restrict__ input,
            float *__restrict__ output,
            int width,
            int height,
            float sigma_color)
```

Definition at line 395 of file gpu_renderer.cu.

7.9.3.7 gpu_realtime_init()

```
void gpu_realtime_init (
            const Scene & scene,
            int width,
            int height)
```

Definition at line 917 of file gpu_renderer.cu.

7.9.3.8 gpu_realtime_render_frame()

```
float gpu_realtime_render_frame (
            const Scene & scene)
```

Definition at line 948 of file gpu_renderer.cu.

7.9.3.9 gpu_realtime_reset_accum()

```
void gpu_realtime_reset_accum ()
```

Definition at line 1012 of file gpu_renderer.cu.

7.9.3.10 gpu_realtime_tonemap()

```
int gpu_realtime_tonemap (
            unsigned char * argb_host,
            int width,
            int height)
```

Definition at line 997 of file gpu_renderer.cu.

7.9.3.11 gpu_render_scene()

```
void gpu_render_scene (
            const Scene & scene,
            ImageBuffer & image,
            float & kernel_ms)
```

Definition at line 788 of file gpu_renderer.cu.

7.9.3.12 gpu_render_scene_denoised()

```
void gpu_render_scene_denoised (
            const Scene & scene,
            ImageBuffer & image,
            float & kernel_ms,
            float sigma_color)
```

Definition at line 861 of file gpu_renderer.cu.

7.9.3.13 grng_float()

```
__device__ float grng_float (
            GRng & state)
```

Definition at line 22 of file gpu_renderer.cu.

7.9.3.14 grng_init()

```
__device__ void grng_init (
            GRng & state,
            uint32_t tid,
            uint32_t seed)
```

Definition at line 18 of file gpu_renderer.cu.

7.9.3.15 intersect_plane()

```
__device__ bool intersect_plane (
            const GPlane & pl,
            const GRay & r,
            float t_min,
            float t_max,
            GHitRecord & rec)
```

Definition at line 94 of file gpu_renderer.cu.

7.9.3.16 intersect_sphere()

```
__device__ bool intersect_sphere (
            const GSphere & s,
            const GRay & r,
            float t_min,
            float t_max,
            GHitRecord & rec)
```

Definition at line 47 of file gpu_renderer.cu.

7.9.3.17 intersect_sphere_soa()

```
__device__ bool intersect_sphere_soa (
            const GSphereSoA & soa,
            int idx,
            const GRay & r,
            float t_min,
            float t_max,
            GHitRecord & rec)
```

Definition at line 70 of file gpu_renderer.cu.

7.9.3.18 random_in_unit_disk()

```
__device__ GVec3 random_in_unit_disk (
            GRng & rng)
```

Definition at line 37 of file gpu_renderer.cu.

7.9.3.19 random_in_unit_sphere()

```
__device__ GVec3 random_in_unit_sphere (
            GRng & rng)
```

Definition at line 26 of file gpu_renderer.cu.

7.9.3.20 random_unit_vector()

```
__device__ GVec3 random_unit_vector (
            GRng & rng)
```

Definition at line 33 of file gpu_renderer.cu.

7.9.3.21 ray_color()

```
__device__ GVec3 ray_color (
            GRay r,
            const GBVHNode * bvh,
            int num_bvh,
            const GSphereSoA & soa,
            const GSphere * spheres,
            int num_spheres,
            const GPlane * planes,
            int num_planes,
            const GMaterial * materials,
            int max_depth,
            GRng & rng,
            const GBVHNode * sh_bvh,
            int sh_count)
```

Definition at line 220 of file gpu_renderer.cu.

7.9.3.22 render_kernel()

```
__global__ void render_kernel (
            float *__restrict__ framebuffer,
            int width,
            int height,
            const GBVHNode *__restrict__ bvh,
            int num_bvh,
            const GSphere *__restrict__ spheres,
            int num_spheres,
            const GPlane *__restrict__ planes,
            int num_planes,
            const GMaterial *__restrict__ materials,
            const float *__restrict__ soa_cx,
            const float *__restrict__ soa_cy,
            const float *__restrict__ soa_cz,
            const float *__restrict__ soa_r,
            const int *__restrict__ soa_mat)
```

Definition at line 288 of file gpu_renderer.cu.

7.9.3.23 scatter_mat()

```
__device__ bool scatter_mat (
            const GMaterial & mat,
            const GRay & r_in,
            const GHitRecord & rec,
            GVec3 & attenuation,
            GRay & scattered,
            GRng & rng)
```

Definition at line 177 of file gpu_renderer.cu.

7.9.3.24 tonemap_kernel()

```
__global__ void tonemap_kernel (
            const float *__restrict__ linear_fb,
            unsigned char *__restrict__ argb_out,
            int width,
            int height,
            int total_samples)
```

Definition at line 358 of file gpu_renderer.cu.

7.9.3.25 trace_brute()

```
__device__ bool trace_brute (
            const GSphere * spheres,
            int num_spheres,
            const GPlane * planes,
            int num_planes,
            const GRay & r,
            float t_min,
            float t_max,
            GHitRecord & rec)
```

Definition at line 154 of file gpu_renderer.cu.

7.9.3.26 trace_bvh()

```
__device__ bool trace_bvh (
            const GBVHNode *__restrict__ nodes,
            int num_nodes,
            const GSphereSoA & soa,
            const GPlane *__restrict__ planes,
            const GRay & r,
            float t_min,
            float t_max,
            GHitRecord & rec,
            const GBVHNode *__restrict__ sh_bvh,
            int sh_count)
```

Definition at line 112 of file gpu_renderer.cu.

7.9.4 Variable Documentation

7.9.4.1 d_camera_bytes

```
__constant__ char d_camera_bytes[sizeof(GCamera)]
```

Definition at line 285 of file gpu_renderer.cu.

7.9.4.2 s_renderer

```
GpuRenderer* s_renderer = nullptr
```

Definition at line 786 of file gpu_renderer.cu.

7.9.4.3 s_rt_acc_grid

```
int s_rt_acc_grid = 0  [static]
```

Definition at line 944 of file gpu_renderer.cu.

7.9.4.4 s_rt_accum_count

```
int s_rt_accum_count = 0  [static]
```

Definition at line 914 of file gpu_renderer.cu.

7.9.4.5 s_rt_accum_fb

```
float* s_rt_accum_fb = nullptr  [static]
```

Definition at line 910 of file gpu_renderer.cu.

7.9.4.6 s_rt_argb

```
unsigned char* s_rt_argb = nullptr  [static]
```

Definition at line 911 of file gpu_renderer.cu.

7.9.4.7 s_rt_argb_host

```
unsigned char* s_rt_argb_host = nullptr  [static]
```

Definition at line 912 of file gpu_renderer.cu.

7.9.4.8 s_rt_block

```
dim3 s_rt_block [static]
```

Definition at line 943 of file gpu_renderer.cu.

7.9.4.9 s_rt_config_done

```
bool s_rt_config_done = false [static]
```

Definition at line 946 of file gpu_renderer.cu.

7.9.4.10 s_rt_frame_fb

```
float* s_rt_frame_fb = nullptr [static]
```

Definition at line 909 of file gpu_renderer.cu.

7.9.4.11 s_rt_grid

```
dim3 s_rt_grid [static]
```

Definition at line 943 of file gpu_renderer.cu.

7.9.4.12 s_rt_h

```
int s_rt_h = 0 [static]
```

Definition at line 913 of file gpu_renderer.cu.

7.9.4.13 s_rt_start

```
cudaEvent_t s_rt_start = nullptr [static]
```

Definition at line 945 of file gpu_renderer.cu.

7.9.4.14 s_rt_stop

```
cudaEvent_t s_rt_stop = nullptr [static]
```

Definition at line 945 of file gpu_renderer.cu.

7.9.4.15 s_rt_stream

```
cudaStream_t s_rt_stream = nullptr [static]
```

Definition at line 915 of file gpu_renderer.cu.

7.9.4.16 s_rt_w

```
int s_rt_w = 0  [static]
```

Definition at line 913 of file gpu_renderer.cu.

7.10 gpu_renderer.cu

Go to the documentation of this file.

```
00001 // GPU Renderer : tout le code CUDA dans un seul fichier .cu
00002 // Evite la separable compilation (problemes de linkage RDC sur MSVC)
00003
00004 #include "gpu/gpu_types.cuh"
00005 #include "gpu/gpu_renderer.h"
00006 #include "scene/primitives.cuh"
00007 #include "scene/material.cuh"
00008 #include <cstdio>
00009 #include <cmath>
00010 #include <vector>
00011 #include <curand_kernel.h>
00012
00013 // ============================================================
00014 // GPU RNG : curand (standard NVIDIA)
00015 // ============================================================
00016 typedef curandState GRng;
00017
00018 __device__ void grng_init(GRng& state, uint32_t tid, uint32_t seed) {
00019     curand_init(seed, tid, 0, &state);
00020 }
00021
00022 __device__ float grng_float(GRng& state) {
00023     return curand_uniform(&state);
00024 }
00025
00026 __device__ GVec3 random_in_unit_sphere(GRng& rng) {
00027     while (true) {
00028         GVec3 p(grng_float(rng)*2.0f-1.0f, grng_float(rng)*2.0f-1.0f, grng_float(rng)*2.0f-1.0f);
00029         if (p.length_squared() < 1.0f) return p;
00030     }
00031 }
00032
00033 __device__ GVec3 random_unit_vector(GRng& rng) {
00034     return normalize(random_in_unit_sphere(rng));
00035 }
00036
00037 __device__ GVec3 random_in_unit_disk(GRng& rng) {
00038     while (true) {
00039         GVec3 p(grng_float(rng)*2.0f-1.0f, grng_float(rng)*2.0f-1.0f, 0.0f);
00040         if (p.length_squared() < 1.0f) return p;
00041     }
00042 }
00043
00044 // ============================================================
00045 // Intersections
00046 // ============================================================
00047 __device__ bool intersect_sphere(const GSphere& s, const GRay& r, float t_min, float t_max,
      GHitRecord& rec) {
00048     GVec3 oc = r.orig - s.center;
00049     float a = dot(r.dir, r.dir);
00050     float half_b = dot(oc, r.dir);
00051     float c = oc.length_squared() - s.radius * s.radius;
00052     float disc = half_b * half_b - a * c;
00053     if (disc < 0.0f) return false;
00054     float sqrtd = sqrtf(disc);
00055     float inv_a = __fdividef(1.0f, a); // fast HW reciprocal, avoids full division
00056     float root = (-half_b - sqrtd) * inv_a;
00057     if (root < t_min || root > t_max) {
00058         root = (-half_b + sqrtd) * inv_a;
00059         if (root < t_min || root > t_max) return false;
00060     }
00061     rec.t = root;
00062     rec.p = r.at(root);
00063     float inv_r = __fdividef(1.0f, s.radius);
00064     rec.set_face_normal(r, (rec.p - s.center) * inv_r);
00065     rec.material_id = s.material_id;
00066     return true;
00067 }
00068
```

```
// SoA layout: coalesced loads when warp threads test consecutive spheres
__device__ bool intersect_sphere_soa(const GSphereSoA& soa, int idx,
                                     const GRay& r, float t_min, float t_max, GHitRecord& rec) {
    float sx = soa.cx[idx], sy = soa.cy[idx], sz = soa.cz[idx], sr = soa.r[idx];
    GVec3 oc(r.orig.x - sx, r.orig.y - sy, r.orig.z - sz);
    float a = dot(r.dir, r.dir);
    float half_b = dot(oc, r.dir);
    float c = oc.length_squared() - sr * sr;
    float disc = half_b * half_b - a * c;
    if (disc < 0.0f) return false;
    float sqrtd = sqrtf(disc);
    float inv_a = __fdividef(1.0f, a);
    float root = (-half_b - sqrtd) * inv_a;
    if (root < t_min || root > t_max) {
        root = (-half_b + sqrtd) * inv_a;
        if (root < t_min || root > t_max) return false;
    }
    rec.t = root;
    rec.p = r.at(root);
    float inv_r = __fdividef(1.0f, sr);
    rec.set_face_normal(r, GVec3(rec.p.x - sx, rec.p.y - sy, rec.p.z - sz) * inv_r);
    rec.material_id = soa.mat[idx];
    return true;
}

__device__ bool intersect_plane(const GPlane& pl, const GRay& r, float t_min, float t_max, GHitRecord& rec) {
    float denom = dot(r.dir, pl.normal);
    if (fabsf(denom) < 1e-6f) return false;
    float t = dot(pl.point - r.orig, pl.normal) * __fdividef(1.0f, denom);
    if (t < t_min || t > t_max) return false;
    rec.t = t;
    rec.p = r.at(t);
    rec.set_face_normal(r, pl.normal);
    rec.material_id = pl.material_id;
    return true;
}

// ============================================================
// BVH traversal iteratif (stack fixe, front-to-back)
// ============================================================
// Top BVH nodes in shared mem: accessed by nearly all rays in the block
#define BVH_SHARED_NODES 7  // 3 premiers niveaux = 1+2+4 = 7 nodes

__device__ bool trace_bvh(const GBVHNode* __restrict__ nodes, int num_nodes,
                          const GSphereSoA& soa,
                          const GPlane* __restrict__ planes,
                          const GRay& r, float t_min, float t_max,
                          GHitRecord& rec,
                          const GBVHNode* __restrict__ sh_bvh, int sh_count) {
    bool hit_anything = false;
    float closest = t_max;
    int stack[16];
    int sp = 0;
    stack[sp++] = 0;

    while (sp > 0) {
        int idx = stack[--sp];
        const GBVHNode& node = (idx < sh_count) ? sh_bvh[idx] : nodes[idx];

        if (!node.bbox.hit(r, t_min, closest))
            continue;

        if (node.num_prims > 0) {
            #pragma unroll 4
            for (uint16_t i = 0; i < node.num_prims; i++) {
                GHitRecord tmp;
                // SoA: each float[] is contiguous -> coalesced reads
                if (intersect_sphere_soa(soa, node.data + i, r, t_min, closest, tmp)) {
                    hit_anything = true; closest = tmp.t; rec = tmp;
                }
            }
        } else {
            int near_child = idx + 1;
            int far_child = node.data;
            float dir_comp = (node.split_axis == 0) ? r.dir.x :
                             (node.split_axis == 1) ? r.dir.y : r.dir.z;
            if (dir_comp < 0.0f) { int tmp = near_child; near_child = far_child; far_child = tmp; }
            stack[sp++] = far_child;
            stack[sp++] = near_child;
        }
    }
    return hit_anything;
}

// Brute force fallback
__device__ bool trace_brute(const GSphere* spheres, int num_spheres,
```

```
                            const GPlane* planes, int num_planes,
                            const GRay& r, float t_min, float t_max,
                            GHitRecord& rec) {
    bool hit = false;
    float closest = t_max;
    GHitRecord tmp;
    for (int i = 0; i < num_spheres; i++) {
        if (intersect_sphere(spheres[i], r, t_min, closest, tmp)) {
            hit = true; closest = tmp.t; rec = tmp;
        }
    }
    for (int i = 0; i < num_planes; i++) {
        if (intersect_plane(planes[i], r, t_min, closest, tmp)) {
            hit = true; closest = tmp.t; rec = tmp;
        }
    }
    return hit;
}

// ============================================================
// Material scatter
// ============================================================
__device__ bool scatter_mat(const GMaterial& mat, const GRay& r_in, const GHitRecord& rec,
                            GVec3& attenuation, GRay& scattered, GRng& rng) {
    switch (mat.type) {
    case GMAT_LAMBERTIAN: {
        GVec3 dir = rec.normal + random_unit_vector(rng);
        if (dir.near_zero()) dir = rec.normal;
        scattered = GRay(rec.p, dir);
        attenuation = mat.albedo;
        return true;
    }
    case GMAT_METAL: {
        GVec3 reflected = reflect(normalize(r_in.dir), rec.normal);
        reflected = reflected + mat.param * random_in_unit_sphere(rng);
        scattered = GRay(rec.p, reflected);
        attenuation = mat.albedo;
        return dot(scattered.dir, rec.normal) > 0.0f;
    }
    case GMAT_DIELECTRIC: {
        attenuation = mat.albedo; // tinted glass support
        float ri = rec.front_face ? (1.0f / mat.param) : mat.param;
        GVec3 unit_dir = normalize(r_in.dir);
        float cos_theta = fminf(dot(-unit_dir, rec.normal), 1.0f);
        float sin_theta = sqrtf(1.0f - cos_theta * cos_theta);
        float r0 = (1.0f - ri) / (1.0f + ri); r0 = r0 * r0;
        // manual pow5: avoids expensive powf (exp+log chain)
        float omc = 1.0f - cos_theta;
        float omc2 = omc * omc;
        float schlick = r0 + (1.0f - r0) * omc2 * omc2 * omc;
        GVec3 direction;
        if (ri * sin_theta > 1.0f || schlick > grng_float(rng))
            direction = reflect(unit_dir, rec.normal);
        else
            direction = refract(unit_dir, rec.normal, ri);
        scattered = GRay(rec.p, direction);
        return true;
    }
    }
    return false;
}

// ============================================================
// Ray color iteratif
// ============================================================
__device__ GVec3 ray_color(GRay r,
                           const GBVHNode* bvh, int num_bvh,
                           const GSphereSoA& soa,
                           const GSphere* spheres, int num_spheres,
                           const GPlane* planes, int num_planes,
                           const GMaterial* materials,
                           int max_depth, GRng& rng,
                           const GBVHNode* sh_bvh, int sh_count) {
    GVec3 throughput(1.0f, 1.0f, 1.0f);

    for (int d = 0; d < max_depth; d++) {
        GHitRecord rec;
        bool hit;
        if (num_bvh > 0) {
            // BVH pour les spheres + brute force pour les plans (peu nombreux)
            float closest = 1e30f;
            hit = trace_bvh(bvh, num_bvh, soa, planes, r, 0.001f, closest, rec, sh_bvh, sh_count);
            if (hit) closest = rec.t;
            // Tester les plans en brute force
            GHitRecord tmp;
            for (int i = 0; i < num_planes; i++) {
                if (intersect_plane(planes[i], r, 0.001f, closest, tmp)) {
```

```
                    hit = true; closest = tmp.t; rec = tmp;
                }
            }
        } else {
            hit = trace_brute(spheres, num_spheres, planes, num_planes, r, 0.001f, 1e30f, rec);
        }
        if (!hit) {
            GVec3 unit_dir = normalize(r.dir);
            float a = 0.5f * (unit_dir.y + 1.0f);
            GVec3 sky(1.0f - 0.5f*a, 1.0f - 0.3f*a, 1.0f);
            return throughput * sky;
        }
        const GMaterial& mat = materials[rec.material_id];
        if (mat.type == GMAT_EMISSIVE)
            return throughput * mat.albedo * mat.param;

        GVec3 attenuation;
        GRay scattered;
        if (!scatter_mat(mat, r, rec, attenuation, scattered, rng))
            return GVec3(0, 0, 0);
        throughput = throughput * attenuation;

        // Russian roulette: unbiased path termination after depth 3
        float max_comp = fmaxf(throughput.x, fmaxf(throughput.y, throughput.z));
        if (max_comp < 0.001f)
            break;

        if (d > 3) {
            if (grng_float(rng) > max_comp)
                break;
            throughput = throughput * (1.0f / max_comp);
        }

        r = scattered;
    }
    return GVec3(0, 0, 0);
}

// ============================================================
// Kernel : 1 thread = 1 pixel, blocs 16x16
// ============================================================
// Camera in constant memory: broadcast to all threads, dedicated cache
// Stored as raw bytes to avoid dynamic initialization issues
__constant__ char d_camera_bytes[sizeof(GCamera)];
#define d_camera (*(const GCamera*)d_camera_bytes)

__global__ void render_kernel(float* __restrict__ framebuffer, int width, int height,
                              const GBVHNode* __restrict__ bvh, int num_bvh,
                              const GSphere* __restrict__ spheres, int num_spheres,
                              const GPlane* __restrict__ planes, int num_planes,
                              const GMaterial* __restrict__ materials,
                              const float* __restrict__ soa_cx, const float* __restrict__ soa_cy,
                              const float* __restrict__ soa_cz, const float* __restrict__ soa_r,
                              const int* __restrict__ soa_mat) {
    // Cooperative load of top BVH nodes into shared mem
    __shared__ GBVHNode sh_bvh[BVH_SHARED_NODES];
    int tid_flat = threadIdx.y * blockDim.x + threadIdx.x;
    if (tid_flat < BVH_SHARED_NODES && tid_flat < num_bvh) {
        sh_bvh[tid_flat] = bvh[tid_flat];
    }
    __syncthreads(); // Attendre que tous les top nodes soient en shared mem

    int sh_count = (num_bvh < BVH_SHARED_NODES) ? num_bvh : BVH_SHARED_NODES;

    int x = blockIdx.x * blockDim.x + threadIdx.x;
    int y = blockIdx.y * blockDim.y + threadIdx.y;
    if (x >= width || y >= height) return;

    GRng rng;
    grng_init(rng, y * width + x, 42);

    GVec3 pixel_color(0, 0, 0);
    int spp = d_camera.samples_per_pixel;
    float inv_spp = 1.0f / (float)spp;

    for (int s = 0; s < spp; s++) {
        float u_off = grng_float(rng) - 0.5f;
        float v_off = grng_float(rng) - 0.5f;
        GVec3 pixel_sample = d_camera.pixel00_loc
                           + ((float)x + u_off) * d_camera.pixel_delta_u
                           + ((float)y + v_off) * d_camera.pixel_delta_v;
        GVec3 ray_origin = d_camera.origin;
        if (d_camera.defocus_angle > 0.0f) {
            GVec3 pd = random_in_unit_disk(rng);
            ray_origin = d_camera.origin + pd.x * d_camera.defocus_disk_u + pd.y *
d_camera.defocus_disk_v;
        }
```

```
        GRay r(ray_origin, pixel_sample - ray_origin);
        GSphereSoA soa;
        soa.cx = const_cast<float*>(soa_cx); soa.cy = const_cast<float*>(soa_cy);
        soa.cz = const_cast<float*>(soa_cz); soa.r = const_cast<float*>(soa_r);
        soa.mat = const_cast<int*>(soa_mat); soa.count = num_spheres;
        pixel_color += ray_color(r, bvh, num_bvh, soa, spheres, num_spheres, planes, num_planes,
                                 materials, d_camera.max_depth, rng, sh_bvh, sh_count);
    }

    pixel_color *= inv_spp;

    // Planar layout (RRR...GGG...BBB): stride-1 writes for coalescing
    int pixel_idx = y * width + x;
    int total_pixels = width * height;
    // Linear HDR output, gamma/tonemap done later
    framebuffer[pixel_idx]                  = fmaxf(pixel_color.x, 0.0f);
    framebuffer[pixel_idx + total_pixels]   = fmaxf(pixel_color.y, 0.0f);
    framebuffer[pixel_idx + total_pixels*2] = fmaxf(pixel_color.z, 0.0f);
}

// ============================================================
// Tonemap GPU kernel : lineaire -> ACES -> gamma -> ARGB 8-bit
// Evite le transfert float D->H + tonemap CPU
// ============================================================
__device__ float aces_tonemap_gpu(float x) {
    float a = x * (2.51f * x + 0.03f);
    float b = x * (2.43f * x + 0.59f) + 0.14f;
    return fmaxf(0.0f, fminf(1.0f, a / b));
}

__global__ void tonemap_kernel(const float* __restrict__ linear_fb,
                               unsigned char* __restrict__ argb_out,
                               int width, int height, int total_samples) {
    int x = blockIdx.x * blockDim.x + threadIdx.x;
    int y = blockIdx.y * blockDim.y + threadIdx.y;
    if (x >= width || y >= height) return;

    int px = y * width + x;
    int tp = width * height;
    float inv_n = 1.0f / (float)total_samples;

    float r = sqrtf(aces_tonemap_gpu(fmaxf(linear_fb[px] * inv_n, 0.0f)));
    float g = sqrtf(aces_tonemap_gpu(fmaxf(linear_fb[px + tp] * inv_n, 0.0f)));
    float b = sqrtf(aces_tonemap_gpu(fmaxf(linear_fb[px + tp*2] * inv_n, 0.0f)));

    int idx = px * 4;
    argb_out[idx + 0] = (unsigned char)(fminf(b, 1.0f) * 255.0f);
    argb_out[idx + 1] = (unsigned char)(fminf(g, 1.0f) * 255.0f);
    argb_out[idx + 2] = (unsigned char)(fminf(r, 1.0f) * 255.0f);
    argb_out[idx + 3] = 255;
}

// ============================================================
// Accumulation kernel : ajoute le rendu au buffer d'accumulation
// ============================================================
__global__ void accumulate_kernel(const float* __restrict__ frame_fb,
                                  float* __restrict__ accum_fb,
                                  int total_pixels) {
    int idx = blockIdx.x * blockDim.x + threadIdx.x;
    if (idx >= total_pixels * 3) return;
    accum_fb[idx] += frame_fb[idx];
}

// ============================================================
// GPU Denoise kernel : filtre bilateral 3x3 sur le framebuffer lineaire
// Lisse le bruit en preservant les contours (difference de couleur)
// ============================================================
__global__ void denoise_bilateral_kernel(
    const float* __restrict__ input,
    float* __restrict__ output,
    int width, int height, float sigma_color)
{
    int x = blockIdx.x * blockDim.x + threadIdx.x;
    int y = blockIdx.y * blockDim.y + threadIdx.y;
    if (x >= width || y >= height) return;

    int tp = width * height;
    int px = y * width + x;
    float cr = input[px], cg = input[px+tp], cb = input[px+tp*2];

    float sum_r = 0, sum_g = 0, sum_b = 0, wsum = 0;
    float inv_sigma2 = 1.0f / (2.0f * sigma_color * sigma_color);

    for (int dy = -1; dy <= 1; dy++) {
        int ny = y + dy;
        if (ny < 0 || ny >= height) continue;
        for (int dx = -1; dx <= 1; dx++) {
```

```
            int nx = x + dx;
            if (nx < 0 || nx >= width) continue;
            int npx = ny * width + nx;
            float nr = input[npx], ng = input[npx+tp], nb = input[npx+tp*2];
            float dr = cr-nr, dg = cg-ng, db = cb-nb;
            float color_dist = dr*dr + dg*dg + db*db;
            float w = expf(-color_dist * inv_sigma2);
            sum_r += nr * w; sum_g += ng * w; sum_b += nb * w;
            wsum += w;
        }
    }
    float inv_w = 1.0f / (wsum + 1e-8f);
    output[px] = sum_r * inv_w;
    output[px+tp] = sum_g * inv_w;
    output[px+tp*2] = sum_b * inv_w;
}

// ============================================================
// GpuRenderer host code
// ============================================================
#define CUDA_CHECK(call) do { \
    cudaError_t err = (call); \
    if (err != cudaSuccess) { \
        fprintf(stderr, "CUDA error at %s:%d: %s\n", __FILE__, __LINE__, cudaGetErrorString(err)); \
        return; \
    } \
} while(0)

GpuRenderer::GpuRenderer() {}
GpuRenderer::~GpuRenderer() { free_scene(); free_framebuffer(); }

void GpuRenderer::free_scene() {
    if (d_spheres_) { cudaFree(d_spheres_); d_spheres_ = nullptr; }
    if (d_soa_cx_) { cudaFree(d_soa_cx_); d_soa_cx_ = nullptr; }
    if (d_soa_cy_) { cudaFree(d_soa_cy_); d_soa_cy_ = nullptr; }
    if (d_soa_cz_) { cudaFree(d_soa_cz_); d_soa_cz_ = nullptr; }
    if (d_soa_r_) { cudaFree(d_soa_r_); d_soa_r_ = nullptr; }
    if (d_soa_mat_) { cudaFree(d_soa_mat_); d_soa_mat_ = nullptr; }
    if (d_planes_) { cudaFree(d_planes_); d_planes_ = nullptr; }
    if (d_materials_) { cudaFree(d_materials_); d_materials_ = nullptr; }
    if (d_bvh_nodes_) { cudaFree(d_bvh_nodes_); d_bvh_nodes_ = nullptr; }
}

void GpuRenderer::free_framebuffer() {
    if (d_framebuffer_) { cudaFree(d_framebuffer_); d_framebuffer_ = nullptr; }
    fb_width_ = fb_height_ = 0;
}

void GpuRenderer::ensure_framebuffer(int width, int height) {
    if (fb_width_ == width && fb_height_ == height && d_framebuffer_) return;
    free_framebuffer();
    fb_width_ = width; fb_height_ = height;
    cudaMalloc(&d_framebuffer_, width * height * 3 * sizeof(float));
}

void GpuRenderer::upload_scene(const GSphere* spheres, int num_spheres,
                               const GPlane* planes, int num_planes,
                               const GMaterial* materials, int num_materials,
                               const GBVHNode* bvh_nodes, int num_bvh_nodes) {
    free_scene();
    num_spheres_ = num_spheres;
    num_planes_ = num_planes;
    num_materials_ = num_materials;
    num_bvh_nodes_ = num_bvh_nodes;

    if (num_spheres > 0) {
        CUDA_CHECK(cudaMalloc(&d_spheres_, num_spheres * sizeof(GSphere)));
        CUDA_CHECK(cudaMemcpy(d_spheres_, spheres, num_spheres * sizeof(GSphere),
cudaMemcpyHostToDevice));

        // SoA upload: split fields for coalesced access in kernel
        std::vector<float> cx(num_spheres), cy(num_spheres), cz(num_spheres), radii(num_spheres);
        std::vector<int> mats(num_spheres);
        for (int i = 0; i < num_spheres; i++) {
            cx[i] = spheres[i].center.x; cy[i] = spheres[i].center.y; cz[i] = spheres[i].center.z;
            radii[i] = spheres[i].radius; mats[i] = spheres[i].material_id;
        }
        CUDA_CHECK(cudaMalloc(&d_soa_cx_, num_spheres * sizeof(float)));
        CUDA_CHECK(cudaMalloc(&d_soa_cy_, num_spheres * sizeof(float)));
        CUDA_CHECK(cudaMalloc(&d_soa_cz_, num_spheres * sizeof(float)));
        CUDA_CHECK(cudaMalloc(&d_soa_r_, num_spheres * sizeof(float)));
        CUDA_CHECK(cudaMalloc(&d_soa_mat_, num_spheres * sizeof(int)));
        CUDA_CHECK(cudaMemcpy(d_soa_cx_, cx.data(), num_spheres * sizeof(float),
cudaMemcpyHostToDevice));
        CUDA_CHECK(cudaMemcpy(d_soa_cy_, cy.data(), num_spheres * sizeof(float),
cudaMemcpyHostToDevice));
        CUDA_CHECK(cudaMemcpy(d_soa_cz_, cz.data(), num_spheres * sizeof(float),
```

```
cudaMemcpyHostToDevice));
        CUDA_CHECK(cudaMemcpy(d_soa_r_, radii.data(), num_spheres * sizeof(float), cudaMemcpyHostToDevice));
        CUDA_CHECK(cudaMemcpy(d_soa_mat_, mats.data(), num_spheres * sizeof(int), cudaMemcpyHostToDevice));
    }
    if (num_planes > 0) {
        CUDA_CHECK(cudaMalloc(&d_planes_, num_planes * sizeof(GPlane)));
        CUDA_CHECK(cudaMemcpy(d_planes_, planes, num_planes * sizeof(GPlane), cudaMemcpyHostToDevice));
    }
    if (num_materials > 0) {
        CUDA_CHECK(cudaMalloc(&d_materials_, num_materials * sizeof(GMaterial)));
        CUDA_CHECK(cudaMemcpy(d_materials_, materials, num_materials * sizeof(GMaterial), cudaMemcpyHostToDevice));
    }
    if (num_bvh_nodes > 0) {
        CUDA_CHECK(cudaMalloc(&d_bvh_nodes_, num_bvh_nodes * sizeof(GBVHNode)));
        CUDA_CHECK(cudaMemcpy(d_bvh_nodes_, bvh_nodes, num_bvh_nodes * sizeof(GBVHNode), cudaMemcpyHostToDevice));

        // L2 cache persistence pour BVH (Ampere CC 8.6+)
        // Les top nodes sont accedes par quasi tous les rayons
        size_t bvh_bytes = num_bvh_nodes * sizeof(GBVHNode);
        cudaDeviceSetLimit(cudaLimitPersistingL2CacheSize, bvh_bytes);
    }
}

void GpuRenderer::render(const GCamera& camera, float* h_output, int width, int height) {
    ensure_framebuffer(width, height);

    // Persistent stream for kernel/transfer overlap
    static cudaStream_t s_stream = nullptr;
    if (!s_stream) {
        cudaStreamCreate(&s_stream);

        // L2 cache persistence : BVH nodes en persistent L2 (Ampere CC 8.6+)
        if (d_bvh_nodes_ && num_bvh_nodes_ > 0) {
            cudaStreamAttrValue attr = {};
            attr.accessPolicyWindow.base_ptr = d_bvh_nodes_;
            attr.accessPolicyWindow.num_bytes = num_bvh_nodes_ * sizeof(GBVHNode);
            attr.accessPolicyWindow.hitRatio = 1.0f;
            attr.accessPolicyWindow.hitProp = cudaAccessPropertyPersisting;
            attr.accessPolicyWindow.missProp = cudaAccessPropertyStreaming;
            cudaStreamSetAttribute(s_stream, cudaStreamAttributeAccessPolicyWindow, &attr);
        }
    }

    // Upload camera en constant memory
    cudaMemcpyToSymbol(d_camera_bytes, &camera, sizeof(GCamera));

    // Block size via occupancy API
    int flat_block_size = 0, min_grid = 0;
    cudaOccupancyMaxPotentialBlockSize(&min_grid, &flat_block_size, render_kernel, 0, 0);
    int bx = 16, by = flat_block_size / bx;
    if (by < 1) by = 1;
    if (by > 16) by = 16;
    dim3 block_size(bx, by);
    dim3 grid_size((width + block_size.x - 1) / block_size.x,
                   (height + block_size.y - 1) / block_size.y);

    cudaEvent_t start, stop;
    cudaEventCreate(&start);
    cudaEventCreate(&stop);

    cudaEventRecord(start, s_stream);

    render_kernel<<grid_size, block_size, 0, s_stream>>(
        d_framebuffer_, width, height,
        d_bvh_nodes_, num_bvh_nodes_,
        d_spheres_, num_spheres_,
        d_planes_, num_planes_,
        d_materials_,
        d_soa_cx_, d_soa_cy_, d_soa_cz_, d_soa_r_, d_soa_mat_
    );

    cudaError_t err = cudaGetLastError();
    if (err != cudaSuccess) {
        fprintf(stderr, "Kernel error: %s\n", cudaGetErrorString(err));
        cudaEventDestroy(start); cudaEventDestroy(stop);
        return;
    }

    // Transfert async D->H sur le meme stream (sequentialise apres le kernel)
    // h_output DOIT etre en memoire pinned (cudaMallocHost) pour l'async
    cudaMemcpyAsync(h_output, d_framebuffer_, width * height * 3 * sizeof(float),
                    cudaMemcpyDeviceToHost, s_stream);
```

```

    cudaEventRecord(stop, s_stream);
    // Synchroniser le stream (attendre kernel + transfert)
    cudaStreamSynchronize(s_stream);

    cudaEventElapsedTime(&last_ms_, start, stop);
    cudaEventDestroy(start);
    cudaEventDestroy(stop);
}

// ============================================================
// Bridge: conversion CPU Scene -> GPU + rendu
// ============================================================
static void convert_scene(const Scene& scene,
                          std::vector<GSphere>& out_spheres,
                          std::vector<GPlane>& out_planes,
                          std::vector<GMaterial>& out_materials,
                          GCamera& out_camera) {
    std::vector<const Material*> mat_ptrs;
    auto get_mat = [&](const Material* m) -> int {
        for (int i = 0; i < (int)mat_ptrs.size(); i++)
            if (mat_ptrs[i] == m) return i;
        int idx = (int)mat_ptrs.size();
        mat_ptrs.push_back(m);
        GMaterial gm;
        switch (m->type()) {
        case Material::Type::Lambertian: {
            auto* lam = static_cast<const Lambertian*>(m);
            gm.type = GMAT_LAMBERTIAN;
            gm.albedo = GVec3((float)lam->albedo().x(), (float)lam->albedo().y(), (float)lam->albedo().z());
            break;
        }
        case Material::Type::Metal: {
            auto* met = static_cast<const Metal*>(m);
            gm.type = GMAT_METAL;
            gm.albedo = GVec3((float)met->albedo().x(), (float)met->albedo().y(), (float)met->albedo().z());
            gm.param = (float)met->fuzz();
            break;
        }
        case Material::Type::Dielectric: {
            auto* die = static_cast<const Dielectric*>(m);
            gm.type = GMAT_DIELECTRIC;
            gm.albedo = GVec3(1, 1, 1);
            gm.param = (float)die->refraction_index();
            break;
        }
        case Material::Type::Emissive: {
            auto* em = static_cast<const Emissive*>(m);
            gm.type = GMAT_EMISSIVE;
            gm.albedo = GVec3((float)em->emit_color().x(), (float)em->emit_color().y(), (float)em->emit_color().z());
            gm.param = (float)em->intensity();
            break;
        }
        default: gm.type = GMAT_LAMBERTIAN; gm.albedo = GVec3(0.5f, 0.5f, 0.5f); break;
        }
        out_materials.push_back(gm);
        return idx;
    };

    for (const auto& obj : scene.world.objects()) {
        if (auto* sp = dynamic_cast<const Sphere*>(obj.get())) {
            GSphere gs;
            gs.center = GVec3((float)sp->center().x(), (float)sp->center().y(), (float)sp->center().z());
            gs.radius = (float)sp->radius();
            HitRecord rec;
            Ray test_ray(sp->center() + Vec3(0, sp->radius() + 1, 0), Vec3(0, -1, 0));
            gs.material_id = (sp->hit(test_ray, Interval(0.001, 1000.0), rec) && rec.material)
                             ? get_mat(rec.material) : 0;
            out_spheres.push_back(gs);
        } else if (auto* pl = dynamic_cast<const Plane*>(obj.get())) {
            GPlane gp;
            HitRecord rec;
            Ray test_ray(Point3(0, 100, 0), Vec3(0, -1, 0));
            if (pl->hit(test_ray, Interval(0.001, 10000.0), rec) && rec.material) {
                gp.point = GVec3((float)rec.p.x(), (float)rec.p.y(), (float)rec.p.z());
                gp.normal = GVec3((float)rec.normal.x(), (float)rec.normal.y(), (float)rec.normal.z());
                gp.material_id = get_mat(rec.material);
            }
            out_planes.push_back(gp);
        } else {
            static bool warned = false;
            if (!warned) {
```

```
                fprintf(stderr, "Warning: GPU ne supporte que Sphere/Plane. Triangles/Cylindres ignores.\n");
                fprintf(stderr, "         Utilisez --scene showcase, final, ou three_spheres pour le GPU.\n");
                warned = true;
            }
        }
    }

    const Camera& cam = scene.camera;
    int w = cam.get_image_width(), h = cam.get_image_height();
    double theta = cam.vfov * 3.14159265358979323846 / 180.0;
    double vp_h = 2.0 * std::tan(theta / 2.0) * cam.focus_dist;
    double vp_w = vp_h * ((double)w / h);
    Vec3 wd = unit_vector(cam.lookfrom - cam.lookat);
    Vec3 ud = unit_vector(cross(cam.vup, wd));
    Vec3 vd = cross(wd, ud);
    Vec3 vu = vp_w * ud, vv = vp_h * (-vd);
    Vec3 pdu = vu / (double)w, pdv = vv / (double)h;
    Point3 p00 = cam.lookfrom - cam.focus_dist * wd - vu/2.0 - vv/2.0 + 0.5*(pdu+pdv);
    double dr = cam.focus_dist * std::tan((cam.defocus_angle/2.0)*3.14159265358979323846/180.0);

    out_camera.origin = GVec3((float)cam.lookfrom.x(), (float)cam.lookfrom.y(), (float)cam.lookfrom.z());
    out_camera.pixel00_loc = GVec3((float)p00.x(), (float)p00.y(), (float)p00.z());
    out_camera.pixel_delta_u = GVec3((float)pdu.x(), (float)pdu.y(), (float)pdu.z());
    out_camera.pixel_delta_v = GVec3((float)pdv.x(), (float)pdv.y(), (float)pdv.z());
    out_camera.defocus_angle = (float)cam.defocus_angle;
    out_camera.defocus_disk_u = GVec3((float)(ud.x()*dr), (float)(ud.y()*dr), (float)(ud.z()*dr));
    out_camera.defocus_disk_v = GVec3((float)(vd.x()*dr), (float)(vd.y()*dr), (float)(vd.z()*dr));
    out_camera.image_width = w; out_camera.image_height = h;
    out_camera.samples_per_pixel = cam.samples_per_pixel;
    out_camera.max_depth = cam.max_depth;
}

// Camera-only conversion (pas de re-conversion des objets/materiaux)
static void convert_camera_only(const Camera& cam, GCamera& out) {
    int w = cam.get_image_width(), h = cam.get_image_height();
    double theta = cam.vfov * 3.14159265358979323846 / 180.0;
    double vp_h = 2.0 * std::tan(theta / 2.0) * cam.focus_dist;
    double vp_w = vp_h * ((double)w / h);
    Vec3 wd = unit_vector(cam.lookfrom - cam.lookat);
    Vec3 ud = unit_vector(cross(cam.vup, wd));
    Vec3 vd = cross(wd, ud);
    Vec3 vu = vp_w * ud, vv = vp_h * (-vd);
    Vec3 pdu = vu / (double)w, pdv = vv / (double)h;
    Point3 p00 = cam.lookfrom - cam.focus_dist * wd - vu/2.0 - vv/2.0 + 0.5*(pdu+pdv);
    double dr = cam.focus_dist * std::tan((cam.defocus_angle/2.0)*3.14159265358979323846/180.0);

    out.origin = GVec3((float)cam.lookfrom.x(), (float)cam.lookfrom.y(), (float)cam.lookfrom.z());
    out.pixel00_loc = GVec3((float)p00.x(), (float)p00.y(), (float)p00.z());
    out.pixel_delta_u = GVec3((float)pdu.x(), (float)pdu.y(), (float)pdu.z());
    out.pixel_delta_v = GVec3((float)pdv.x(), (float)pdv.y(), (float)pdv.z());
    out.defocus_angle = (float)cam.defocus_angle;
    out.defocus_disk_u = GVec3((float)(ud.x()*dr), (float)(ud.y()*dr), (float)(ud.z()*dr));
    out.defocus_disk_v = GVec3((float)(vd.x()*dr), (float)(vd.y()*dr), (float)(vd.z()*dr));
    out.image_width = w; out.image_height = h;
    out.samples_per_pixel = cam.samples_per_pixel;
    out.max_depth = cam.max_depth;
}

// ============================================================
// Simple GPU BVH builder (on spheres only, planes are tested brute-force in leaves)
// ============================================================
struct GBuildInfo { int sphere_idx; GAABB bbox; GVec3 centroid; };

static void build_gpu_bvh(std::vector<GBuildInfo>& info, int start, int end,
                          std::vector<GBVHNode>& nodes, int num_planes) {
    int my_idx = (int)nodes.size();
    nodes.push_back(GBVHNode());

    GAABB bounds;
    for (int i = start; i < end; i++) {
        bounds.mn.x = fminf(bounds.mn.x, info[i].bbox.mn.x);
        bounds.mn.y = fminf(bounds.mn.y, info[i].bbox.mn.y);
        bounds.mn.z = fminf(bounds.mn.z, info[i].bbox.mn.z);
        bounds.mx.x = fmaxf(bounds.mx.x, info[i].bbox.mx.x);
        bounds.mx.y = fmaxf(bounds.mx.y, info[i].bbox.mx.y);
        bounds.mx.z = fmaxf(bounds.mx.z, info[i].bbox.mx.z);
    }
    nodes[my_idx].bbox = bounds;

    int n = end - start;
    if (n <= 4) {
        // Leaf : stores sphere indices [start, end)
        nodes[my_idx].data = (uint32_t)start;
        nodes[my_idx].num_prims = (uint16_t)n;
```

```
        nodes[my_idx].split_axis = 0;
        nodes[my_idx].prim_type = 0; // spheres
        return;
    }

    // Find longest axis of centroid bounds
    GVec3 cmin(1e30f,1e30f,1e30f), cmax(-1e30f,-1e30f,-1e30f);
    for (int i = start; i < end; i++) {
        cmin.x = fminf(cmin.x, info[i].centroid.x);
        cmin.y = fminf(cmin.y, info[i].centroid.y);
        cmin.z = fminf(cmin.z, info[i].centroid.z);
        cmax.x = fmaxf(cmax.x, info[i].centroid.x);
        cmax.y = fmaxf(cmax.y, info[i].centroid.y);
        cmax.z = fmaxf(cmax.z, info[i].centroid.z);
    }
    float dx = cmax.x-cmin.x, dy = cmax.y-cmin.y, dz = cmax.z-cmin.z;
    int axis = (dx > dy && dx > dz) ? 0 : (dy > dz) ? 1 : 2;

    // Sort by centroid on axis
    std::sort(info.begin()+start, info.begin()+end,
        [axis](const GBuildInfo& a, const GBuildInfo& b) {
            float va = (axis==0)?a.centroid.x:(axis==1)?a.centroid.y:a.centroid.z;
            float vb = (axis==0)?b.centroid.x:(axis==1)?b.centroid.y:b.centroid.z;
            return va < vb;
        });

    int mid = start + n / 2;

    nodes[my_idx].split_axis = (uint8_t)axis;
    nodes[my_idx].num_prims = 0; // interior
    nodes[my_idx].prim_type = 0;

    // Left child (immediately after)
    build_gpu_bvh(info, start, mid, nodes, num_planes);
    // Right child
    nodes[my_idx].data = (uint32_t)nodes.size();
    build_gpu_bvh(info, mid, end, nodes, num_planes);
}

// Renderer global persistant
GpuRenderer* s_renderer = nullptr;

void gpu_render_scene(const Scene& scene, ImageBuffer& image, float& kernel_ms) {
    // Tout est cache entre les appels : scene, BVH, renderer, framebuffer pinned
    // Renderer global pour acces depuis gpu_realtime_render_frame
    static const Scene* s_last_scene = nullptr;
    static float* s_fb = nullptr;
    static int s_fb_w = 0, s_fb_h = 0;

    GCamera gcam;

    // Re-upload scene seulement si elle a change
    if (!s_renderer || s_last_scene != &scene) {
        std::vector<GSphere> spheres;
        std::vector<GPlane> planes;
        std::vector<GMaterial> materials;
        convert_scene(scene, spheres, planes, materials, gcam);

        std::vector<GBVHNode> bvh_nodes;
        if (spheres.size() > 8) {
            std::vector<GBuildInfo> info(spheres.size());
            for (int i = 0; i < (int)spheres.size(); i++) {
                info[i].sphere_idx = i;
                float r = spheres[i].radius;
                GVec3 c = spheres[i].center;
                info[i].bbox = GAABB(GVec3(c.x-r, c.y-r, c.z-r), GVec3(c.x+r, c.y+r, c.z+r));
                info[i].centroid = c;
            }
            build_gpu_bvh(info, 0, (int)info.size(), bvh_nodes, (int)planes.size());
            std::vector<GSphere> ordered(spheres.size());
            for (int i = 0; i < (int)info.size(); i++)
                ordered[i] = spheres[info[i].sphere_idx];
            spheres = std::move(ordered);
            fprintf(stderr, "GPU BVH: %d nodes for %d spheres\n", (int)bvh_nodes.size(),
(int)spheres.size());
        }

        delete s_renderer;
        s_renderer = new GpuRenderer();
        s_renderer->upload_scene(spheres.data(), (int)spheres.size(),
                                 planes.data(), (int)planes.size(),
                                 materials.data(), (int)materials.size(),
                                 bvh_nodes.empty() ? nullptr : bvh_nodes.data(),
                                 (int)bvh_nodes.size());
        s_last_scene = &scene;
    } else {
        // Scene deja uploadee : recalculer seulement la camera
```

```
        convert_camera_only(scene.camera, gcam);
    }

    int w = gcam.image_width, h = gcam.image_height;

    // Framebuffer pinned persistant
    if (!s_fb || s_fb_w != w || s_fb_h != h) {
        if (s_fb) cudaFreeHost(s_fb);
        cudaMallocHost(&s_fb, w * h * 3 * sizeof(float));
        s_fb_w = w; s_fb_h = h;
    }

    s_renderer->render(gcam, s_fb, w, h);
    kernel_ms = s_renderer->last_render_ms();

    // Copie rapide vers ImageBuffer
    int total_px = w * h;
    for (int y = 0; y < h; y++)
        for (int x = 0; x < w; x++) {
            int px = y * w + x;
            image.set_pixel(x, y, Color(
                fminf(s_fb[px], 1.0f),
                fminf(s_fb[px + total_px], 1.0f),
                fminf(s_fb[px + total_px*2], 1.0f)));
        }
}

// ============================================================
// GPU render + bilateral denoise
void gpu_render_scene_denoised(const Scene& scene, ImageBuffer& image,
                               float& kernel_ms, float sigma_color) {
    // Render normalement (remplit s_renderer->d_framebuffer_)
    gpu_render_scene(scene, image, kernel_ms);

    // Le framebuffer device est dans s_renderer->d_framebuffer_
    if (!s_renderer) return;
    int w = image.width(), h = image.height();
    size_t fb_bytes = w * h * 3 * sizeof(float);

    // Buffer temporaire pour le resultat denoise
    static float* d_denoise_buf = nullptr;
    static int d_denoise_size = 0;
    if (!d_denoise_buf || d_denoise_size != w * h) {
        if (d_denoise_buf) cudaFree(d_denoise_buf);
        cudaMalloc(&d_denoise_buf, fb_bytes);
        d_denoise_size = w * h;
    }

    // Denoise sur le d_framebuffer_ du renderer (deja sur GPU)
    dim3 block(16, 16);
    dim3 grid((w+15)/16, (h+15)/16);
    denoise_bilateral_kernel<<grid, block>>(
        s_renderer->d_framebuffer_, d_denoise_buf, w, h, sigma_color);
    cudaDeviceSynchronize();

    // Copier le resultat denoise vers host (pinned)
    float* h_buf = nullptr;
    cudaMallocHost(&h_buf, fb_bytes);
    cudaMemcpy(h_buf, d_denoise_buf, fb_bytes, cudaMemcpyDeviceToHost);

    // Ecrire dans l'ImageBuffer
    int tp = w * h;
    for (int y = 0; y < h; y++)
        for (int x = 0; x < w; x++) {
            int px = y * w + x;
            image.set_pixel(x, y, Color(
                fminf(h_buf[px], 1.0f),
                fminf(h_buf[px+tp], 1.0f),
                fminf(h_buf[px+tp*2], 1.0f)));
        }
    cudaFreeHost(h_buf);
}

// ============================================================
// GPU-side realtime progressive rendering
// Tout reste sur le GPU : render -> accumulate -> tonemap -> transfer ARGB
// ============================================================
static float* s_rt_frame_fb = nullptr;    // framebuffer du frame courant (lineaire)
static float* s_rt_accum_fb = nullptr;    // accumulation (lineaire, somme)
static unsigned char* s_rt_argb = nullptr; // ARGB 8-bit pour affichage
static unsigned char* s_rt_argb_host = nullptr; // pinned host buffer
static int s_rt_w = 0, s_rt_h = 0;
static int s_rt_accum_count = 0;
static cudaStream_t s_rt_stream = nullptr;

void gpu_realtime_init(const Scene& scene, int width, int height) {
    s_rt_w = width; s_rt_h = height;
```

```
    size_t px = width * height;
    size_t fb_bytes = px * 3 * sizeof(float);

    if (s_rt_frame_fb) cudaFree(s_rt_frame_fb);
    if (s_rt_accum_fb) cudaFree(s_rt_accum_fb);
    if (s_rt_argb) cudaFree(s_rt_argb);
    if (s_rt_argb_host) cudaFreeHost(s_rt_argb_host);

    cudaMalloc(&s_rt_frame_fb, fb_bytes);
    cudaMalloc(&s_rt_accum_fb, fb_bytes);
    cudaMalloc(&s_rt_argb, px * 4);
    cudaMallocHost(&s_rt_argb_host, px * 4);
    cudaMemset(s_rt_accum_fb, 0, fb_bytes);
    s_rt_accum_count = 0;

    if (!s_rt_stream) cudaStreamCreate(&s_rt_stream);

    // Force scene upload
    ImageBuffer dummy(width, height);
    float km = 0;
    gpu_render_scene(scene, dummy, km);
}

// Cached per-session (pas recalcule a chaque frame)
static dim3 s_rt_block, s_rt_grid;
static int s_rt_acc_grid = 0;
static cudaEvent_t s_rt_start = nullptr, s_rt_stop = nullptr;
static bool s_rt_config_done = false;

float gpu_realtime_render_frame(const Scene& scene) {
    GCamera gcam;
    convert_camera_only(scene.camera, gcam);
    cudaMemcpyToSymbol(d_camera_bytes, &gcam, sizeof(GCamera));

    extern GpuRenderer* s_renderer;
    if (!s_renderer) return 0;

    // Config une seule fois (block size, grid, events)
    if (!s_rt_config_done) {
        s_renderer->ensure_framebuffer(s_rt_w, s_rt_h);
        int flat_block = 0, min_grid = 0;
        cudaOccupancyMaxPotentialBlockSize(&min_grid, &flat_block, render_kernel, 0, 0);
        int bx = 16, by = flat_block / bx;
        if (by < 1) by = 1; if (by > 16) by = 16;
        s_rt_block = dim3(bx, by);
        s_rt_grid = dim3((s_rt_w + bx - 1) / bx, (s_rt_h + by - 1) / by);
        int total_elems = s_rt_w * s_rt_h * 3;
        s_rt_acc_grid = (total_elems + 255) / 256;
        cudaEventCreate(&s_rt_start);
        cudaEventCreate(&s_rt_stop);
        s_rt_config_done = true;
    }

    cudaEventRecord(s_rt_start, s_rt_stream);

    // Render + accumulate, tout sur le stream (pas de sync entre les deux)
    render_kernel<<<s_rt_grid, s_rt_block, 0, s_rt_stream>>>(
        s_rt_frame_fb, s_rt_w, s_rt_h,
        s_renderer->d_bvh_nodes_, s_renderer->num_bvh_nodes_,
        s_renderer->d_spheres_, s_renderer->num_spheres_,
        s_renderer->d_planes_, s_renderer->num_planes_,
        s_renderer->d_materials_,
        s_renderer->d_soa_cx_, s_renderer->d_soa_cy_,
        s_renderer->d_soa_cz_, s_renderer->d_soa_r_, s_renderer->d_soa_mat_);

    accumulate_kernel<<<s_rt_acc_grid, 256, 0, s_rt_stream>>>(
        s_rt_frame_fb, s_rt_accum_fb, s_rt_w * s_rt_h);

    cudaEventRecord(s_rt_stop, s_rt_stream);
    // PAS de cudaStreamSynchronize ici - le tonemap synchronisera
    s_rt_accum_count++;

    float ms = 0;
    cudaEventSynchronize(s_rt_stop);
    cudaEventElapsedTime(&ms, s_rt_start, s_rt_stop);
    return ms;
}

int gpu_realtime_tonemap(unsigned char* argb_host, int width, int height) {
    dim3 block(16, 16);
    dim3 grid((width + 15) / 16, (height + 15) / 16);

    tonemap_kernel<<<grid, block, 0, s_rt_stream>>>(
        s_rt_accum_fb, s_rt_argb, width, height, s_rt_accum_count);

    // Transfer ARGB seulement (4 bytes/pixel au lieu de 12 bytes/pixel float)
    cudaMemcpyAsync(argb_host, s_rt_argb, width * height * 4,
```

```
                    cudaMemcpyDeviceToHost, s_rt_stream);
    cudaStreamSynchronize(s_rt_stream);

    return s_rt_accum_count;
}

void gpu_realtime_reset_accum() {
    if (s_rt_accum_fb)
        cudaMemset(s_rt_accum_fb, 0, s_rt_w * s_rt_h * 3 * sizeof(float));
    s_rt_accum_count = 0;
}
```

7.11 src/gpu/gpu_renderer.h File Reference

```
#include "scene/scene.h"
#include "utils/image.h"
```

Classes

- class GpuRenderer

Functions

- void gpu_render_scene (const Scene &scene, ImageBuffer &image, float &kernel_ms)
- void gpu_render_scene_denoised (const Scene &scene, ImageBuffer &image, float &kernel_ms, float sigma_color=0.1f)
- void gpu_realtime_init (const Scene &scene, int width, int height)
- float gpu_realtime_render_frame (const Scene &scene)
- int gpu_realtime_tonemap (unsigned char ∗argb_host, int width, int height)
- void gpu_realtime_reset_accum ()

7.11.1 Function Documentation

7.11.1.1 gpu_realtime_init()

```
void gpu_realtime_init (
            const Scene & scene,
            int width,
            int height)
```

Definition at line 917 of file gpu_renderer.cu.

7.11.1.2 gpu_realtime_render_frame()

```
float gpu_realtime_render_frame (
            const Scene & scene)
```

Definition at line 36 of file gpu_renderer.cu.

7.11.1.3 gpu_realtime_reset_accum()

```
void gpu_realtime_reset_accum ()
```

Definition at line 1012 of file gpu_renderer.cu.

7.11.1.4 gpu_realtime_tonemap()

```
int gpu_realtime_tonemap (
            unsigned char * argb_host,
            int width,
            int height)
```

Definition at line 997 of file gpu_renderer.cu.

7.11.1.5 gpu_render_scene()

```
void gpu_render_scene (
            const Scene & scene,
            ImageBuffer & image,
            float & kernel_ms)
```

Definition at line 788 of file gpu_renderer.cu.

7.11.1.6 gpu_render_scene_denoised()

```
void gpu_render_scene_denoised (
            const Scene & scene,
            ImageBuffer & image,
            float & kernel_ms,
            float sigma_color = 0.1f)
```

Definition at line 37 of file gpu_renderer.cu.

7.12 gpu_renderer.h

Go to the documentation of this file.

```
#ifndef RT_GPU_RENDERER_H
#define RT_GPU_RENDERER_H

// GPU renderer interface - includable by MSVC (no CUDA qualifiers in this header)
// The actual GpuRenderer class internals are in gpu_renderer.cu

#include "scene/scene.h"
#include "utils/image.h"

// Forward declarations for GPU types (defined in gpu_types.cuh, only visible to nvcc)
struct GSphere;
struct GPlane;
struct GMaterial;
struct GBVHNode;
struct GCamera;

// Host-side GPU renderer interface
class GpuRenderer {
public:
```

```
    GpuRenderer();
    ~GpuRenderer();

    // Upload scene data to GPU (call once)
    void upload_scene(const GSphere* spheres, int num_spheres,
                      const GPlane* planes, int num_planes,
                      const GMaterial* materials, int num_materials,
                      const GBVHNode* bvh_nodes, int num_bvh_nodes);

    // Render to host buffer (RGB float, 3 floats per pixel)
    void render(const GCamera& camera, float* h_output, int width, int height);

    // Get last render time in ms
    float last_render_ms() const { return last_ms_; }

    // Public accessors for realtime GPU rendering
    friend float gpu_realtime_render_frame(const Scene&);
    friend void gpu_render_scene_denoised(const Scene&, ImageBuffer&, float&, float);

private:
    GSphere* d_spheres_ = nullptr;
    // SoA spheres
    float* d_soa_cx_ = nullptr;
    float* d_soa_cy_ = nullptr;
    float* d_soa_cz_ = nullptr;
    float* d_soa_r_ = nullptr;
    int* d_soa_mat_ = nullptr;
    GPlane* d_planes_ = nullptr;
    GMaterial* d_materials_ = nullptr;
    GBVHNode* d_bvh_nodes_ = nullptr;
    float* d_framebuffer_ = nullptr;

    int num_spheres_ = 0;
    int num_planes_ = 0;
    int num_materials_ = 0;
    int num_bvh_nodes_ = 0;
    int fb_width_ = 0;
    int fb_height_ = 0;

    float last_ms_ = 0.0f;

public:
    void ensure_framebuffer(int width, int height);
private:
    void free_scene();
    void free_framebuffer();
};

// Bridge function: converts CPU Scene -> GPU data + renders
// Defined in gpu_renderer.cu (compiled by nvcc)
void gpu_render_scene(const Scene& scene, ImageBuffer& image,
                      float& kernel_ms);

// GPU render + bilateral denoise sur GPU
void gpu_render_scene_denoised(const Scene& scene, ImageBuffer& image,
                               float& kernel_ms, float sigma_color = 0.1f);

// GPU-side progressive rendering (no host transfer per frame)
// Initialise les buffers GPU pour l'accumulation
void gpu_realtime_init(const Scene& scene, int width, int height);
// Rend 1 frame et accumule sur GPU, retourne le kernel time
float gpu_realtime_render_frame(const Scene& scene);
// Tonemap GPU -> host ARGB buffer, retourne total samples
int gpu_realtime_tonemap(unsigned char* argb_host, int width, int height);
// Reset l'accumulation (quand camera bouge)
void gpu_realtime_reset_accum();

#endif // RT_GPU_RENDERER_H
```

7.13 src/gpu/gpu_types.cuh File Reference

```
#include <math.h>
#include <cstdint>
```

Classes

- struct GRay

- struct GAABB
- struct GMaterial
- struct GSphere
- struct GSphereSoA
- struct GPlane
- struct GHitRecord
- struct GBVHNode
- struct GCamera

Macros

- #define RT_GPU_TYPES_CUH

Enumerations

- enum GMaterialType : uint8_t { GMAT_LAMBERTIAN = 0 , GMAT_METAL = 1 , GMAT_DIELECTRIC = 2 , GMAT_EMISSIVE = 3 }

Functions

- struct __align__ (16) GVec3
- __host__ __device__ GVec3 operator∗ (float t, const GVec3 &v)
- __host__ __device__ float dot (const GVec3 &a, const GVec3 &b)
- __host__ __device__ GVec3 cross (const GVec3 &a, const GVec3 &b)
- __host__ __device__ GVec3 normalize (const GVec3 &v)
- __host__ __device__ GVec3 reflect (const GVec3 &v, const GVec3 &n)
- __host__ __device__ GVec3 refract (const GVec3 &uv, const GVec3 &n, float etai_over_etat)

7.13.1 Macro Definition Documentation

7.13.1.1 RT_GPU_TYPES_CUH

```
#define RT_GPU_TYPES_CUH
```

Definition at line 2 of file gpu_types.cuh.

7.13.2 Enumeration Type Documentation

7.13.2.1 GMaterialType

```
enum GMaterialType : uint8_t
```

Enumerator

GMAT_LAMBERTIAN	
GMAT_METAL	

GMAT_DIELECTRIC	
GMAT_EMISSIVE	

Definition at line 102 of file gpu_types.cuh.

7.13.3 Function Documentation

7.13.3.1 __align__()

```
struct __align__ (
            16 )
```

Definition at line 1 of file gpu_types.cuh.

7.13.3.2 cross()

```
__host__ __device__ GVec3 cross (
            const GVec3 & a,
            const GVec3 & b)  [inline]
```

Definition at line 35 of file gpu_types.cuh.

7.13.3.3 dot()

```
__host__ __device__ float dot (
            const GVec3 & a,
            const GVec3 & b)  [inline]
```

Definition at line 34 of file gpu_types.cuh.

7.13.3.4 normalize()

```
__host__ __device__ GVec3 normalize (
            const GVec3 & v)  [inline]
```

Definition at line 38 of file gpu_types.cuh.

7.13.3.5 operator∗()

```
__host__ __device__ GVec3 operator* (
            float t,
            const GVec3 & v)  [inline]
```

Definition at line 33 of file gpu_types.cuh.

7.13.3.6 reflect()

```
__host__ __device__ GVec3 reflect (
            const GVec3 & v,
            const GVec3 & n)  [inline]
```

Definition at line 46 of file gpu_types.cuh.

7.13.3.7 refract()

```
__host__ __device__ GVec3 refract (
            const GVec3 & uv,
            const GVec3 & n,
            float etai_over_etat)  [inline]
```

Definition at line 49 of file gpu_types.cuh.

7.14 gpu_types.cuh

Go to the documentation of this file.

```
00001 #ifndef RT_GPU_TYPES_CUH
00002 #define RT_GPU_TYPES_CUH
00003
00004 #include <math.h>
00005 #include <cstdint>
00006
00007 // ============================================================
00008 // GPU Vec3 (float, POD, no vtable)
00009 // ============================================================
00010 struct __align__(16) GVec3 {
00011     float x, y, z;
00012     float _pad; // padding to 16 bytes for coalesced 128-bit loads
00013
00014     __host__ __device__ GVec3() : x(0), y(0), z(0), _pad(0) {}
00015     __host__ __device__ GVec3(float x_, float y_, float z_) : x(x_), y(y_), z(z_), _pad(0) {}
00016
00017     __host__ __device__ GVec3 operator+(const GVec3& v) const { return {x+v.x, y+v.y, z+v.z}; }
00018     __host__ __device__ GVec3 operator-(const GVec3& v) const { return {x-v.x, y-v.y, z-v.z}; }
00019     __host__ __device__ GVec3 operator*(float t) const { return {x*t, y*t, z*t}; }
00020     __host__ __device__ GVec3 operator*(const GVec3& v) const { return {x*v.x, y*v.y, z*v.z}; }
00021     __host__ __device__ GVec3 operator/(float t) const { float inv = 1.0f/t; return {x*inv, y*inv,
      z*inv}; }
00022     __host__ __device__ GVec3 operator-() const { return {-x, -y, -z}; }
00023     __host__ __device__ GVec3& operator+=(const GVec3& v) { x+=v.x; y+=v.y; z+=v.z; return *this; }
00024     __host__ __device__ GVec3& operator*=(float t) { x*=t; y*=t; z*=t; return *this; }
00025
00026     __host__ __device__ float length_squared() const { return x*x + y*y + z*z; }
00027     __host__ __device__ float length() const { return sqrtf(length_squared()); }
00028     __host__ __device__ bool near_zero() const {
00029         return fabsf(x) < 1e-6f && fabsf(y) < 1e-6f && fabsf(z) < 1e-6f;
00030     }
00031 };
00032
00033 __host__ __device__ inline GVec3 operator*(float t, const GVec3& v) { return {t*v.x, t*v.y, t*v.z}; }
00034 __host__ __device__ inline float dot(const GVec3& a, const GVec3& b) { return a.x*b.x + a.y*b.y +
      a.z*b.z; }
00035 __host__ __device__ inline GVec3 cross(const GVec3& a, const GVec3& b) {
00036     return {a.y*b.z - a.z*b.y, a.z*b.x - a.x*b.z, a.x*b.y - a.y*b.x};
00037 }
00038 __host__ __device__ inline GVec3 normalize(const GVec3& v) {
00039     float len_sq = v.length_squared();
00040     if (len_sq > 0.0f) {
00041         float inv_len = rsqrtf(len_sq); // rsqrtf = hardware 1/sqrt, plus rapide que sqrtf+div
00042         return v * inv_len;
00043     }
00044     return v;
00045 }
00046 __host__ __device__ inline GVec3 reflect(const GVec3& v, const GVec3& n) {
00047     return v - 2.0f * dot(v, n) * n;
00048 }
```

```
__host__ __device__ inline GVec3 refract(const GVec3& uv, const GVec3& n, float etai_over_etat) {
    float cos_theta = fminf(dot(-uv, n), 1.0f);
    GVec3 r_perp = etai_over_etat * (uv + cos_theta * n);
    GVec3 r_parallel = -sqrtf(fabsf(1.0f - r_perp.length_squared())) * n;
    return r_perp + r_parallel;
}

// ============================================================
// GPU Ray (with precomputed inverse direction)
// ============================================================
struct GRay {
    GVec3 orig;
    GVec3 dir;
    GVec3 inv_dir;
    GVec3 orig_mul_inv; // precomputed -orig * inv_dir for FMA AABB test

    __host__ __device__ GRay() {}
    __host__ __device__ GRay(const GVec3& o, const GVec3& d)
        : orig(o), dir(d),
          inv_dir(1.0f/d.x, 1.0f/d.y, 1.0f/d.z),
          orig_mul_inv(-o.x/d.x, -o.y/d.y, -o.z/d.z) {}

    __host__ __device__ GVec3 at(float t) const { return orig + dir * t; }
};

// ============================================================
// GPU AABB
// ============================================================
struct GAABB {
    GVec3 mn, mx; // min, max

    __host__ __device__ GAABB() : mn(1e30f, 1e30f, 1e30f), mx(-1e30f, -1e30f, -1e30f) {}
    __host__ __device__ GAABB(const GVec3& a, const GVec3& b) : mn(a), mx(b) {}

    __host__ __device__ bool hit(const GRay& r, float t_min, float t_max) const {
        // FMA-based slab test (Opt#7 level 4 from RTIOW CUDA article)
        // fmaf(invDir, bound, origMulInv) = invDir*bound + (-orig*invDir) = (bound-orig)*invDir
        float tx0 = fmaf(r.inv_dir.x, mn.x, r.orig_mul_inv.x);
        float tx1 = fmaf(r.inv_dir.x, mx.x, r.orig_mul_inv.x);
        float ty0 = fmaf(r.inv_dir.y, mn.y, r.orig_mul_inv.y);
        float ty1 = fmaf(r.inv_dir.y, mx.y, r.orig_mul_inv.y);
        float tz0 = fmaf(r.inv_dir.z, mn.z, r.orig_mul_inv.z);
        float tz1 = fmaf(r.inv_dir.z, mx.z, r.orig_mul_inv.z);

        float tEnter = fmaxf(fmaxf(fminf(tx0,tx1), fminf(ty0,ty1)), fmaxf(fminf(tz0,tz1), t_min));
        float tExit  = fminf(fminf(fmaxf(tx0,tx1), fmaxf(ty0,ty1)), fminf(fmaxf(tz0,tz1), t_max));
        return tEnter <= tExit;
    }
};

// ============================================================
// GPU Material (tagged union, pas de vtable)
// ============================================================
enum GMaterialType : uint8_t { GMAT_LAMBERTIAN = 0, GMAT_METAL = 1, GMAT_DIELECTRIC = 2, GMAT_EMISSIVE = 3 };

struct GMaterial {
    GVec3 albedo;
    float param;           // fuzz pour metal, refraction_index pour dielectric
    GMaterialType type;

    __host__ __device__ GMaterial()
        : albedo(0.5f, 0.5f, 0.5f), param(0.0f), type(GMAT_LAMBERTIAN) {}
};

// ============================================================
// GPU Sphere
// ============================================================
struct GSphere {
    GVec3 center;
    float radius;
    int material_id;
};

// SoA layout pour spheres (acces coalescents sur GPU)
struct GSphereSoA {
    float* cx;  // center.x[]
    float* cy;  // center.y[]
    float* cz;  // center.z[]
    float* r;   // radius[]
    int* mat;   // material_id[]
    int count;
};

// ============================================================
// GPU Plane
// ============================================================
```

```
struct GPlane {
    GVec3 point;
    GVec3 normal;
    int material_id;
};

// ============================================================
// GPU HitRecord
// ============================================================
struct GHitRecord {
    GVec3 p;
    GVec3 normal;
    float t;
    int material_id;
    bool front_face;

    __host__ __device__ void set_face_normal(const GRay& r, const GVec3& outward_normal) {
        front_face = dot(r.dir, outward_normal) < 0.0f;
        normal = front_face ? outward_normal : -outward_normal;
    }
};

// ============================================================
// GPU Flat BVH Node
// ============================================================
struct GBVHNode {
    GAABB bbox;
    uint32_t data;        // leaf: primitives_offset | interior: second_child_offset
    uint16_t num_prims;   // 0 = interior
    uint8_t split_axis;
    uint8_t prim_type;    // 0 = sphere, 1 = plane (for leaves)
};

// ============================================================
// GPU Camera
// ============================================================
struct GCamera {
    GVec3 origin;
    GVec3 pixel00_loc;
    GVec3 pixel_delta_u;
    GVec3 pixel_delta_v;
    GVec3 defocus_disk_u;
    GVec3 defocus_disk_v;
    float defocus_angle;
    int image_width;
    int image_height;
    int samples_per_pixel;
    int max_depth;
};

#endif // RT_GPU_TYPES_CUH
```

7.15 src/hybrid/hybrid_renderer.cu File Reference

```
#include "hybrid/hybrid_renderer.h"
#include "gpu/gpu_renderer.h"
#include "gpu/gpu_types.cuh"
#include "scene/primitives.cuh"
#include "scene/material.cuh"
#include "scene/bvh_builder.h"
#include "cpu/cpu_renderer.h"
#include "math/random.cuh"
#include "math/interval.cuh"
#include <thread>
#include <atomic>
#include <chrono>
#include <cstdio>
#include <vector>
#include <cmath>
```

Functions

- void hybrid_render_scene (const Scene &scene, ImageBuffer &image, float gpu_fraction, float &cpu_time↩ _ms, float &gpu_time_ms)

7.15.1 Function Documentation

7.15.1.1 hybrid_render_scene()

```
void hybrid_render_scene (
            const Scene & scene,
            ImageBuffer & image,
            float gpu_fraction,
            float & cpu_time_ms,
            float & gpu_time_ms)
```

Definition at line 22 of file hybrid_renderer.cu.

7.16 hybrid_renderer.cu

Go to the documentation of this file.

```
00001 // Hybrid CPU+GPU renderer avec work stealing par tuiles
00002 // Le GPU rend un batch de tuiles (la fraction haute de l'image)
00003 // Le CPU rend les tuiles restantes en multi-thread
00004 // Les deux travaillent en parallele
00005
00006 #include "hybrid/hybrid_renderer.h"
00007 #include "gpu/gpu_renderer.h"
00008 #include "gpu/gpu_types.cuh"
00009 #include "scene/primitives.cuh"
00010 #include "scene/material.cuh"
00011 #include "scene/bvh_builder.h"
00012 #include "cpu/cpu_renderer.h"
00013 #include "math/random.cuh"
00014 #include "math/interval.cuh"
00015 #include <thread>
00016 #include <atomic>
00017 #include <chrono>
00018 #include <cstdio>
00019 #include <vector>
00020 #include <cmath>
00021
00022 void hybrid_render_scene(const Scene& scene, ImageBuffer& image,
00023                          float gpu_fraction,
00024                          float& cpu_time_ms, float& gpu_time_ms) {
00025     int w = image.width();
00026     int h = image.height();
00027     constexpr int TILE_SIZE = 32;
00028
00029     int tiles_x = (w + TILE_SIZE - 1) / TILE_SIZE;
00030     int tiles_y = (h + TILE_SIZE - 1) / TILE_SIZE;
00031     int total_tiles = tiles_x * tiles_y;
00032
00033     // Repartition : GPU prend les premieres tuiles, CPU le reste
00034     int gpu_tiles = (int)(total_tiles * gpu_fraction);
00035     if (gpu_tiles < 0) gpu_tiles = 0;
00036     if (gpu_tiles > total_tiles) gpu_tiles = total_tiles;
00037     int cpu_start_tile = gpu_tiles;
00038
00039     // Calcul des lignes GPU (tuiles completes depuis le haut)
00040     int gpu_rows = 0;
00041     if (gpu_tiles > 0) {
00042         int gpu_tile_rows = (gpu_tiles + tiles_x - 1) / tiles_x;
00043         gpu_rows = gpu_tile_rows * TILE_SIZE;
00044         if (gpu_rows > h) gpu_rows = h;
00045         gpu_tiles = gpu_tile_rows * tiles_x; // ajuster au nombre reel
00046         cpu_start_tile = gpu_tiles;
00047     }
```

```

    fprintf(stderr, "Hybrid: GPU %d tuiles (lignes 0-%d), CPU %d tuiles (lignes %d-%d), total %d\n",
            gpu_tiles, gpu_rows, total_tiles - cpu_start_tile, gpu_rows, h, total_tiles);

    // --- GPU thread : rend la partie haute ---
    std::thread gpu_thread;
    float gpu_kernel_ms = 0;

    if (gpu_rows > 0) {
        gpu_thread = std::thread([&]() {
            auto t0 = std::chrono::high_resolution_clock::now();

            // GPU rend toute l'image (plus simple que de modifier le kernel)
            // On ne copie que les lignes GPU dans l'ImageBuffer
            float km = 0;
            gpu_render_scene(scene, image, km);
            gpu_kernel_ms = km;

            auto t1 = std::chrono::high_resolution_clock::now();
            gpu_time_ms = (float)std::chrono::duration<double, std::milli>(t1 - t0).count();
        });
    }

    // --- CPU : rend les tuiles restantes avec work stealing ---
    if (cpu_start_tile < total_tiles) {
        auto t0 = std::chrono::high_resolution_clock::now();

        CpuRenderer cpu_renderer(scene, image);
        cpu_renderer.enable_bvh();

        int num_threads = (int)std::thread::hardware_concurrency();
        if (num_threads <= 0) num_threads = 4;

        int cpu_total = total_tiles - cpu_start_tile;
        std::atomic<int> next_tile(0);

        auto worker = [&]() {
            while (true) {
                int local_idx = next_tile.fetch_add(1);
                if (local_idx >= cpu_total) break;

                int tile_idx = cpu_start_tile + local_idx;
                int tx = tile_idx % tiles_x;
                int ty = tile_idx / tiles_x;
                int x0 = tx * TILE_SIZE;
                int y0 = ty * TILE_SIZE;
                int x1 = std::min(x0 + TILE_SIZE, w);
                int y1 = std::min(y0 + TILE_SIZE, h);

                cpu_renderer.render_tile(x0, y0, x1, y1);
            }
        };

        std::vector<std::thread> threads;
        for (int i = 0; i < num_threads; i++)
            threads.emplace_back(worker);
        for (auto& t : threads)
            t.join();

        auto t1 = std::chrono::high_resolution_clock::now();
        cpu_time_ms = (float)std::chrono::duration<double, std::milli>(t1 - t0).count();
    }

    if (gpu_thread.joinable())
        gpu_thread.join();

    fprintf(stderr, "Hybrid: CPU %.1fms, GPU kernel %.1fms (total GPU %.1fms)\n",
            cpu_time_ms, gpu_kernel_ms, gpu_time_ms);
}
```

7.17 src/hybrid/hybrid_renderer.h File Reference

```
#include "scene/scene.h"
#include "utils/image.h"
```

Functions

- void hybrid_render_scene (const Scene &scene, ImageBuffer &image, float gpu_fraction, float &cpu_time↩ _ms, float &gpu_time_ms)

7.17.1 Function Documentation

7.17.1.1 hybrid_render_scene()

```
void hybrid_render_scene (
            const Scene & scene,
            ImageBuffer & image,
            float gpu_fraction,
            float & cpu_time_ms,
            float & gpu_time_ms)
```

Definition at line 22 of file hybrid_renderer.cu.

7.18 hybrid_renderer.h

Go to the documentation of this file.

```
00001 #ifndef RT_HYBRID_RENDERER_H
00002 #define RT_HYBRID_RENDERER_H
00003
00004 #include "scene/scene.h"
00005 #include "utils/image.h"
00006
00007 // Rendu hybride CPU+GPU avec work stealing par tuiles
00008 // GPU et CPU consomment des tuiles depuis une queue partagee
00009 void hybrid_render_scene(const Scene& scene, ImageBuffer& image,
00010                          float gpu_fraction,  // fraction initiale pour le GPU
00011                          float& cpu_time_ms, float& gpu_time_ms);
00012
00013 #endif // RT_HYBRID_RENDERER_H
```

7.19 src/main.cpp File Reference

```
#include "scene/scene.h"
#include "utils/image.h"
#include "utils/denoise.h"
#include "cpu/cpu_renderer.h"
#include <iostream>
#include <string>
#include <chrono>
#include <cstdlib>
```

Functions

- int main (int argc, char *argv[])

7.19.1 Function Documentation

7.19.1.1 main()

```
int main (
            int argc,
            char * argv[])
```

Definition at line 16 of file main.cpp.

7.20 main.cpp

Go to the documentation of this file.

```
#include "scene/scene.h"
#include "utils/image.h"
#include "utils/denoise.h"
#include "cpu/cpu_renderer.h"
#include <iostream>
#include <string>
#include <chrono>
#include <cstdlib>

#ifdef RT_CUDA_ENABLED
#include "gpu/gpu_renderer.h"
#include "gpu/gpu_realtime.h"
#include "hybrid/hybrid_renderer.h"
#endif

int main(int argc, char* argv[]) {
    std::string scene_name = "three_spheres";
    std::string output_file = "output/render.ppm";
    int width = 0;
    int spp = 0;
    int threads = 0;
    bool use_bvh = true;
    bool single_thread = false;
    bool use_gpu = false;
    bool use_hybrid = false;
    bool use_realtime = false;
    bool use_denoise = false;
    bool use_denoise_gpu = false;
    float gpu_frac = 0.7f;

    for (int i = 1; i < argc; ++i) {
        std::string arg = argv[i];
        if (arg == "--scene" && i + 1 < argc) scene_name = argv[++i];
        else if (arg == "--output" && i + 1 < argc) output_file = argv[++i];
        else if (arg == "--width" && i + 1 < argc) width = std::atoi(argv[++i]);
        else if (arg == "--spp" && i + 1 < argc) spp = std::atoi(argv[++i]);
        else if (arg == "--threads" && i + 1 < argc) threads = std::atoi(argv[++i]);
        else if (arg == "--no-bvh") use_bvh = false;
        else if (arg == "--single-thread") single_thread = true;
        else if (arg == "--gpu") use_gpu = true;
        else if (arg == "--hybrid") use_hybrid = true;
        else if (arg == "--realtime") use_realtime = true;
        else if (arg == "--denoise") use_denoise = true;
        else if (arg == "--denoise-gpu") use_denoise_gpu = true;
        else if (arg == "--gpu-frac" && i + 1 < argc) gpu_frac = (float)std::atof(argv[++i]);
        else if (arg == "--help") {
            std::cout « "Usage: raytracer [options]\n"
                      « "  --scene <name>       Scene: three_spheres, final\n"
                      « "  --output <file>      Output PPM file\n"
                      « "  --width <pixels>     Image width\n"
                      « "  --spp <samples>      Samples per pixel\n"
                      « "  --threads <n>        Number of CPU threads (0=auto)\n"
                      « "  --no-bvh             Disable BVH\n"
                      « "  --single-thread      Single-threaded CPU\n"
                      « "  --gpu                Use GPU (CUDA) renderer\n"
                      « "  --hybrid             Use hybrid CPU+GPU renderer\n"
                      « "  --gpu-frac <0-1>     GPU fraction in hybrid mode (default: 0.7)\n";
            return 0;
        }
    }

    Scene scene;
    if (scene_name == "final")
        scene = Scene::build_final_scene();
    else if (scene_name == "triangle")
        scene = Scene::build_triangle_scene();
    else if (scene_name == "cornell")
        scene = Scene::build_cornell_box();
    else if (scene_name == "showcase")
        scene = Scene::build_showcase();
    else if (scene_name == "hall")
        scene = Scene::build_hall();
    else if (scene_name == "galaxy")
        scene = Scene::build_galaxy();
    else if (scene_name == "minimal")
        scene = Scene::build_minimal();
    else
        scene = Scene::build_three_spheres();

    if (width > 0) {
        scene.camera.image_width = width;
        scene.camera.initialize();
```

```
    }
    if (spp > 0)
        scene.camera.samples_per_pixel = spp;

    int img_w = scene.camera.get_image_width();
    int img_h = scene.camera.get_image_height();
    ImageBuffer image(img_w, img_h);

    std::cerr « "Rendering " « img_w « "x" « img_h
              « " (" « scene.camera.samples_per_pixel « " spp, depth "
              « scene.camera.max_depth « ")";

#ifdef RT_CUDA_ENABLED
    if (use_realtime) {
        std::cerr « " [REALTIME GPU]\n";
        int rt_w = (width > 0) ? width : 640;
        int rt_h = (int)(rt_w / scene.camera.aspect_ratio);
        scene.camera.image_width = rt_w;
        scene.camera.initialize();
        gpu_realtime_render(scene, rt_w, rt_h);
        return 0;
    }

    if (use_hybrid) {
        std::cerr « " [HYBRID gpu=" « (int)(gpu_frac*100) « "%]\n";
        float cpu_ms = 0, gpu_ms = 0;
        auto start = std::chrono::high_resolution_clock::now();
        hybrid_render_scene(scene, image, gpu_frac, cpu_ms, gpu_ms);
        auto end = std::chrono::high_resolution_clock::now();
        double elapsed = std::chrono::duration<double>(end - start).count();
        std::cerr « "Total: " « elapsed « "s\n";
        image.write_ppm(output_file);
        std::cerr « "Saved to: " « output_file « "\n";
        return 0;
    }

    if (use_gpu) {
        std::cerr « (use_denoise_gpu ? " [GPU+DENOISE]\n" : " [GPU]\n");
        float kernel_ms = 0;
        auto start = std::chrono::high_resolution_clock::now();
        if (use_denoise_gpu)
            gpu_render_scene_denoised(scene, image, kernel_ms);
        else
            gpu_render_scene(scene, image, kernel_ms);
        auto end = std::chrono::high_resolution_clock::now();
        double elapsed = std::chrono::duration<double>(end - start).count();
        std::cerr « "GPU kernel: " « kernel_ms « " ms\n";
        std::cerr « "Total (incl. transfer): " « elapsed « "s\n";
        image.write_ppm(output_file);
        std::cerr « "Saved to: " « output_file « "\n";
        return 0;
    }
#else
    if (use_gpu || use_hybrid) {
        std::cerr « "\nError: GPU not compiled. Rebuild with -DENABLE_CUDA=ON\n";
        return 1;
    }
#endif

    std::cerr « (use_bvh ? " [BVH]" : "") « "\n";
    CpuRenderer renderer(scene, image);
    if (use_bvh) {
        std::cerr « "Building BVH...\n";
        renderer.enable_bvh();
    }

    auto start = std::chrono::high_resolution_clock::now();
    if (single_thread)
        renderer.render();
    else
        renderer.render_threaded(threads);
    auto end = std::chrono::high_resolution_clock::now();

    double elapsed = std::chrono::duration<double>(end - start).count();
    std::cerr « "Render time: " « elapsed « "s\n";
    if (use_denoise) {
        std::cerr « "Denoising...\n";
        denoise_bilateral(image, 3, 0.1);
    }
    image.write_ppm(output_file);
    std::cerr « "Saved to: " « output_file « "\n";

    return 0;
}
```

7.21 src/math/aabb.cuh File Reference

```
#include "math/vec3.cuh"
#include "math/ray.cuh"
#include "math/interval.cuh"
#include <algorithm>
#include <cmath>
```

Classes

- class AABB

Macros

- #define RT_MATH_AABB_CUH

Functions

- RT_HOSTDEV AABB surrounding_box (const AABB &a, const AABB &b)

7.21.1 Macro Definition Documentation

7.21.1.1 RT_MATH_AABB_CUH

```
#define RT_MATH_AABB_CUH
```

Definition at line 8 of file aabb.cuh.

7.21.2 Function Documentation

7.21.2.1 surrounding_box()

```
RT_HOSTDEV AABB surrounding_box (
            const AABB & a,
            const AABB & b)  [inline]
```

Definition at line 107 of file aabb.cuh.

7.22 aabb.cuh

Go to the documentation of this file.

```
// aabb.cuh - Axis-Aligned Bounding Box pour l'acceleration BVH.
// Chaque noeud BVH stocke une AABB ; la traversee teste le rayon
// contre la boite avant de descendre dans les enfants.
// Le test slab (Williams et al. 2005) utilise inv_dir precalcule
// dans Ray pour transformer 6 divisions en 6 multiplications.

#ifndef RT_MATH_AABB_CUH
#define RT_MATH_AABB_CUH

#include "math/vec3.cuh"
#include "math/ray.cuh"
#include "math/interval.cuh"
#include <algorithm>
#include <cmath>

class AABB {
public:
    Interval x, y, z; // bornes sur chaque axe

    RT_HOSTDEV AABB() {} // intervalles vides par defaut

    RT_HOSTDEV AABB(const Interval& x, const Interval& y, const Interval& z)
        : x(x), y(y), z(z) {}

    // Construction depuis deux coins opposes (ordre quelconque)
    RT_HOSTDEV AABB(const Point3& a, const Point3& b) {
        x = (a[0] <= b[0]) ? Interval(a[0], b[0]) : Interval(b[0], a[0]);
        y = (a[1] <= b[1]) ? Interval(a[1], b[1]) : Interval(b[1], a[1]);
        z = (a[2] <= b[2]) ? Interval(a[2], b[2]) : Interval(b[2], a[2]);
    }

    // Acces generique par indice d'axe (0=x, 1=y, 2=z)
    // Utilise par le BVH pour choisir l'axe de split
    RT_HOSTDEV const Interval& axis_interval(int n) const {
        if (n == 1) return y;
        if (n == 2) return z;
        return x;
    }

    RT_HOSTDEV Point3 min_point() const { return Point3(x.min, y.min, z.min); }
    RT_HOSTDEV Point3 max_point() const { return Point3(x.max, y.max, z.max); }

    // Test slab de Williams et al. : intersection rayon-boite en 6 mul + 6 cmp.
    // inv_dir precalcule dans Ray transforme (bound - origin) / dir en multiplication.
    // On resserre progressivement [t_min, t_max] par axe ; si l'intervalle
    // devient vide (t_max < t_min), le rayon rate la boite -> early exit.
    RT_HOSTDEV bool hit(const Ray& r, double t_min, double t_max) const {
        const Point3& ro = r.orig;
        const Vec3& inv = r.inv_dir;

        // Axe X
        double tx1 = (x.min - ro[0]) * inv[0];
        double tx2 = (x.max - ro[0]) * inv[0];
        if (tx1 > tx2) { double tmp = tx1; tx1 = tx2; tx2 = tmp; }
        if (tx1 > t_min) t_min = tx1;
        if (tx2 < t_max) t_max = tx2;
        if (t_max < t_min) return false;

        // Axe Y
        double ty1 = (y.min - ro[1]) * inv[1];
        double ty2 = (y.max - ro[1]) * inv[1];
        if (ty1 > ty2) { double tmp = ty1; ty1 = ty2; ty2 = tmp; }
        if (ty1 > t_min) t_min = ty1;
        if (ty2 < t_max) t_max = ty2;
        if (t_max < t_min) return false;

        // Axe Z
        double tz1 = (z.min - ro[2]) * inv[2];
        double tz2 = (z.max - ro[2]) * inv[2];
        if (tz1 > tz2) { double tmp = tz1; tz1 = tz2; tz2 = tmp; }
        if (tz1 > t_min) t_min = tz1;
        if (tz2 < t_max) t_max = tz2;
        if (t_max < t_min) return false;

        return true;
    }

    // Surcharge pour les tests qui passent un Interval
    RT_HOSTDEV bool hit(const Ray& r, Interval ray_t) const {
        return hit(r, ray_t.min, ray_t.max);
    }

```

```
    // Surface totale de la boite : utilisee par la SAH (Surface Area Heuristic)
    // pour estimer le cout de split lors de la construction du BVH.
    // Cout ~ surface_enfant / surface_parent * nb_primitives_enfant
    RT_HOSTDEV double surface_area() const {
        double dx = x.size(), dy = y.size(), dz = z.size();
        return 2.0 * (dx*dy + dy*dz + dz*dx);
    }

    // Axe le plus long : heuristique de split quand SAH n'est pas utilisee
    RT_HOSTDEV int longest_axis() const {
        double sx = x.size(), sy = y.size(), sz = z.size();
        if (sx > sy) return (sx > sz) ? 0 : 2;
        return (sy > sz) ? 1 : 2;
    }

    static const AABB empty;
    static const AABB universe;
};

inline const AABB AABB::empty = AABB(Interval::empty, Interval::empty, Interval::empty);
inline const AABB AABB::universe = AABB(Interval::universe, Interval::universe, Interval::universe);

// Union de deux AABB : construit la plus petite boite englobant les deux.
// Utilise recursivement lors de la construction bottom-up du BVH.
inline RT_HOSTDEV AABB surrounding_box(const AABB& a, const AABB& b) {
    return AABB(
        Interval(std::fmin(a.x.min, b.x.min), std::fmax(a.x.max, b.x.max)),
        Interval(std::fmin(a.y.min, b.y.min), std::fmax(a.y.max, b.y.max)),
        Interval(std::fmin(a.z.min, b.z.min), std::fmax(a.z.max, b.z.max))
    );
}

#endif // RT_MATH_AABB_CUH
```

7.23 src/math/interval.cuh File Reference

```
#include <cmath>
#include <limits>
#include "math/vec3.cuh"
```

Classes

- class Interval

Macros

- #define RT_MATH_INTERVAL_CUH

7.23.1 Macro Definition Documentation

7.23.1.1 RT_MATH_INTERVAL_CUH

```
#define RT_MATH_INTERVAL_CUH
```

Definition at line 6 of file interval.cuh.

7.24 interval.cuh

Go to the documentation of this file.

```
// interval.cuh - Intervalle ferme [min, max] sur les reels.
// Utilise partout : parametrage t valide des rayons, bornes AABB,
// clamping des couleurs avant tone mapping.

#ifndef RT_MATH_INTERVAL_CUH
#define RT_MATH_INTERVAL_CUH

#include <cmath>
#include <limits>

#include "math/vec3.cuh" // pour RT_HOSTDEV

class Interval {
public:
    double min, max;

    // Par defaut : intervalle vide (min > max, aucun t valide)
    RT_HOSTDEV Interval()
        : min(+std::numeric_limits<double>::infinity()),
          max(-std::numeric_limits<double>::infinity()) {}

    RT_HOSTDEV Interval(double min, double max) : min(min), max(max) {}

    RT_HOSTDEV double size() const { return max - min; }

    // Inclusion large (bornes incluses)
    RT_HOSTDEV bool contains(double x) const { return min <= x && x <= max; }

    // Inclusion stricte (bornes exclues) : utilise pour ignorer les self-intersections
    // a t~0 (shadow acne) en testant t dans ]epsilon, +inf[
    RT_HOSTDEV bool surrounds(double x) const { return min < x && x < max; }

    RT_HOSTDEV double clamp(double x) const {
        if (x < min) return min;
        if (x > max) return max;
        return x;
    }

    static const Interval empty;
    static const Interval universe;
};

// Singletons inline (C++17) : evitent les problemes d'ordre d'initialisation
inline const Interval Interval::empty    = Interval(+std::numeric_limits<double>::infinity(),
                                                    -std::numeric_limits<double>::infinity());
inline const Interval Interval::universe = Interval(-std::numeric_limits<double>::infinity(),
                                                    +std::numeric_limits<double>::infinity());

#endif // RT_MATH_INTERVAL_CUH
```

7.25 src/math/random.cuh File Reference

```
#include "math/vec3.cuh"
#include <random>
```

Macros

- #define RT_MATH_RANDOM_CUH

Functions

- double random_double ()
- double random_double (double min, double max)
- Vec3 random_vec3 ()

- Vec3 random_vec3 (double min, double max)
- Vec3 random_in_unit_sphere ()
- Vec3 random_unit_vector ()
- Vec3 random_on_hemisphere (const Vec3 &normal)
- Vec3 random_in_unit_disk ()

7.25.1 Macro Definition Documentation

7.25.1.1 RT_MATH_RANDOM_CUH

```
#define RT_MATH_RANDOM_CUH
```

Definition at line 6 of file random.cuh.

7.25.2 Function Documentation

7.25.2.1 random_double() [1/2]

```
double random_double ()  [inline]
```

Definition at line 13 of file random.cuh.

7.25.2.2 random_double() [2/2]

```
double random_double (
            double min,
            double max)  [inline]
```

Definition at line 19 of file random.cuh.

7.25.2.3 random_in_unit_disk()

```
Vec3 random_in_unit_disk ()  [inline]
```

Definition at line 59 of file random.cuh.

7.25.2.4 random_in_unit_sphere()

```
Vec3 random_in_unit_sphere ()  [inline]
```

Definition at line 33 of file random.cuh.

7.25.2.5 random_on_hemisphere()

```
Vec3 random_on_hemisphere (
            const Vec3 & normal)  [inline]
```

Definition at line 50 of file random.cuh.

7.25.2.6 random_unit_vector()

```
Vec3 random_unit_vector ()  [inline]
```

Definition at line 44 of file random.cuh.

7.25.2.7 random_vec3() [1/2]

```
Vec3 random_vec3 ()  [inline]
```

Definition at line 23 of file random.cuh.

7.25.2.8 random_vec3() [2/2]

```
Vec3 random_vec3 (
            double min,
            double max)  [inline]
```

Definition at line 27 of file random.cuh.

7.26 random.cuh

Go to the documentation of this file.

```
00001 // random.cuh - Fonctions aleatoires CPU pour le Monte Carlo.
00002 // thread_local : chaque thread du pool CPU a son propre generateur,
00003 // pas de mutex ni contention. Le GPU utilise curand a la place (gpu_renderer.cu).
00004
00005 #ifndef RT_MATH_RANDOM_CUH
00006 #define RT_MATH_RANDOM_CUH
00007
00008 #include "math/vec3.cuh"
00009 #include <random>
00010
00011 // Mersenne Twister 19937 : periode 2^19937-1, qualite statistique suffisante
00012 // pour le path tracing. Seed via random_device (entropie hardware si disponible).
00013 inline double random_double() {
00014     thread_local std::mt19937 gen(std::random_device{}());
00015     thread_local std::uniform_real_distribution<double> dist(0.0, 1.0);
00016     return dist(gen);
00017 }
00018
00019 inline double random_double(double min, double max) {
00020     return min + (max - min) * random_double();
00021 }
00022
00023 inline Vec3 random_vec3() {
00024     return Vec3(random_double(), random_double(), random_double());
00025 }
00026
00027 inline Vec3 random_vec3(double min, double max) {
00028     return Vec3(random_double(min, max), random_double(min, max), random_double(min, max));
00029 }
00030
```

```
// Rejection sampling dans le cube [-1,1]^3 : ~52% d'acceptance.
// Plus simple et aussi rapide que Box-Muller pour 3 composantes.
inline Vec3 random_in_unit_sphere() {
    while (true) {
        Vec3 p = random_vec3(-1.0, 1.0);
        if (p.length_squared() < 1.0)
            return p;
    }
}

// Direction aleatoire uniforme sur la sphere unite :
// normaliser un point uniforme dans la boule donne une distribution
// uniforme sur la surface (distribution de Lambert pour le diffus).
inline Vec3 random_unit_vector() {
    return unit_vector(random_in_unit_sphere());
}

// Hemisphere cosinus-pondere implicite : si le vecteur pointe
// du mauvais cote de la normale, on le retourne.
inline Vec3 random_on_hemisphere(const Vec3& normal) {
    Vec3 on_sphere = random_unit_vector();
    if (dot(on_sphere, normal) > 0.0)
        return on_sphere;
    return -on_sphere;
}

// Disque unite : utilise pour le depth of field (DOF).
// L'ouverture de la lentille est echantillonnee uniformement sur un disque.
inline Vec3 random_in_unit_disk() {
    while (true) {
        Vec3 p(random_double(-1.0, 1.0), random_double(-1.0, 1.0), 0.0);
        if (p.length_squared() < 1.0)
            return p;
    }
}

#endif // RT_MATH_RANDOM_CUH
```

7.27 src/math/ray.cuh File Reference

```
#include "math/vec3.cuh"
```

Classes

- class Ray

Macros

- #define RT_MATH_RAY_CUH

7.27.1 Macro Definition Documentation

7.27.1.1 RT_MATH_RAY_CUH

```
#define RT_MATH_RAY_CUH
```

Definition at line 7 of file ray.cuh.

7.28 ray.cuh

Go to the documentation of this file.

```
// ray.cuh - Rayon parametrique P(t) = origin + t * direction.
// Precalcule inv_dir et dir_sign a la construction pour accelerer
// le test slab AABB (Williams et al. 2005) : on evite 6 divisions par
// test d'intersection boite, critique dans la traversee BVH.

#ifndef RT_MATH_RAY_CUH
#define RT_MATH_RAY_CUH

#include "math/vec3.cuh"

class Ray {
public:
    Point3 orig;
    Vec3 dir;
    Vec3 inv_dir;       // 1/direction, precalcule (Williams et al.)
    int dir_sign[3];    // 0 si positif, 1 si negatif : indexation directe min/max AABB

    RT_HOSTDEV Ray() {}
    RT_HOSTDEV Ray(const Point3& origin, const Vec3& direction)
        : orig(origin), dir(direction)
    {
        // Precalcul une seule fois par rayon, reutilise a chaque noeud BVH
        inv_dir = Vec3(1.0 / direction.x(), 1.0 / direction.y(), 1.0 / direction.z());
        dir_sign[0] = (inv_dir.x() < 0) ? 1 : 0;
        dir_sign[1] = (inv_dir.y() < 0) ? 1 : 0;
        dir_sign[2] = (inv_dir.z() < 0) ? 1 : 0;
    }

    RT_HOSTDEV const Point3& origin() const { return orig; }
    RT_HOSTDEV const Vec3& direction() const { return dir; }

    // Point le long du rayon a la distance t
    RT_HOSTDEV Point3 at(double t) const {
        return orig + t * dir;
    }
};

#endif // RT_MATH_RAY_CUH
```

7.29 src/math/vec3.cuh File Reference

```
#include <cmath>
#include <iostream>
```

Classes

- class Vec3

Macros

- #define RT_MATH_VEC3_CUH
- #define RT_HOSTDEV

Typedefs

- using Point3 = Vec3
- using Color = Vec3

Functions

- RT_HOSTDEV Vec3 operator+ (const Vec3 &u, const Vec3 &v)
- RT_HOSTDEV Vec3 operator- (const Vec3 &u, const Vec3 &v)
- RT_HOSTDEV Vec3 operator∗ (const Vec3 &u, const Vec3 &v)
- RT_HOSTDEV Vec3 operator∗ (double t, const Vec3 &v)
- RT_HOSTDEV Vec3 operator∗ (const Vec3 &v, double t)
- RT_HOSTDEV Vec3 operator/ (const Vec3 &v, double t)
- RT_HOSTDEV double dot (const Vec3 &u, const Vec3 &v)
- RT_HOSTDEV Vec3 cross (const Vec3 &u, const Vec3 &v)
- RT_HOSTDEV Vec3 unit_vector (const Vec3 &v)
- RT_HOSTDEV Vec3 reflect (const Vec3 &v, const Vec3 &n)
- RT_HOSTDEV Vec3 refract (const Vec3 &uv, const Vec3 &n, double etai_over_etat)
- std::ostream & operator<< (std::ostream &out, const Vec3 &v)

7.29.1 Macro Definition Documentation

7.29.1.1 RT_HOSTDEV

```
#define RT_HOSTDEV
```

Definition at line 16 of file vec3.cuh.

7.29.1.2 RT_MATH_VEC3_CUH

```
#define RT_MATH_VEC3_CUH
```

Definition at line 6 of file vec3.cuh.

7.29.2 Typedef Documentation

7.29.2.1 Color

```
using Color = Vec3
```

Definition at line 72 of file vec3.cuh.

7.29.2.2 Point3

```
using Point3 = Vec3
```

Definition at line 71 of file vec3.cuh.

7.29.3 Function Documentation

7.29.3.1 cross()

```
RT_HOSTDEV Vec3 cross (
            const Vec3 & u,
            const Vec3 & v)  [inline]
```

Definition at line 108 of file vec3.cuh.

7.29.3.2 dot()

```
RT_HOSTDEV double dot (
            const Vec3 & u,
            const Vec3 & v)  [inline]
```

Definition at line 104 of file vec3.cuh.

7.29.3.3 operator∗() [1/3]

```
RT_HOSTDEV Vec3 operator* (
            const Vec3 & u,
            const Vec3 & v)  [inline]
```

Definition at line 86 of file vec3.cuh.

7.29.3.4 operator∗() [2/3]

```
RT_HOSTDEV Vec3 operator* (
            const Vec3 & v,
            double t)  [inline]
```

Definition at line 94 of file vec3.cuh.

7.29.3.5 operator∗() [3/3]

```
RT_HOSTDEV Vec3 operator* (
            double t,
            const Vec3 & v)  [inline]
```

Definition at line 90 of file vec3.cuh.

7.29.3.6 operator+()

```
RT_HOSTDEV Vec3 operator+ (
            const Vec3 & u,
            const Vec3 & v)  [inline]
```

Definition at line 76 of file vec3.cuh.

7.29.3.7 operator-()

```
RT_HOSTDEV Vec3 operator- (
            const Vec3 & u,
            const Vec3 & v)  [inline]
```

Definition at line 80 of file vec3.cuh.

7.29.3.8 operator/()

```
RT_HOSTDEV Vec3 operator/ (
            const Vec3 & v,
            double t)  [inline]
```

Definition at line 98 of file vec3.cuh.

7.29.3.9 operator<<()

```
std::ostream & operator<< (
            std::ostream & out,
            const Vec3 & v)  [inline]
```

Definition at line 135 of file vec3.cuh.

7.29.3.10 reflect()

```
RT_HOSTDEV Vec3 reflect (
            const Vec3 & v,
            const Vec3 & n)  [inline]
```

Definition at line 121 of file vec3.cuh.

7.29.3.11 refract()

```
RT_HOSTDEV Vec3 refract (
            const Vec3 & uv,
            const Vec3 & n,
            double etai_over_etat)  [inline]
```

Definition at line 128 of file vec3.cuh.

7.29.3.12 unit_vector()

```
RT_HOSTDEV Vec3 unit_vector (
            const Vec3 & v)  [inline]
```

Definition at line 116 of file vec3.cuh.

7.30 vec3.cuh

Go to the documentation of this file.

```
// vec3.cuh - Vecteur 3D double precision, brique de base du path tracer.
// Header-only : force l'inlining partout (hot path intersections + shading).
// RT_HOSTDEV permet la compilation croisee CPU (MSVC) / GPU (nvcc).

#ifndef RT_MATH_VEC3_CUH
#define RT_MATH_VEC3_CUH

#include <cmath>
#include <iostream>

// Macro de portabilite CPU/GPU : declenche __host__ __device__ uniquement
// quand nvcc compile le fichier, transparent pour MSVC.
#ifdef __CUDACC__
#define RT_HOSTDEV __host__ __device__
#else
#define RT_HOSTDEV
#endif

class Vec3 {
public:
    double e[3]; // stockage brut, pas de padding

    RT_HOSTDEV Vec3() : e{0.0, 0.0, 0.0} {}
    RT_HOSTDEV Vec3(double x, double y, double z) : e{x, y, z} {}

    RT_HOSTDEV double x() const { return e[0]; }
    RT_HOSTDEV double y() const { return e[1]; }
    RT_HOSTDEV double z() const { return e[2]; }

    RT_HOSTDEV Vec3 operator-() const { return Vec3(-e[0], -e[1], -e[2]); }
    RT_HOSTDEV double operator[](int i) const { return e[i]; }
    RT_HOSTDEV double& operator[](int i) { return e[i]; }

    RT_HOSTDEV Vec3& operator+=(const Vec3& v) {
        e[0] += v.e[0]; e[1] += v.e[1]; e[2] += v.e[2];
        return *this;
    }

    RT_HOSTDEV Vec3& operator-=(const Vec3& v) {
        e[0] -= v.e[0]; e[1] -= v.e[1]; e[2] -= v.e[2];
        return *this;
    }

    RT_HOSTDEV Vec3& operator*=(double t) {
        e[0] *= t; e[1] *= t; e[2] *= t;
        return *this;
    }

    // Multiplication par l'inverse : evite 3 divisions
    RT_HOSTDEV Vec3& operator/=(double t) {
        return *this *= (1.0 / t);
    }

    RT_HOSTDEV double length_squared() const {
        return e[0]*e[0] + e[1]*e[1] + e[2]*e[2];
    }

    RT_HOSTDEV double length() const {
        return std::sqrt(length_squared());
    }

    // Detection de vecteur quasi-nul apres scatter :
    // evite les NaN en normalisant un vecteur degenere
    RT_HOSTDEV bool near_zero() const {
        constexpr double s = 1e-8;
        return (std::fabs(e[0]) < s) && (std::fabs(e[1]) < s) && (std::fabs(e[2]) < s);
    }
};

// Alias semantiques : meme type sous-jacent, intention differente
using Point3 = Vec3;
using Color = Vec3;

// --- Operateurs arithmetiques ---

inline RT_HOSTDEV Vec3 operator+(const Vec3& u, const Vec3& v) {
    return Vec3(u.e[0]+v.e[0], u.e[1]+v.e[1], u.e[2]+v.e[2]);
}

inline RT_HOSTDEV Vec3 operator-(const Vec3& u, const Vec3& v) {
    return Vec3(u.e[0]-v.e[0], u.e[1]-v.e[1], u.e[2]-v.e[2]);
}
```

```

// Produit composante par composante (Hadamard) :
// sert a moduler la couleur par l'albedo du materiau
inline RT_HOSTDEV Vec3 operator*(const Vec3& u, const Vec3& v) {
    return Vec3(u.e[0]*v.e[0], u.e[1]*v.e[1], u.e[2]*v.e[2]);
}

inline RT_HOSTDEV Vec3 operator*(double t, const Vec3& v) {
    return Vec3(t*v.e[0], t*v.e[1], t*v.e[2]);
}

inline RT_HOSTDEV Vec3 operator*(const Vec3& v, double t) {
    return t * v;
}

inline RT_HOSTDEV Vec3 operator/(const Vec3& v, double t) {
    return (1.0/t) * v;
}

// --- Fonctions geometriques ---

inline RT_HOSTDEV double dot(const Vec3& u, const Vec3& v) {
    return u.e[0]*v.e[0] + u.e[1]*v.e[1] + u.e[2]*v.e[2];
}

inline RT_HOSTDEV Vec3 cross(const Vec3& u, const Vec3& v) {
    return Vec3(
        u.e[1]*v.e[2] - u.e[2]*v.e[1],
        u.e[2]*v.e[0] - u.e[0]*v.e[2],
        u.e[0]*v.e[1] - u.e[1]*v.e[0]
    );
}

inline RT_HOSTDEV Vec3 unit_vector(const Vec3& v) {
    return v / v.length();
}

// Reflexion speculaire : v incident, n normale unitaire
inline RT_HOSTDEV Vec3 reflect(const Vec3& v, const Vec3& n) {
    return v - 2.0 * dot(v, n) * n;
}

// Refraction via loi de Snell-Descartes.
// uv doit etre unitaire. etai_over_etat = n1/n2 (indices de refraction).
// Decomposition en composante perpendiculaire + parallele a la normale.
inline RT_HOSTDEV Vec3 refract(const Vec3& uv, const Vec3& n, double etai_over_etat) {
    double cos_theta = std::fmin(dot(-uv, n), 1.0);
    Vec3 r_out_perp = etai_over_etat * (uv + cos_theta * n);
    Vec3 r_out_parallel = -std::sqrt(std::fabs(1.0 - r_out_perp.length_squared())) * n;
    return r_out_perp + r_out_parallel;
}

inline std::ostream& operator«(std::ostream& out, const Vec3& v) {
    return out « v.e[0] « ' ' « v.e[1] « ' ' « v.e[2];
}

#endif // RT_MATH_VEC3_CUH
```

7.31 src/scene/bvh_builder.cpp File Reference

```
#include "scene/bvh_builder.h"
#include "scene/primitives.cuh"
#include <algorithm>
#include <limits>
```

7.32 bvh_builder.cpp

Go to the documentation of this file.

```
#include "scene/bvh_builder.h"
#include "scene/primitives.cuh"
#include <algorithm>
#include <limits>
```

```

// ============================================================
// BVHAccel - Optimized flat BVH
// SAH construction, iterative traversal, front-to-back order
// ============================================================

BVHAccel::BVHAccel(const HittableList& list) {
    auto prims = list.objects();
    if (prims.empty()) return;

    // Build primitive info
    std::vector<BuildPrimInfo> prim_info(prims.size());
    for (size_t i = 0; i < prims.size(); ++i) {
        prim_info[i].prim_index = i;
        prim_info[i].bbox = prims[i]->bounding_box();
        prim_info[i].centroid = (prim_info[i].bbox.min_point() + prim_info[i].bbox.max_point()) * 0.5;
    }

    // Build tree (using new for stable pointers)
    int total_nodes = 0;
    BuildNode* root = recursive_build(prim_info, 0, static_cast<int>(prims.size()),
                                      prims, total_nodes);

    // Flatten to contiguous array
    nodes_.resize(total_nodes);
    int offset = 0;
    flatten_tree(root, offset);

    // Free build tree
    delete_tree(root);
}

BVHAccel::BuildNode* BVHAccel::recursive_build(
    std::vector<BuildPrimInfo>& prim_info,
    int start, int end,
    std::vector<std::shared_ptr<Hittable»& src_prims,
    int& total_nodes)
{
    BuildNode* node = new BuildNode();
    total_nodes++;

    // Compute bounds of all primitives in range
    AABB bounds;
    for (int i = start; i < end; ++i)
        bounds = surrounding_box(bounds, prim_info[i].bbox);

    int n_prims = end - start;

    // Small leaf
    if (n_prims <= 2) {
        int first_offset = static_cast<int>(ordered_prims_.size());
        for (int i = start; i < end; ++i)
            ordered_prims_.push_back(src_prims[prim_info[i].prim_index]);
        node->bbox = bounds;
        node->first_prim_offset = first_offset;
        node->num_primitives = n_prims;
        return node;
    }

    // Compute centroid bounds
    AABB centroid_bounds;
    for (int i = start; i < end; ++i) {
        Point3 c = prim_info[i].centroid;
        centroid_bounds = surrounding_box(centroid_bounds, AABB(c, c));
    }

    int axis = centroid_bounds.longest_axis();
    double axis_extent = centroid_bounds.axis_interval(axis).size();

    // Degenerate: all centroids at same position
    if (axis_extent < 1e-8) {
        int first_offset = static_cast<int>(ordered_prims_.size());
        for (int i = start; i < end; ++i)
            ordered_prims_.push_back(src_prims[prim_info[i].prim_index]);
        node->bbox = bounds;
        node->first_prim_offset = first_offset;
        node->num_primitives = n_prims;
        return node;
    }

    // SAH bucket evaluation
    constexpr int N_BUCKETS = 12;
    struct Bucket { int count = 0; AABB bounds; };
    Bucket buckets[N_BUCKETS];

    double cmin = centroid_bounds.axis_interval(axis).min;
    double cmax = centroid_bounds.axis_interval(axis).max;
```

```

    for (int i = start; i < end; ++i) {
        int b = static_cast<int>(N_BUCKETS *
            ((prim_info[i].centroid[axis] - cmin) / (cmax - cmin)));
        if (b >= N_BUCKETS) b = N_BUCKETS - 1;
        buckets[b].count++;
        buckets[b].bounds = surrounding_box(buckets[b].bounds, prim_info[i].bbox);
    }

    // Evaluate cost for each split position
    constexpr double T_TRAV = 0.125;
    constexpr double T_ISECT = 1.0;
    double parent_area = bounds.surface_area();
    double best_cost = std::numeric_limits<double>::infinity();
    int best_split = -1;

    // Forward sweep: accumulate left side
    double cost_left_area[N_BUCKETS - 1];
    int cost_left_count[N_BUCKETS - 1];
    {
        AABB accum;
        int count = 0;
        for (int i = 0; i < N_BUCKETS - 1; ++i) {
            accum = surrounding_box(accum, buckets[i].bounds);
            count += buckets[i].count;
            cost_left_area[i] = accum.surface_area();
            cost_left_count[i] = count;
        }
    }
    // Backward sweep: accumulate right side and find best
    {
        AABB accum;
        int count = 0;
        for (int i = N_BUCKETS - 1; i >= 1; --i) {
            accum = surrounding_box(accum, buckets[i].bounds);
            count += buckets[i].count;
            double cost = T_TRAV + T_ISECT *
                (cost_left_count[i-1] * cost_left_area[i-1] +
                 count * accum.surface_area()) / parent_area;
            if (cost < best_cost) {
                best_cost = cost;
                best_split = i;
            }
        }
    }

    // Compare with leaf cost
    double leaf_cost = T_ISECT * n_prims;
    if (best_cost >= leaf_cost && n_prims <= 16) {
        // Leaf is cheaper
        int first_offset = static_cast<int>(ordered_prims_.size());
        for (int i = start; i < end; ++i)
            ordered_prims_.push_back(src_prims[prim_info[i].prim_index]);
        node->bbox = bounds;
        node->first_prim_offset = first_offset;
        node->num_primitives = n_prims;
        return node;
    }

    // Partition primitives at best split
    auto mid_it = std::partition(
        prim_info.begin() + start, prim_info.begin() + end,
        [&](const BuildPrimInfo& pi) {
            int b = static_cast<int>(N_BUCKETS *
                ((pi.centroid[axis] - cmin) / (cmax - cmin)));
            if (b >= N_BUCKETS) b = N_BUCKETS - 1;
            return b < best_split;
        });
    int mid = static_cast<int>(std::distance(prim_info.begin(), mid_it));

    if (mid == start || mid == end)
        mid = start + n_prims / 2;

    // Recurse
    node->children[0] = recursive_build(prim_info, start, mid, src_prims, total_nodes);
    node->children[1] = recursive_build(prim_info, mid, end, src_prims, total_nodes);
    node->bbox = surrounding_box(node->children[0]->bbox, node->children[1]->bbox);
    node->split_axis = axis;
    node->num_primitives = 0;
    return node;
}

int BVHAccel::flatten_tree(BuildNode* node, int& offset) {
    FlatBVHNode& flat = nodes_[offset];
    flat.bbox = node->bbox;
    int my_offset = offset++;

```

```
    if (node->num_primitives > 0) {
        // Leaf
        flat.primitives_offset = static_cast<uint32_t>(node->first_prim_offset);
        flat.num_primitives = static_cast<uint16_t>(node->num_primitives);
    } else {
        // Interior
        flat.split_axis = static_cast<uint8_t>(node->split_axis);
        flat.num_primitives = 0;
        // Left child is immediately next (offset)
        flatten_tree(node->children[0], offset);
        // Right child offset stored in node
        flat.second_child_offset = static_cast<uint32_t>(
            flatten_tree(node->children[1], offset));
    }
    return my_offset;
}

void BVHAccel::delete_tree(BuildNode* node) {
    if (!node) return;
    delete_tree(node->children[0]);
    delete_tree(node->children[1]);
    delete node;
}

bool BVHAccel::hit(const Ray& r, Interval ray_t, HitRecord& rec) const {
    if (nodes_.empty()) return false;

    bool hit_anything = false;
    double closest = ray_t.max;

    // Iterative traversal with explicit stack
    constexpr int MAX_STACK = 64;
    int stack[MAX_STACK];
    int stack_ptr = 0;
    stack[stack_ptr++] = 0;

    while (stack_ptr > 0) {
        int node_idx = stack[--stack_ptr];
        const FlatBVHNode& node = nodes_[node_idx];

        if (!node.bbox.hit(r, ray_t.min, closest))
            continue;

        if (node.num_primitives > 0) {
            // Leaf: test all primitives
            for (uint16_t i = 0; i < node.num_primitives; ++i) {
                if (ordered_prims_[node.primitives_offset + i]->hit(
                        r, Interval(ray_t.min, closest), rec)) {
                    hit_anything = true;
                    closest = rec.t;
                }
            }
        } else {
            // Interior: push children in front-to-back order
            // Left child = node_idx + 1, right child = second_child_offset
            int near_child = node_idx + 1;
            int far_child = node.second_child_offset;

            // If ray direction is negative along split axis, swap
            if (r.dir_sign[node.split_axis]) {
                int tmp = near_child; near_child = far_child; far_child = tmp;
            }

            // Push far first (popped last = traversed last)
            stack[stack_ptr++] = far_child;
            stack[stack_ptr++] = near_child;
        }
    }
    return hit_anything;
}

AABB BVHAccel::bounding_box() const {
    if (nodes_.empty()) return AABB::empty;
    return nodes_[0].bbox;
}

// ============================================================
// Legacy BVHNode (for test backward compatibility)
// ============================================================

BVHNode::BVHNode(const HittableList& list) {
    auto objects = list.objects();
    *this = BVHNode(objects, 0, objects.size());
}

BVHNode::BVHNode(std::vector<std::shared_ptr<Hittable»& objects, size_t start, size_t end) {
    AABB span_bbox;
```

```
    for (size_t i = start; i < end; ++i)
        span_bbox = surrounding_box(span_bbox, objects[i]->bounding_box());
    bbox_ = span_bbox;

    size_t object_span = end - start;

    if (object_span == 1) {
        left_ = right_ = objects[start];
    } else if (object_span == 2) {
        left_ = objects[start];
        right_ = objects[start + 1];
    } else {
        int axis = bbox_.longest_axis();
        split_axis_ = axis;

        std::sort(objects.begin() + start, objects.begin() + end,
            [axis](const std::shared_ptr<Hittable>& a, const std::shared_ptr<Hittable>& b) {
                return a->bounding_box().axis_interval(axis).min
                     < b->bounding_box().axis_interval(axis).min;
            });

        size_t mid = start + object_span / 2;
        left_ = std::make_shared<BVHNode>(objects, start, mid);
        right_ = std::make_shared<BVHNode>(objects, mid, end);
    }
}

bool BVHNode::hit(const Ray& r, Interval ray_t, HitRecord& rec) const {
    if (!bbox_.hit(r, ray_t))
        return false;

    // Front-to-back based on ray direction
    Hittable* first = (r.dir_sign[split_axis_] == 0) ? left_.get() : right_.get();
    Hittable* second = (r.dir_sign[split_axis_] == 0) ? right_.get() : left_.get();

    bool hit_near = first->hit(r, ray_t, rec);
    bool hit_far = second->hit(r,
        Interval(ray_t.min, hit_near ? rec.t : ray_t.max), rec);

    return hit_near || hit_far;
}

AABB BVHNode::bounding_box() const {
    return bbox_;
}
```

7.33 src/scene/bvh_builder.h File Reference

```
#include "scene/bvh_node.cuh"
#include <vector>
#include <memory>
```

Classes

- class BVHAccel
- struct BVHAccel::BuildPrimInfo
- struct BVHAccel::BuildNode

7.34 bvh_builder.h

Go to the documentation of this file.

```
#ifndef RT_SCENE_BVH_BUILDER_H
#define RT_SCENE_BVH_BUILDER_H

#include "scene/bvh_node.cuh"
#include <vector>
#include <memory>
```

```

class HittableList;

// Optimized flat BVH: SAH construction, iterative traversal, front-to-back order
class BVHAccel : public Hittable {
public:
    BVHAccel(const HittableList& list);

    bool hit(const Ray& r, Interval ray_t, HitRecord& rec) const override;
    AABB bounding_box() const override;

private:
    std::vector<FlatBVHNode> nodes_;
    std::vector<std::shared_ptr<Hittable>> ordered_prims_;

    struct BuildPrimInfo {
        size_t prim_index;
        AABB bbox;
        Point3 centroid;
    };

    struct BuildNode {
        AABB bbox;
        BuildNode* children[2] = {nullptr, nullptr};
        int split_axis = 0;
        int first_prim_offset = 0;
        int num_primitives = 0;
    };

    BuildNode* recursive_build(
        std::vector<BuildPrimInfo>& prim_info,
        int start, int end,
        std::vector<std::shared_ptr<Hittable>>& src_prims,
        int& total_nodes);

    int flatten_tree(BuildNode* node, int& offset);
    void delete_tree(BuildNode* node);
};

#endif // RT_SCENE_BVH_BUILDER_H
```

7.35 src/scene/bvh_node.cuh File Reference

```
#include "math/aabb.cuh"
#include "scene/primitives.cuh"
#include <cstdint>
#include <memory>
#include <vector>
```

Classes

- struct FlatBVHNode
- class BVHNode

Macros

- #define RT_SCENE_BVH_NODE_CUH

7.35.1 Macro Definition Documentation

7.35.1.1 RT_SCENE_BVH_NODE_CUH

```
#define RT_SCENE_BVH_NODE_CUH
```

Definition at line 2 of file bvh_node.cuh.

7.36 bvh_node.cuh

Go to the documentation of this file.

```
#ifndef RT_SCENE_BVH_NODE_CUH
#define RT_SCENE_BVH_NODE_CUH

#include "math/aabb.cuh"
#include "scene/primitives.cuh"
#include <cstdint>
#include <memory>
#include <vector>

// Compact flat BVH node for cache-friendly iterative traversal
struct FlatBVHNode {
    AABB bbox;
    union {
        uint32_t primitives_offset;   // leaf: index into ordered primitives
        uint32_t second_child_offset; // interior: index of right child (left = this+1)
    };
    uint16_t num_primitives; // 0 = interior node
    uint8_t split_axis;      // for front-to-back traversal
    uint8_t pad;
};

// Legacy recursive BVH (kept for test compatibility)
class BVHNode : public Hittable {
public:
    BVHNode(std::vector<std::shared_ptr<Hittable»& objects, size_t start, size_t end);
    BVHNode(const class HittableList& list);

    bool hit(const Ray& r, Interval ray_t, HitRecord& rec) const override;
    AABB bounding_box() const override;

private:
    std::shared_ptr<Hittable> left_;
    std::shared_ptr<Hittable> right_;
    AABB bbox_;
    int split_axis_ = 0;
};

#endif // RT_SCENE_BVH_NODE_CUH
```

7.37 src/scene/camera.cpp File Reference

```
#include "scene/camera.cuh"
#include "math/random.cuh"
#include <cmath>
#include <algorithm>
```

7.38 camera.cpp

Go to the documentation of this file.

```
#include "scene/camera.cuh"
#include "math/random.cuh"
#include <cmath>
#include <algorithm>

void Camera::initialize() {
    image_height_ = std::max(1, static_cast<int>(image_width / aspect_ratio));

    center_ = lookfrom;

    // Viewport dimensions
    double theta = vfov * 3.14159265358979323846 / 180.0;
    double h = std::tan(theta / 2.0);
    double viewport_height = 2.0 * h * focus_dist;
    double viewport_width = viewport_height * (static_cast<double>(image_width) / image_height_);

    // Camera coordinate frame
```

```
    w_ = unit_vector(lookfrom - lookat);
    u_ = unit_vector(cross(vup, w_));
    v_ = cross(w_, u_);

    // Vectors across the horizontal and down the vertical viewport edges
    Vec3 viewport_u = viewport_width * u_;
    Vec3 viewport_v = viewport_height * (-v_);

    // Pixel deltas
    pixel_delta_u_ = viewport_u / static_cast<double>(image_width);
    pixel_delta_v_ = viewport_v / static_cast<double>(image_height_);

    // Upper left pixel location
    Point3 viewport_upper_left = center_ - focus_dist * w_
                                 - viewport_u / 2.0
                                 - viewport_v / 2.0;
    pixel100_loc_ = viewport_upper_left + 0.5 * (pixel_delta_u_ + pixel_delta_v_);

    // Defocus disk basis vectors
    double defocus_radius = focus_dist * std::tan((defocus_angle / 2.0) * 3.14159265358979323846 /
180.0);
    defocus_disk_u_ = u_ * defocus_radius;
    defocus_disk_v_ = v_ * defocus_radius;
}

Ray Camera::get_ray(int i, int j) const {
    Vec3 offset = sample_square();
    Point3 pixel_sample = pixel100_loc_
                        + ((i + offset.x()) * pixel_delta_u_)
                        + ((j + offset.y()) * pixel_delta_v_);

    Point3 ray_origin = (defocus_angle <= 0.0) ? center_ : defocus_disk_sample();
    Vec3 ray_direction = pixel_sample - ray_origin;

    return Ray(ray_origin, ray_direction);
}

Vec3 Camera::sample_square() const {
    return Vec3(random_double() - 0.5, random_double() - 0.5, 0.0);
}

Point3 Camera::defocus_disk_sample() const {
    Vec3 p = random_in_unit_disk();
    return center_ + (p[0] * defocus_disk_u_) + (p[1] * defocus_disk_v_);
}
```

7.39 src/scene/camera.cuh File Reference

```
#include "math/vec3.cuh"
#include "math/ray.cuh"
```

Classes

- class Camera

Macros

- #define RT_SCENE_CAMERA_CUH

7.39.1 Macro Definition Documentation

7.39.1.1 RT_SCENE_CAMERA_CUH

```
#define RT_SCENE_CAMERA_CUH
```

Definition at line 2 of file camera.cuh.

7.40 camera.cuh

Go to the documentation of this file.

```
#ifndef RT_SCENE_CAMERA_CUH
#define RT_SCENE_CAMERA_CUH

#include "math/vec3.cuh"
#include "math/ray.cuh"

class Camera {
public:
    // Configuration (set before calling initialize())
    double aspect_ratio = 16.0 / 9.0;
    int image_width = 400;
    int samples_per_pixel = 100;
    int max_depth = 50;

    double vfov = 90.0;        // vertical field of view in degrees
    Point3 lookfrom = Point3(0, 0, 0);
    Point3 lookat = Point3(0, 0, -1);
    Vec3 vup = Vec3(0, 1, 0);

    double defocus_angle = 0.0;  // variation angle of rays through each pixel
    double focus_dist = 10.0;    // distance from lookfrom to the plane of perfect focus

    // Call once after setting configuration
    void initialize();

    // Get image dimensions
    int get_image_width() const { return image_width; }
    int get_image_height() const { return image_height_; }

    // Generate a ray for pixel (i, j) with random sub-pixel offset
    Ray get_ray(int i, int j) const;

private:
    int image_height_;
    Point3 center_;
    Point3 pixel00_loc_;    // location of pixel (0,0)
    Vec3 pixel_delta_u_;    // offset to pixel to the right
    Vec3 pixel_delta_v_;    // offset to pixel below
    Vec3 u_, v_, w_;        // camera basis vectors
    Vec3 defocus_disk_u_;
    Vec3 defocus_disk_v_;

    Vec3 sample_square() const;
    Point3 defocus_disk_sample() const;
};

#endif // RT_SCENE_CAMERA_CUH
```

7.41 src/scene/material.cpp File Reference

```
#include "scene/material.cuh"
#include "scene/primitives.cuh"
#include "math/random.cuh"
#include <cmath>
#include <algorithm>
```

7.42 material.cpp

Go to the documentation of this file.

```
#include "scene/material.cuh"
#include "scene/primitives.cuh"
#include "math/random.cuh"
#include <cmath>
#include <algorithm>

```

```
// ============================================================
// Lambertian
// ============================================================

Lambertian::Lambertian(const Color& albedo) : albedo_(albedo) {
    type_ = Type::Lambertian;
}

bool Lambertian::scatter(const Ray& /*r_in*/, const HitRecord& rec,
                         Color& attenuation, Ray& scattered) const {
    Vec3 scatter_direction = rec.normal + random_unit_vector();

    // Catch degenerate scatter direction
    if (scatter_direction.near_zero())
        scatter_direction = rec.normal;

    scattered = Ray(rec.p, scatter_direction);
    attenuation = albedo_;
    return true;
}

// ============================================================
// Metal
// ============================================================

Metal::Metal(const Color& albedo, double fuzz)
    : albedo_(albedo), fuzz_(std::min(fuzz, 1.0)) {
    type_ = Type::Metal;
}

bool Metal::scatter(const Ray& r_in, const HitRecord& rec,
                    Color& attenuation, Ray& scattered) const {
    Vec3 reflected = reflect(unit_vector(r_in.direction()), rec.normal);
    reflected = reflected + fuzz_ * random_in_unit_sphere();
    scattered = Ray(rec.p, reflected);
    attenuation = albedo_;
    return (dot(scattered.direction(), rec.normal) > 0.0);
}

// ============================================================
// Dielectric
// ============================================================

Dielectric::Dielectric(double refraction_index)
    : refraction_index_(refraction_index) {
    type_ = Type::Dielectric;
}

bool Dielectric::scatter(const Ray& r_in, const HitRecord& rec,
                         Color& attenuation, Ray& scattered) const {
    attenuation = Color(1.0, 1.0, 1.0);
    double ri = rec.front_face ? (1.0 / refraction_index_) : refraction_index_;

    Vec3 unit_dir = unit_vector(r_in.direction());
    double cos_theta = std::fmin(dot(-unit_dir, rec.normal), 1.0);
    double sin_theta = std::sqrt(1.0 - cos_theta * cos_theta);

    bool cannot_refract = ri * sin_theta > 1.0;
    Vec3 direction;

    if (cannot_refract || reflectance(cos_theta, ri) > random_double())
        direction = reflect(unit_dir, rec.normal);
    else
        direction = refract(unit_dir, rec.normal, ri);

    scattered = Ray(rec.p, direction);
    return true;
}

double Dielectric::reflectance(double cosine, double refraction_index) {
    // Schlick's approximation
    double r0 = (1.0 - refraction_index) / (1.0 + refraction_index);
    r0 = r0 * r0;
    double omc = 1.0 - cosine;
    double omc2 = omc * omc;
    return r0 + (1.0 - r0) * omc2 * omc2 * omc;
}

// ============================================================
// Checker (damier procedural)
// ============================================================

Checker::Checker(const Color& color1, const Color& color2, double scale)
    : color1_(color1), color2_(color2), scale_(scale) {
    type_ = Type::Checker;
}

```

```
bool Checker::scatter(const Ray& /*r_in*/, const HitRecord& rec,
                      Color& attenuation, Ray& scattered) const {
    // Pattern damier base sur la position du hit
    double inv_scale = 1.0 / scale_;
    int ix = (int)std::floor(rec.p.x() * inv_scale);
    int iy = (int)std::floor(rec.p.y() * inv_scale);
    int iz = (int)std::floor(rec.p.z() * inv_scale);
    bool is_even = ((ix + iy + iz) % 2 == 0);

    attenuation = is_even ? color1_ : color2_;

    Vec3 scatter_direction = rec.normal + random_unit_vector();
    if (scatter_direction.near_zero()) scatter_direction = rec.normal;
    scattered = Ray(rec.p, scatter_direction);
    return true;
}

// ============================================================
// Emissive (area light)
// ============================================================

Emissive::Emissive(const Color& emit_color, double intensity)
    : emit_(emit_color), intensity_(intensity) {
    type_ = Type::Emissive;
}

bool Emissive::scatter(const Ray& /*r_in*/, const HitRecord& /*rec*/,
                       Color& /*attenuation*/, Ray& /*scattered*/) const {
    return false; // emissive ne scatter pas, juste emet
}
```

7.43 src/scene/material.cuh File Reference

```
#include "math/vec3.cuh"
#include "math/ray.cuh"
#include <cstdint>
```

Classes

- class Material
- class Lambertian
- class Metal
- class Dielectric
- class Checker
- class Emissive

Macros

- #define RT_SCENE_MATERIAL_CUH

7.43.1 Macro Definition Documentation

7.43.1.1 RT_SCENE_MATERIAL_CUH

```
#define RT_SCENE_MATERIAL_CUH
```

Definition at line 2 of file material.cuh.

7.44 material.cuh

Go to the documentation of this file.

```
#ifndef RT_SCENE_MATERIAL_CUH
#define RT_SCENE_MATERIAL_CUH

#include "math/vec3.cuh"
#include "math/ray.cuh"
#include <cstdint>

struct HitRecord; // forward declaration

// --- Base Material class ---

class Material {
public:
    // Type tag pour dispatch rapide sans vtable
    enum class Type : uint8_t { Lambertian, Metal, Dielectric, Checker, Emissive, Unknown };

    virtual ~Material() = default;

    virtual bool scatter(const Ray& r_in, const HitRecord& rec,
                         Color& attenuation, Ray& scattered) const = 0;

    Type type() const { return type_; }

protected:
    Type type_ = Type::Unknown;
};

// --- Lambertian ---

class Lambertian : public Material {
public:
    explicit Lambertian(const Color& albedo);

    bool scatter(const Ray& r_in, const HitRecord& rec,
                 Color& attenuation, Ray& scattered) const override;

    const Color& albedo() const { return albedo_; }

private:
    Color albedo_;
};

// --- Metal ---

class Metal : public Material {
public:
    Metal(const Color& albedo, double fuzz);

    bool scatter(const Ray& r_in, const HitRecord& rec,
                 Color& attenuation, Ray& scattered) const override;

    const Color& albedo() const { return albedo_; }
    double fuzz() const { return fuzz_; }

private:
    Color albedo_;
    double fuzz_;
};

// --- Dielectric ---

class Dielectric : public Material {
public:
    explicit Dielectric(double refraction_index);

    bool scatter(const Ray& r_in, const HitRecord& rec,
                 Color& attenuation, Ray& scattered) const override;

    double refraction_index() const { return refraction_index_; }

    // Schlick's approximation for reflectance (public for devirtualized scatter)
    static double reflectance(double cosine, double refraction_index);

private:
    double refraction_index_;
};

// --- Checker (damier procedural) ---

class Checker : public Material {
public:
    Checker(const Color& color1, const Color& color2, double scale = 1.0);
```

```

    bool scatter(const Ray& r_in, const HitRecord& rec,
                 Color& attenuation, Ray& scattered) const override;

    const Color& color1() const { return color1_; }
    const Color& color2() const { return color2_; }
    double scale() const { return scale_; }

private:
    Color color1_, color2_;
    double scale_;
};

// --- Emissive (area light) ---

class Emissive : public Material {
public:
    Emissive(const Color& emit_color, double intensity = 1.0);

    bool scatter(const Ray& r_in, const HitRecord& rec,
                 Color& attenuation, Ray& scattered) const override;

    Color emitted() const { return emit_ * intensity_; }
    const Color& emit_color() const { return emit_; }
    double intensity() const { return intensity_; }

private:
    Color emit_;
    double intensity_;
};

#endif // RT_SCENE_MATERIAL_CUH
```

7.45 src/scene/primitives.cpp File Reference

```
#include "scene/primitives.cuh"
#include <cmath>
```

7.46 primitives.cpp

Go to the documentation of this file.

```
#include "scene/primitives.cuh"
#include <cmath>

// ============================================================
// Sphere
// ============================================================

Sphere::Sphere(const Point3& center, double radius, const Material* material)
    : center_(center), radius_(radius), material_(material) {}

bool Sphere::hit(const Ray& r, Interval ray_t, HitRecord& rec) const {
    Vec3 oc = r.origin() - center_;
    double a = r.direction().length_squared();
    double half_b = dot(oc, r.direction());
    double c = oc.length_squared() - radius_ * radius_;

    double discriminant = half_b * half_b - a * c;
    if (discriminant < 0.0)
        return false;

    double sqrtd = std::sqrt(discriminant);

    // Find nearest root in the acceptable range
    double inv_a = 1.0 / a;
    double root = (-half_b - sqrtd) * inv_a;
    if (!ray_t.surrounds(root)) {
        root = (-half_b + sqrtd) * inv_a;
        if (!ray_t.surrounds(root))
            return false;
    }

    rec.t = root;
```

```
    rec.p = r.at(rec.t);
    Vec3 outward_normal = (rec.p - center_) * (1.0 / radius_);
    rec.set_face_normal(r, outward_normal);
    rec.material = material_;

    return true;
}

AABB Sphere::bounding_box() const {
    Vec3 rvec(radius_, radius_, radius_);
    return AABB(center_ - rvec, center_ + rvec);
}

// ============================================================
// Plane
// ============================================================

Plane::Plane(const Point3& point, const Vec3& normal, const Material* material)
    : point_(point), normal_(unit_vector(normal)), material_(material) {}

bool Plane::hit(const Ray& r, Interval ray_t, HitRecord& rec) const {
    double denom = dot(r.direction(), normal_);

    // Ray parallel to plane
    if (std::fabs(denom) < 1e-8)
        return false;

    double t = dot(point_ - r.origin(), normal_) / denom;

    if (!ray_t.surrounds(t))
        return false;

    rec.t = t;
    rec.p = r.at(t);
    rec.set_face_normal(r, normal_);
    rec.material = material_;

    return true;
}

AABB Plane::bounding_box() const {
    // Infinite plane: use a very large bounding box
    // Slightly thicken along normal to avoid zero-volume AABB
    constexpr double big = 1e4;
    constexpr double eps = 0.0001;
    Point3 lo = point_ - Vec3(big, big, big);
    Point3 hi = point_ + Vec3(big, big, big);
    // Ensure non-zero thickness along normal direction
    for (int i = 0; i < 3; ++i) {
        if (std::fabs(normal_[i]) > 0.99) {
            lo.e[i] = point_[i] - eps;
            hi.e[i] = point_[i] + eps;
        }
    }
    return AABB(lo, hi);
}

// ============================================================
// Cylinder (surface quadrique, axe Y)
// Equation : (px - cx)^2 + (pz - cz)^2 = r^2, base_.y <= py <= base_.y + height_
// ============================================================

Cylinder::Cylinder(const Point3& base, double radius, double height, const Material* material)
    : base_(base), radius_(radius), height_(height), material_(material) {}

bool Cylinder::hit(const Ray& r, Interval ray_t, HitRecord& rec) const {
    // Intersection rayon-cylindre infini (axe Y)
    double ox = r.origin().x() - base_.x();
    double oz = r.origin().z() - base_.z();
    double dx = r.direction().x();
    double dz = r.direction().z();

    double a = dx*dx + dz*dz;
    double half_b = ox*dx + oz*dz;
    double c = ox*ox + oz*oz - radius_*radius_;

    double disc = half_b*half_b - a*c;
    if (disc < 0.0) return false;

    double sqrtd = std::sqrt(disc);
    double root = (-half_b - sqrtd) / a;

    // Tester les deux racines
    for (int i = 0; i < 2; i++) {
        if (i == 1) root = (-half_b + sqrtd) / a;
        if (!ray_t.surrounds(root)) continue;

```

```
        double y = r.origin().y() + root * r.direction().y();
        if (y < base_.y() || y > base_.y() + height_) continue;

        rec.t = root;
        rec.p = r.at(root);
        // Normale du cylindre : direction radiale (pas de composante Y)
        Vec3 outward_normal = Vec3(rec.p.x() - base_.x(), 0, rec.p.z() - base_.z()) / radius_;
        rec.set_face_normal(r, outward_normal);
        rec.material = material_;
        return true;
    }
    return false;
}

AABB Cylinder::bounding_box() const {
    return AABB(
        Point3(base_.x() - radius_, base_.y(), base_.z() - radius_),
        Point3(base_.x() + radius_, base_.y() + height_, base_.z() + radius_)
    );
}

// ============================================================
// Triangle (Moller-Trumbore)
// ============================================================

Triangle::Triangle(const Point3& v0, const Point3& v1, const Point3& v2, const Material* material)
    : v0_(v0), v1_(v1), v2_(v2), material_(material) {
    normal_ = unit_vector(cross(v1 - v0, v2 - v0));
}

bool Triangle::hit(const Ray& r, Interval ray_t, HitRecord& rec) const {
    Vec3 edge1 = v1_ - v0_;
    Vec3 edge2 = v2_ - v0_;
    Vec3 h = cross(r.direction(), edge2);
    double a = dot(edge1, h);

    if (std::fabs(a) < 1e-8)
        return false; // rayon parallele au triangle

    double f = 1.0 / a;
    Vec3 s = r.origin() - v0_;
    double u = f * dot(s, h);
    if (u < 0.0 || u > 1.0)
        return false;

    Vec3 q = cross(s, edge1);
    double v = f * dot(r.direction(), q);
    if (v < 0.0 || u + v > 1.0)
        return false;

    double t = f * dot(edge2, q);
    if (!ray_t.surrounds(t))
        return false;

    rec.t = t;
    rec.p = r.at(t);
    rec.set_face_normal(r, normal_);
    rec.material = material_;
    return true;
}

AABB Triangle::bounding_box() const {
    Point3 lo(
        std::fmin(v0_.x(), std::fmin(v1_.x(), v2_.x())) - 0.0001,
        std::fmin(v0_.y(), std::fmin(v1_.y(), v2_.y())) - 0.0001,
        std::fmin(v0_.z(), std::fmin(v1_.z(), v2_.z())) - 0.0001
    );
    Point3 hi(
        std::fmax(v0_.x(), std::fmax(v1_.x(), v2_.x())) + 0.0001,
        std::fmax(v0_.y(), std::fmax(v1_.y(), v2_.y())) + 0.0001,
        std::fmax(v0_.z(), std::fmax(v1_.z(), v2_.z())) + 0.0001
    );
    return AABB(lo, hi);
}

// ============================================================
// HittableList
// ============================================================

void HittableList::add(std::shared_ptr<Hittable> object) {
    bbox_ = surrounding_box(bbox_, object->bounding_box());
    objects_.push_back(std::move(object));
}

void HittableList::clear() {
    objects_.clear();
}
```

```

bool HittableList::hit(const Ray& r, Interval ray_t, HitRecord& rec) const {
    HitRecord temp_rec;
    bool hit_anything = false;
    double closest_so_far = ray_t.max;

    for (const auto& object : objects_) {
        if (object->hit(r, Interval(ray_t.min, closest_so_far), temp_rec)) {
            hit_anything = true;
            closest_so_far = temp_rec.t;
            rec = temp_rec;
        }
    }

    return hit_anything;
}

AABB HittableList::bounding_box() const {
    return bbox_;
}
```

7.47 src/scene/primitives.cuh File Reference

```
#include "math/vec3.cuh"
#include "math/ray.cuh"
#include "math/interval.cuh"
#include "math/aabb.cuh"
#include <memory>
#include <vector>
```

Classes

- struct HitRecord
- class Hittable
- class Sphere
- class Plane
- class Cylinder
- class Triangle
- class HittableList

Macros

- #define RT_SCENE_PRIMITIVES_CUH

7.47.1 Macro Definition Documentation

7.47.1.1 RT_SCENE_PRIMITIVES_CUH

```
#define RT_SCENE_PRIMITIVES_CUH
```

Definition at line 2 of file primitives.cuh.

7.48 primitives.cuh

Go to the documentation of this file.

```
#ifndef RT_SCENE_PRIMITIVES_CUH
#define RT_SCENE_PRIMITIVES_CUH

#include "math/vec3.cuh"
#include "math/ray.cuh"
#include "math/interval.cuh"
#include "math/aabb.cuh"
#include <memory>
#include <vector>

class Material;

// --- HitRecord ---

struct HitRecord {
    Point3 p;
    Vec3 normal;
    const Material* material = nullptr;
    double t = 0.0;
    bool front_face = true;

    // Sets normal to always point against the incident ray
    void set_face_normal(const Ray& r, const Vec3& outward_normal) {
        front_face = dot(r.direction(), outward_normal) < 0.0;
        normal = front_face ? outward_normal : -outward_normal;
    }
};

// --- Hittable (abstract base) ---

class Hittable {
public:
    virtual ~Hittable() = default;

    virtual bool hit(const Ray& r, Interval ray_t, HitRecord& rec) const = 0;
    virtual AABB bounding_box() const = 0;
};

// --- Sphere ---

class Sphere : public Hittable {
public:
    Sphere(const Point3& center, double radius, const Material* material);

    bool hit(const Ray& r, Interval ray_t, HitRecord& rec) const override;
    AABB bounding_box() const override;

    const Point3& center() const { return center_; }
    double radius() const { return radius_; }

private:
    Point3 center_;
    double radius_;
    const Material* material_;
};

// --- Plane ---

class Plane : public Hittable {
public:
    Plane(const Point3& point, const Vec3& normal, const Material* material);

    bool hit(const Ray& r, Interval ray_t, HitRecord& rec) const override;
    AABB bounding_box() const override;

private:
    Point3 point_;
    Vec3 normal_;   // stored as unit vector
    const Material* material_;
};

// --- Cylinder (surface quadrique, axe Y, hauteur finie) ---

class Cylinder : public Hittable {
public:
    Cylinder(const Point3& base, double radius, double height, const Material* material);

    bool hit(const Ray& r, Interval ray_t, HitRecord& rec) const override;
    AABB bounding_box() const override;

private:
    Point3 base_;
```

```
    double radius_;
    double height_;
    const Material* material_;
};

// --- Triangle (Moller-Trumbore intersection) ---

class Triangle : public Hittable {
public:
    Triangle(const Point3& v0, const Point3& v1, const Point3& v2, const Material* material);

    bool hit(const Ray& r, Interval ray_t, HitRecord& rec) const override;
    AABB bounding_box() const override;

private:
    Point3 v0_, v1_, v2_;
    Vec3 normal_;
    const Material* material_;
};

// --- HittableList ---

class HittableList : public Hittable {
public:
    HittableList() = default;

    void add(std::shared_ptr<Hittable> object);
    void clear();

    bool hit(const Ray& r, Interval ray_t, HitRecord& rec) const override;
    AABB bounding_box() const override;

    const std::vector<std::shared_ptr<Hittable»& objects() const { return objects_; }

private:
    std::vector<std::shared_ptr<Hittable» objects_;
    AABB bbox_;
};

#endif // RT_SCENE_PRIMITIVES_CUH
```

7.49 src/scene/scene.cpp File Reference

```
#include "scene/scene.h"
#include "math/random.cuh"
#include <cmath>
```

7.50 scene.cpp

Go to the documentation of this file.

```
#include "scene/scene.h"
#include "math/random.cuh"
#include <cmath>

// ============================================================
// Scene "three_spheres" : 3 materiaux sur sol gris
// Plan + 3 spheres + bulle interne = 5 objets
// ============================================================
Scene Scene::build_three_spheres() {
    Scene scene;

    auto* mat_ground = scene.add_material<Lambertian>(Color(0.45, 0.45, 0.42));
    auto* mat_terre = scene.add_material<Lambertian>(Color(0.6, 0.25, 0.1));
    auto* mat_chrome = scene.add_material<Metal>(Color(0.9, 0.9, 0.92), 0.0);
    auto* mat_eau = scene.add_material<Dielectric>(1.33);

    scene.world.add(std::make_shared<Plane>(Point3(0, 0, 0), Vec3(0, 1, 0), mat_ground));

    // Disposition en triangle : avant-gauche, arriere-centre, avant-droite
    scene.world.add(std::make_shared<Sphere>(Point3(-1.5, 0.6, 1.0), 0.6, mat_terre));
    scene.world.add(std::make_shared<Sphere>(Point3(0, 1.1, -1.2), 1.1, mat_eau));
    scene.world.add(std::make_shared<Sphere>(Point3(1.8, 0.5, 0.5), 0.5, mat_chrome));
```

```
    // Bulle dans la sphere d'eau
    scene.world.add(std::make_shared<Sphere>(Point3(0, 1.1, -1.2), -1.0, mat_eau));

    scene.camera.aspect_ratio = 16.0 / 9.0;
    scene.camera.image_width = 800;
    scene.camera.samples_per_pixel = 100;
    scene.camera.max_depth = 50;
    scene.camera.vfov = 40.0;
    scene.camera.lookfrom = Point3(4.0, 2.5, 4.5);
    scene.camera.lookat = Point3(0, 0.4, -0.2);
    scene.camera.vup = Vec3(0, 1, 0);
    scene.camera.defocus_angle = 0.3;
    scene.camera.focus_dist = 6.0;
    scene.camera.initialize();

    return scene;
}

// ============================================================
// Scene "final" : grille concentrique de spheres (stress test)
// Anneaux concentriques autour d'une sphere centrale, hauteurs variees
// ============================================================
Scene Scene::build_final_scene() {
    Scene scene;

    auto* ground = scene.add_material<Lambertian>(Color(0.5, 0.5, 0.48));
    scene.world.add(std::make_shared<Plane>(Point3(0, 0, 0), Vec3(0, 1, 0), ground));

    // Sphere centrale : verre
    auto* m_center = scene.add_material<Dielectric>(1.5);
    scene.world.add(std::make_shared<Sphere>(Point3(0, 1.5, 0), 1.5, m_center));

    // Anneaux concentriques
    int n_rings = 6;
    for (int ring = 1; ring <= n_rings; ++ring) {
        double radius = ring * 2.0;
        int count = ring * 8;
        for (int i = 0; i < count; ++i) {
            double angle = 2.0 * 3.14159265 * i / count + ring * 0.3;
            double x = radius * std::cos(angle);
            double z = radius * std::sin(angle);
            double r = 0.15 + random_double() * 0.15;
            double y = r + random_double() * 0.1 * ring;

            double choose = random_double();
            if (choose < 0.5) {
                Color c(random_double(0.15, 0.85), random_double(0.15, 0.85), random_double(0.15, 0.85));
                auto* mat = scene.add_material<Lambertian>(c);
                scene.world.add(std::make_shared<Sphere>(Point3(x, y, z), r, mat));
            } else if (choose < 0.82) {
                Color c(random_double(0.5, 1.0), random_double(0.5, 1.0), random_double(0.5, 1.0));
                auto* mat = scene.add_material<Metal>(c, random_double(0.0, 0.35));
                scene.world.add(std::make_shared<Sphere>(Point3(x, y, z), r, mat));
            } else {
                auto* mat = scene.add_material<Dielectric>(1.4 + random_double() * 0.3);
                scene.world.add(std::make_shared<Sphere>(Point3(x, y, z), r, mat));
            }
        }
    }

    // 2 grosses spheres decoratives
    auto* m_gold = scene.add_material<Metal>(Color(0.85, 0.65, 0.2), 0.1);
    scene.world.add(std::make_shared<Sphere>(Point3(-4.0, 1.0, 3.0), 1.0, m_gold));

    auto* m_rouge = scene.add_material<Lambertian>(Color(0.7, 0.1, 0.1));
    scene.world.add(std::make_shared<Sphere>(Point3(3.5, 0.8, -4.0), 0.8, m_rouge));

    scene.camera.aspect_ratio = 16.0 / 9.0;
    scene.camera.image_width = 800;
    scene.camera.samples_per_pixel = 50;
    scene.camera.max_depth = 50;
    scene.camera.vfov = 30.0;
    scene.camera.lookfrom = Point3(12, 5, 10);
    scene.camera.lookat = Point3(0, 0.8, 0);
    scene.camera.vup = Vec3(0, 1, 0);
    scene.camera.defocus_angle = 0.3;
    scene.camera.focus_dist = 15.0;
    scene.camera.initialize();

    return scene;
}

// ============================================================
// Scene "triangle" : pyramide + formes geometriques
// ============================================================
Scene Scene::build_triangle_scene() {
```

```
    Scene scene;

    auto* mat_ground = scene.add_material<Checker>(Color(0.8, 0.75, 0.7), Color(0.35, 0.3, 0.3), 1.0);
    auto* mat_red    = scene.add_material<Lambertian>(Color(0.75, 0.12, 0.12));
    auto* mat_teal   = scene.add_material<Lambertian>(Color(0.1, 0.6, 0.55));
    auto* mat_amber  = scene.add_material<Metal>(Color(0.85, 0.7, 0.25), 0.1);
    auto* mat_silver = scene.add_material<Metal>(Color(0.8, 0.82, 0.85), 0.02);
    auto* mat_glass  = scene.add_material<Dielectric>(1.5);

    scene.world.add(std::make_shared<Plane>(Point3(0, 0, 0), Vec3(0, 1, 0), mat_ground));

    // Pyramide
    Point3 a(-0.8, 0, -0.8), b(0.8, 0, -0.8), c(0.8, 0, 0.8), d(-0.8, 0, 0.8);
    Point3 apex(0, 1.6, 0);
    scene.world.add(std::make_shared<Triangle>(a, b, apex, mat_red));
    scene.world.add(std::make_shared<Triangle>(b, c, apex, mat_teal));
    scene.world.add(std::make_shared<Triangle>(c, d, apex, mat_amber));
    scene.world.add(std::make_shared<Triangle>(d, a, apex, mat_silver));
    scene.world.add(std::make_shared<Triangle>(a, c, b, mat_ground));
    scene.world.add(std::make_shared<Triangle>(a, d, c, mat_ground));

    // Sphere verre a cote
    scene.world.add(std::make_shared<Sphere>(Point3(2.2, 0.5, 0.5), 0.5, mat_glass));
    // Cylindre
    scene.world.add(std::make_shared<Cylinder>(Point3(-2.0, 0, 0.5), 0.3, 1.5, mat_silver));

    scene.camera.aspect_ratio = 16.0 / 9.0;
    scene.camera.image_width = 800;
    scene.camera.samples_per_pixel = 100;
    scene.camera.max_depth = 50;
    scene.camera.vfov = 40.0;
    scene.camera.lookfrom = Point3(3.5, 2.5, 4.5);
    scene.camera.lookat = Point3(0, 0.6, 0);
    scene.camera.vup = Vec3(0, 1, 0);
    scene.camera.defocus_angle = 0.0;
    scene.camera.focus_dist = 5.0;
    scene.camera.initialize();

    return scene;
}

// ============================================================
// Scene "cornell" : boite avec 2 spheres
// ============================================================
Scene Scene::build_cornell_box() {
    Scene scene;

    auto* mat_white = scene.add_material<Lambertian>(Color(0.73, 0.73, 0.73));
    auto* mat_red   = scene.add_material<Lambertian>(Color(0.65, 0.05, 0.05));
    auto* mat_green = scene.add_material<Lambertian>(Color(0.12, 0.45, 0.15));
    auto* mat_metal = scene.add_material<Metal>(Color(0.8, 0.85, 0.88), 0.0);
    auto* mat_glass = scene.add_material<Dielectric>(1.5);
    auto* mat_light = scene.add_material<Emissive>(Color(1, 1, 1), 15.0);

    double S = 5.55; // taille de la boite
    // Sol
    scene.world.add(std::make_shared<Triangle>(Point3(0,0,0), Point3(S,0,0), Point3(S,0,S), mat_white));
    scene.world.add(std::make_shared<Triangle>(Point3(0,0,0), Point3(S,0,S), Point3(0,0,S), mat_white));
    // Plafond
    scene.world.add(std::make_shared<Triangle>(Point3(0,S,0), Point3(S,S,S), Point3(S,S,0), mat_white));
    scene.world.add(std::make_shared<Triangle>(Point3(0,S,0), Point3(0,S,S), Point3(S,S,S), mat_white));
    // Fond
    scene.world.add(std::make_shared<Triangle>(Point3(0,0,S), Point3(S,0,S), Point3(S,S,S), mat_white));
    scene.world.add(std::make_shared<Triangle>(Point3(0,0,S), Point3(S,S,S), Point3(0,S,S), mat_white));
    // Gauche rouge
    scene.world.add(std::make_shared<Triangle>(Point3(0,0,0), Point3(0,0,S), Point3(0,S,S), mat_red));
    scene.world.add(std::make_shared<Triangle>(Point3(0,0,0), Point3(0,S,S), Point3(0,S,0), mat_red));
    // Droit vert
    scene.world.add(std::make_shared<Triangle>(Point3(S,0,0), Point3(S,S,0), Point3(S,S,S), mat_green));
    scene.world.add(std::make_shared<Triangle>(Point3(S,0,0), Point3(S,S,S), Point3(S,0,S), mat_green));
    // Lumiere au plafond
    double L = 1.5;
    double cx = S/2, cz = S/2;
    scene.world.add(std::make_shared<Triangle>(
        Point3(cx-L, S-0.01, cz-L), Point3(cx+L, S-0.01, cz-L), Point3(cx+L, S-0.01, cz+L), mat_light));
    scene.world.add(std::make_shared<Triangle>(
        Point3(cx-L, S-0.01, cz-L), Point3(cx+L, S-0.01, cz+L), Point3(cx-L, S-0.01, cz+L), mat_light));
```

```

    scene.world.add(std::make_shared<Sphere>(Point3(1.8, 0.8, 1.9), 0.8, mat_metal));
    scene.world.add(std::make_shared<Sphere>(Point3(3.7, 0.6, 3.5), 0.6, mat_glass));

    scene.camera.aspect_ratio = 1.0;
    scene.camera.image_width = 600;
    scene.camera.samples_per_pixel = 300;
    scene.camera.max_depth = 50;
    scene.camera.vfov = 40.0;
    scene.camera.lookfrom = Point3(S/2, S/2, -8);
    scene.camera.lookat = Point3(S/2, S/2, 0);
    scene.camera.vup = Vec3(0, 1, 0);
    scene.camera.defocus_angle = 0.0;
    scene.camera.focus_dist = 10.0;
    scene.camera.initialize();

    return scene;
}

// ============================================================
// Scene "showcase" : vitrine de toutes les features
// ============================================================
Scene Scene::build_showcase() {
    Scene scene;

    auto* mat_ground = scene.add_material<Checker>(Color(0.9, 0.9, 0.9), Color(0.3, 0.3, 0.35), 1.0);
    auto* mat_mirror   = scene.add_material<Metal>(Color(0.95, 0.95, 0.95), 0.0);
    auto* mat_gold     = scene.add_material<Metal>(Color(0.85, 0.65, 0.13), 0.05);
    auto* mat_copper   = scene.add_material<Metal>(Color(0.72, 0.45, 0.20), 0.15);
    auto* mat_glass    = scene.add_material<Dielectric>(1.5);
    auto* mat_diamond  = scene.add_material<Dielectric>(2.42);
    auto* mat_red      = scene.add_material<Lambertian>(Color(0.9, 0.15, 0.15));
    auto* mat_blue     = scene.add_material<Lambertian>(Color(0.15, 0.35, 0.9));
    auto* mat_green    = scene.add_material<Lambertian>(Color(0.2, 0.85, 0.3));
    auto* mat_orange   = scene.add_material<Lambertian>(Color(0.95, 0.55, 0.1));
    auto* mat_purple   = scene.add_material<Lambertian>(Color(0.6, 0.2, 0.8));
    auto* mat_light    = scene.add_material<Emissive>(Color(1.0, 0.95, 0.9), 4.0);

    scene.world.add(std::make_shared<Plane>(Point3(0, 0, 0), Vec3(0, 1, 0), mat_ground));

    scene.world.add(std::make_shared<Sphere>(Point3(0, 1.2, 0), 1.2, mat_mirror));
    scene.world.add(std::make_shared<Sphere>(Point3(-2.5, 0.8, 0.5), 0.8, mat_glass));
    scene.world.add(std::make_shared<Sphere>(Point3(-2.5, 0.8, 0.5), -0.75, mat_glass));
    scene.world.add(std::make_shared<Sphere>(Point3(2.5, 0.9, -0.3), 0.9, mat_gold));
    scene.world.add(std::make_shared<Sphere>(Point3(-1.0, 0.35, 2.0), 0.35, mat_red));
    scene.world.add(std::make_shared<Sphere>(Point3(-0.2, 0.25, 2.5), 0.25, mat_orange));
    scene.world.add(std::make_shared<Sphere>(Point3(0.7, 0.30, 2.2), 0.30, mat_blue));
    scene.world.add(std::make_shared<Sphere>(Point3(1.5, 0.28, 2.3), 0.28, mat_green));
    scene.world.add(std::make_shared<Sphere>(Point3(1.0, 0.5, -2.0), 0.5, mat_diamond));
    scene.world.add(std::make_shared<Sphere>(Point3(-1.5, 0.4, -1.8), 0.4, mat_copper));
    scene.world.add(std::make_shared<Sphere>(Point3(3.5, 0.35, 1.5), 0.35, mat_purple));
    scene.world.add(std::make_shared<Cylinder>(Point3(-3.5, 0, -1.0), 0.3, 2.0, mat_copper));

    scene.world.add(std::make_shared<Triangle>(Point3(-2,4,-2), Point3(2,4,-2), Point3(2,4,2),
mat_light));
    scene.world.add(std::make_shared<Triangle>(Point3(-2,4,-2), Point3(2,4,2), Point3(-2,4,2),
mat_light));

    scene.camera.aspect_ratio = 16.0 / 9.0;
    scene.camera.image_width = 800;
    scene.camera.samples_per_pixel = 200;
    scene.camera.max_depth = 50;
    scene.camera.vfov = 35.0;
    scene.camera.lookfrom = Point3(4, 2.5, 6);
    scene.camera.lookat = Point3(0, 0.5, 0);
    scene.camera.vup = Vec3(0, 1, 0);
    scene.camera.defocus_angle = 0.4;
    scene.camera.focus_dist = 7.0;
    scene.camera.initialize();
    return scene;
}

// ============================================================
// Scene "hall" : salle avec colonnes
// ============================================================
Scene Scene::build_hall() {
    Scene scene;
    auto* mat_floor = scene.add_material<Checker>(Color(0.95, 0.95, 0.95), Color(0.15, 0.15, 0.2),
2.0);
    auto* mat_wall  = scene.add_material<Lambertian>(Color(0.85, 0.82, 0.78));
    auto* mat_chrome = scene.add_material<Metal>(Color(0.9, 0.9, 0.9), 0.02);
    auto* mat_gold  = scene.add_material<Metal>(Color(0.85, 0.65, 0.13), 0.0);
    auto* mat_glass = scene.add_material<Dielectric>(1.5);
    auto* mat_ruby  = scene.add_material<Lambertian>(Color(0.7, 0.05, 0.1));
    auto* mat_emerald = scene.add_material<Lambertian>(Color(0.05, 0.6, 0.15));
    auto* mat_light = scene.add_material<Emissive>(Color(1.0, 0.98, 0.95), 5.0);

```

```
    scene.world.add(std::make_shared<Plane>(Point3(0, 0, 0), Vec3(0, 1, 0), mat_floor));
    scene.world.add(std::make_shared<Triangle>(Point3(-20,0,-8), Point3(20,0,-8), Point3(20,10,-8), mat_wall));
    scene.world.add(std::make_shared<Triangle>(Point3(-20,0,-8), Point3(20,10,-8), Point3(-20,10,-8), mat_wall));

    for (int i = -3; i <= 3; i++) {
        scene.world.add(std::make_shared<Cylinder>(Point3(i*3.0, 0, -5), 0.25, 4.0, mat_chrome));
        scene.world.add(std::make_shared<Cylinder>(Point3(i*3.0, 0, 3), 0.25, 4.0, mat_chrome));
    }
    scene.world.add(std::make_shared<Sphere>(Point3(0, 1.5, -1), 1.5, mat_gold));
    for (int i = -2; i <= 2; i++)
        scene.world.add(std::make_shared<Sphere>(Point3(i*2.5, 0.4, 1.0), 0.4, mat_glass));
    scene.world.add(std::make_shared<Sphere>(Point3(-4, 0.3, 0), 0.3, mat_ruby));
    scene.world.add(std::make_shared<Sphere>(Point3(4, 0.3, 0), 0.3, mat_emerald));
    scene.world.add(std::make_shared<Triangle>(Point3(-3,4,-3), Point3(3,4,-3), Point3(3,4,1), mat_light));
    scene.world.add(std::make_shared<Triangle>(Point3(-3,4,-3), Point3(3,4,1), Point3(-3,4,1), mat_light));

    scene.camera.aspect_ratio = 16.0 / 9.0; scene.camera.image_width = 800;
    scene.camera.samples_per_pixel = 200; scene.camera.max_depth = 50;
    scene.camera.vfov = 50.0; scene.camera.lookfrom = Point3(0, 2.0, 8);
    scene.camera.lookat = Point3(0, 1.0, -2); scene.camera.vup = Vec3(0, 1, 0);
    scene.camera.defocus_angle = 0.0; scene.camera.focus_dist = 10.0;
    scene.camera.initialize();
    return scene;
}

// ============================================================
// Scene "galaxy" : ~1000 spheres en spirale (stress test)
// ============================================================
Scene Scene::build_galaxy() {
    Scene scene;
    auto* mat_ground = scene.add_material<Checker>(Color(0.1, 0.1, 0.15), Color(0.05, 0.05, 0.08), 3.0);
    scene.world.add(std::make_shared<Plane>(Point3(0, 0, 0), Vec3(0, 1, 0), mat_ground));

    for (int arm = 0; arm < 4; arm++) {
        double arm_off = arm * 3.14159265 / 2.0;
        for (int i = 0; i < 250; i++) {
            double t = i * 0.04, r = 1.0 + t * 2.5, angle = t * 2.5 + arm_off;
            double x = r * std::cos(angle) + random_double(-0.3, 0.3);
            double z = r * std::sin(angle) + random_double(-0.3, 0.3);
            double y = 0.1 + random_double(0.0, 0.3), radius = 0.08 + random_double(0.0, 0.12);
            double mc = random_double();
            const Material* mat;
            if (mc < 0.5) mat = scene.add_material<Lambertian>(random_vec3() * random_vec3());
            else if (mc < 0.8) mat = scene.add_material<Metal>(random_vec3(0.5, 1.0), random_double(0, 0.3));
            else mat = scene.add_material<Dielectric>(1.5);
            scene.world.add(std::make_shared<Sphere>(Point3(x, y, z), radius, mat));
        }
    }
    auto* mat_c = scene.add_material<Metal>(Color(0.95, 0.95, 0.98), 0.0);
    scene.world.add(std::make_shared<Sphere>(Point3(0, 1.5, 0), 1.5, mat_c));
    scene.world.add(std::make_shared<Sphere>(Point3(-3, 0.8, 2), 0.8, scene.add_material<Dielectric>(1.5)));
    scene.world.add(std::make_shared<Sphere>(Point3(3, 0.7, -2), 0.7, scene.add_material<Metal>(Color(0.85, 0.65, 0.13), 0.0)));

    scene.camera.aspect_ratio = 16.0 / 9.0; scene.camera.image_width = 800;
    scene.camera.samples_per_pixel = 100; scene.camera.max_depth = 30;
    scene.camera.vfov = 30.0; scene.camera.lookfrom = Point3(15, 8, 15);
    scene.camera.lookat = Point3(0, 0.5, 0); scene.camera.vup = Vec3(0, 1, 0);
    scene.camera.defocus_angle = 0.3; scene.camera.focus_dist = 20.0;
    scene.camera.initialize();
    return scene;
}

// ============================================================
// Scene "minimal" : noir blanc miroir verre, DOF
// ============================================================
Scene Scene::build_minimal() {
    Scene scene;
    auto* mat_white = scene.add_material<Lambertian>(Color(0.95, 0.95, 0.95));
    auto* mat_black = scene.add_material<Lambertian>(Color(0.02, 0.02, 0.02));
    auto* mat_mirror = scene.add_material<Metal>(Color(0.98, 0.98, 0.98), 0.0);
    auto* mat_glass = scene.add_material<Dielectric>(1.5);

    scene.world.add(std::make_shared<Plane>(Point3(0, 0, 0), Vec3(0, 1, 0), mat_white));
    scene.world.add(std::make_shared<Triangle>(Point3(-10,0,-5), Point3(10,0,-5), Point3(10,10,-5), mat_white));
    scene.world.add(std::make_shared<Triangle>(Point3(-10,0,-5), Point3(10,10,-5), Point3(-10,10,-5), mat_white));

    scene.world.add(std::make_shared<Sphere>(Point3(-2, 1, 0), 1.0, mat_black));
```

```
    scene.world.add(std::make_shared<Sphere>(Point3(0, 1, 0), 1.0, mat_mirror));
    scene.world.add(std::make_shared<Sphere>(Point3(2, 1, 0), 1.0, mat_glass));
    scene.world.add(std::make_shared<Sphere>(Point3(2, 1, 0), -0.95, mat_glass));

    scene.camera.aspect_ratio = 16.0 / 9.0; scene.camera.image_width = 800;
    scene.camera.samples_per_pixel = 300; scene.camera.max_depth = 50;
    scene.camera.vfov = 30.0; scene.camera.lookfrom = Point3(0, 2.5, 8);
    scene.camera.lookat = Point3(0, 0.8, 0); scene.camera.vup = Vec3(0, 1, 0);
    scene.camera.defocus_angle = 0.8; scene.camera.focus_dist = 8.0;
    scene.camera.initialize();
    return scene;
}
```

7.51 src/scene/scene.h File Reference

```
#include "scene/primitives.cuh"
#include "scene/camera.cuh"
#include "scene/material.cuh"
#include <memory>
#include <vector>
```

Classes

- class Scene

7.52 scene.h

Go to the documentation of this file.

```
#ifndef RT_SCENE_SCENE_H
#define RT_SCENE_SCENE_H

#include "scene/primitives.cuh"
#include "scene/camera.cuh"
#include "scene/material.cuh"
#include <memory>
#include <vector>

class Scene {
public:
    Camera camera;
    HittableList world;

    // Scene owns all materials
    template<typename T, typename... Args>
    T* add_material(Args&&... args) {
        auto mat = std::make_unique<T>(std::forward<Args>(args)...);
        T* ptr = mat.get();
        materials_.push_back(std::move(mat));
        return ptr;
    }

    // Factory: three spheres + ground plane
    static Scene build_three_spheres();

    // Factory: final scene with many random spheres
    static Scene build_final_scene();

    // Factory: scene with triangles (pyramide + spheres)
    static Scene build_triangle_scene();

    // Factory: Cornell Box (classique)
    static Scene build_cornell_box();

    // Factory: scene showcase (lumineuse, variee)
    static Scene build_showcase();

    // Factory: colonnes de verre et metal dans une salle
```

```
    static Scene build_hall();

    // Factory: galaxie de spheres (stress test ~1000 objets)
    static Scene build_galaxy();

    // Factory: scene minimaliste noir/blanc
    static Scene build_minimal();

private:
    std::vector<std::unique_ptr<Material» materials_;
};

#endif // RT_SCENE_SCENE_H
```

7.53 src/utils/cuda_utils.h File Reference

7.54 cuda_utils.h

Go to the documentation of this file.

```
#ifndef RT_UTILS_CUDA_UTILS_H
#define RT_UTILS_CUDA_UTILS_H

#ifdef __CUDACC__
#include <cstdio>

#define CUDA_CHECK(call) do { \
    cudaError_t err = (call); \
    if (err != cudaSuccess) { \
        fprintf(stderr, "CUDA error at %s:%d: %s\n", __FILE__, __LINE__, cudaGetErrorString(err)); \
        return; \
    } \
} while(0)

#endif // __CUDACC__

#endif // RT_UTILS_CUDA_UTILS_H
```

7.55 src/utils/denoise.cpp File Reference

```
#include "utils/denoise.h"
#include <cmath>
#include <vector>
```

Functions

- void denoise_bilateral (ImageBuffer &image, int radius, double sigma_color)

7.55.1 Function Documentation

7.55.1.1 denoise_bilateral()

```
void denoise_bilateral (
            ImageBuffer & image,
            int radius,
            double sigma_color)
```

Definition at line 5 of file denoise.cpp.

7.56 denoise.cpp

Go to the documentation of this file.

```
#include "utils/denoise.h"
#include <cmath>
#include <vector>

void denoise_bilateral(ImageBuffer& image, int radius, double sigma_color) {
    int w = image.width(), h = image.height();
    std::vector<Color> output(w * h);
    double sigma_spatial = (double)radius;

    for (int y = 0; y < h; y++) {
        for (int x = 0; x < w; x++) {
            Color center = image.get_pixel(x, y);
            Color sum(0, 0, 0);
            double weight_sum = 0;

            for (int dy = -radius; dy <= radius; dy++) {
                for (int dx = -radius; dx <= radius; dx++) {
                    int nx = x + dx, ny = y + dy;
                    if (nx < 0 || nx >= w || ny < 0 || ny >= h) continue;

                    Color neighbor = image.get_pixel(nx, ny);

                    // Distance spatiale
                    double spatial_dist = std::sqrt((double)(dx*dx + dy*dy));
                    double ws = std::exp(-spatial_dist / (2.0 * sigma_spatial * sigma_spatial));

                    // Difference de couleur
                    double dr = center.x() - neighbor.x();
                    double dg = center.y() - neighbor.y();
                    double db = center.z() - neighbor.z();
                    double color_dist = dr*dr + dg*dg + db*db;
                    double wc = std::exp(-color_dist / (2.0 * sigma_color * sigma_color));

                    double w_total = ws * wc;
                    sum += neighbor * w_total;
                    weight_sum += w_total;
                }
            }

            if (weight_sum > 0)
                output[y * w + x] = sum / weight_sum;
            else
                output[y * w + x] = center;
        }
    }

    // Ecrire les resultats
    for (int y = 0; y < h; y++)
        for (int x = 0; x < w; x++)
            image.set_pixel(x, y, output[y * w + x]);
}
```

7.57 src/utils/denoise.h File Reference

```
#include "utils/image.h"
```

Functions

- void denoise_bilateral (ImageBuffer &image, int radius=3, double sigma_color=0.1)

7.57.1 Function Documentation

7.57.1.1 denoise_bilateral()

```
void denoise_bilateral (
            ImageBuffer & image,
            int radius = 3,
            double sigma_color = 0.1)
```

Definition at line 5 of file denoise.cpp.

7.58 denoise.h

Go to the documentation of this file.

```
00001 #ifndef RT_UTILS_DENOISE_H
00002 #define RT_UTILS_DENOISE_H
00003
00004 #include "utils/image.h"
00005
00006 // Simple bilateral denoise : lisse le bruit tout en preservant les contours
00007 // sigma_spatial : rayon du filtre (en pixels)
00008 // sigma_color : sensibilite aux differences de couleur
00009 void denoise_bilateral(ImageBuffer& image, int radius = 3, double sigma_color = 0.1);
00010
00011 #endif
```

7.59 src/utils/image.cpp File Reference

```
#include "utils/image.h"
#include <fstream>
#include <algorithm>
#include <cmath>
#include <stdexcept>
```

Functions

- static double aces_tonemap (double x)

7.59.1 Function Documentation

7.59.1.1 aces_tonemap()

```
double aces_tonemap (
            double x)  [static]
```

Definition at line 19 of file image.cpp.

7.60 image.cpp

Go to the documentation of this file.

```
#include "utils/image.h"
#include <fstream>
#include <algorithm>
#include <cmath>
#include <stdexcept>

ImageBuffer::ImageBuffer(int width, int height)
    : width_(width), height_(height), pixels_(width * height, Color(0, 0, 0)) {}

void ImageBuffer::set_pixel(int x, int y, const Color& color) {
    pixels_[index(x, y)] = color;
}

Color ImageBuffer::get_pixel(int x, int y) const {
    return pixels_[index(x, y)];
}

// ACES filmic tone mapping + sRGB gamma
static double aces_tonemap(double x) {
    // ACES filmic curve: meilleur rendu que simple sqrt gamma
    double a = x * (2.51 * x + 0.03);
    double b = x * (2.43 * x + 0.59) + 0.14;
    return std::max(0.0, std::min(1.0, a / b));
}

int ImageBuffer::to_byte(double linear) {
    double tonemapped = aces_tonemap(std::max(0.0, linear));
    // sRGB gamma approximation
    double corrected = std::sqrt(tonemapped);
    return static_cast<int>(255.999 * corrected);
}

void ImageBuffer::write_ppm(std::ostream& out) const {
    out « "P3\n" « width_ « ' ' « height_ « "\n255\n";
    for (int y = 0; y < height_; ++y) {
        for (int x = 0; x < width_; ++x) {
            const Color& c = pixels_[index(x, y)];
            out « to_byte(c.x()) « ' '
                « to_byte(c.y()) « ' '
                « to_byte(c.z()) « '\n';
        }
    }
}

void ImageBuffer::write_ppm(const std::string& filename) const {
    std::ofstream file(filename);
    if (!file)
        throw std::runtime_error("Cannot open file: " + filename);
    write_ppm(file);
}

void ImageBuffer::write_ppm_binary(const std::string& filename) const {
    std::ofstream file(filename, std::ios::binary);
    if (!file)
        throw std::runtime_error("Cannot open file: " + filename);

    // Header P6 (toujours en texte)
    file « "P6\n" « width_ « ' ' « height_ « "\n255\n";

    // Pixel data en binaire (3 octets par pixel)
    for (int y = 0; y < height_; ++y) {
        for (int x = 0; x < width_; ++x) {
            const Color& c = pixels_[index(x, y)];
            unsigned char rgb[3] = {
                (unsigned char)to_byte(c.x()),
                (unsigned char)to_byte(c.y()),
                (unsigned char)to_byte(c.z())
            };
            file.write(reinterpret_cast<char*>(rgb), 3);
        }
    }
}
```

7.61 src/utils/image.h File Reference

```
#include "math/vec3.cuh"
#include <vector>
```

```
#include <string>
#include <ostream>
```

Classes

- class ImageBuffer

7.62 image.h

Go to the documentation of this file.

```
#ifndef RT_UTILS_IMAGE_H
#define RT_UTILS_IMAGE_H

#include "math/vec3.cuh"
#include <vector>
#include <string>
#include <ostream>

class ImageBuffer {
public:
    ImageBuffer(int width, int height);

    int width() const { return width_; }
    int height() const { return height_; }

    void set_pixel(int x, int y, const Color& color);
    Color get_pixel(int x, int y) const;

    // Write PPM P3 (text) format
    void write_ppm(std::ostream& out) const;
    void write_ppm(const std::string& filename) const;

    // Write PPM P6 (binary) format - plus compact, plus rapide
    void write_ppm_binary(const std::string& filename) const;

private:
    int width_;
    int height_;
    std::vector<Color> pixels_;

    int index(int x, int y) const { return y * width_ + x; }

    // Apply gamma correction (gamma 2.0 = sqrt)
    static int to_byte(double linear);
};

#endif // RT_UTILS_IMAGE_H
```

7.63 src/utils/timer.h File Reference

```
#include <chrono>
```

Classes

- class Timer

7.64 timer.h

Go to the documentation of this file.

```
#ifndef RT_UTILS_TIMER_H
#define RT_UTILS_TIMER_H

#include <chrono>

class Timer {
public:
    Timer() : start_(std::chrono::high_resolution_clock::now()) {}

    void reset() {
        start_ = std::chrono::high_resolution_clock::now();
    }

    // Returns elapsed time in seconds
    double elapsed() const {
        auto now = std::chrono::high_resolution_clock::now();
        return std::chrono::duration<double>(now - start_).count();
    }

    // Returns elapsed time in milliseconds
    double elapsed_ms() const {
        return elapsed() * 1000.0;
    }

private:
    std::chrono::high_resolution_clock::time_point start_;
};

#endif // RT_UTILS_TIMER_H
```